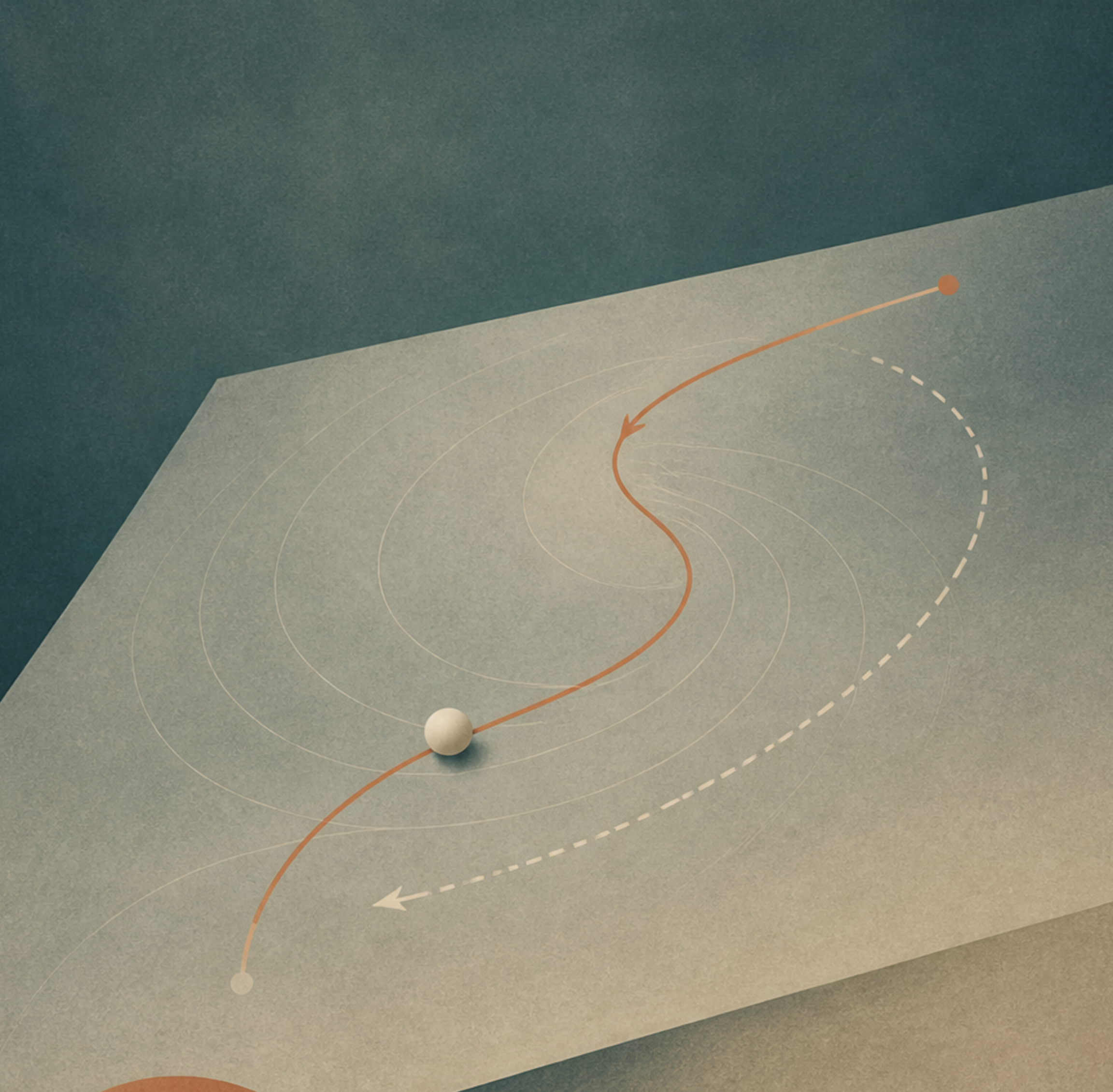
Classical Mechanics

Core track, 2011 Fall Modern Physics Stanford Partial



Leonard Susskind lecture notes

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

ABOUT THESE NOTES

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CHAPTER 1

STATES, REVERSIBILITY, AND THE ORDER OF MOTION

Classical mechanics is the foundation on which the rest of physics is built. This is not merely because it describes particles, planets, and mechanical devices. Its deeper contribution is a framework for asking what a physical state is, how a state changes, what information a law of motion must preserve, and why quantities such as energy and momentum can remain constant. The same questions survive in more abstract form throughout modern physics.

We shall not begin with a falling body or with Newton's equation. Instead, we shall strip dynamics down to its logical skeleton. Once that skeleton is visible in a world with only a few possible states, the passage to continuous motion will be much easier to understand.

1.1 A Stroboscopic World

Ordinary time appears continuous: the second hand of a watch sweeps through every intermediate position. For the moment, imagine instead that the world is illuminated by a strobe. We inspect it only at equally spaced beats,

$$t_n = n\Delta t, \qquad n \in \mathbb{Z}, \tag{1.1}$$

where Δt may be a second, a millisecond, or any other fixed interval. A law of motion is then an update rule

$$q_n \longmapsto q_{n+1}. \tag{1.2}$$

The question "What happens next?" has become literal: given the state at one flash, what will be seen at the following flash?

We also choose systems much simpler than anything in nature. Their purpose is not realism but clarity. The simplest nontrivial example is a coin laid on a table. It has two possible configurations, heads and tails, which we denote by H and T. Its space of possibilities is

$$\Gamma = \{H, T\}. \tag{1.3}$$

Definition 1.1. The *phase space* is the set of all possible states of a system. A *state* is a complete specification of the information needed to determine what happens next.

This definition is operational. A description is not a state merely because we call it one. If it does not contain enough information to determine the next step, it is incomplete and the phase space must be enlarged.

A world with only one state would be still simpler, but nothing could happen there. Its only possible law would send that state back to itself forever. Two states provide the first opportunity for a genuine alternative.

1.2 Two Elementary Laws

One possible law leaves the coin unchanged:

$$H \longmapsto H, \qquad T \longmapsto T. \tag{1.4}$$

We represent the rule by drawing a loop from each state back to itself. Starting at the tail of an arrow and following it to its head tells us the next state.

A second law flips the state at every beat:

$$H \longmapsto T, \qquad T \longmapsto H. \tag{1.5}$$

Starting from heads, the complete history is

$$H, T, H, T, H, T, \ldots. \tag{1.6}$$

Both laws are deterministic. Once the state at one instant is known, the state at every later instant is fixed.

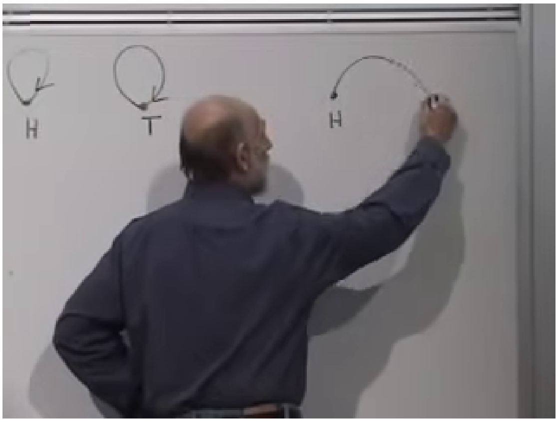

Figure 1.1: The board at 00:06:58. The two self-loops are complete, and the first cross-arrow of the alternating law is being drawn. The reverse arrow is supplied by the spoken rule, not inferred from the unfinished chalk mark.

Lecture frame: 00:06:58.

Figure 1.2: The two reversible laws on a two-state phase space.

Question from the audience. Does determinism already imply that the law is reversible?[a]

Response. Not by itself. Determinism says that the present has a unique future. Reversibility requires the converse as well: the present must have a unique past. The two laws above happen to satisfy both conditions, but a larger state space lets us separate them.

[a] Lecture timestamp: 00:06:43–00:06:50.

1.3 More States and More Orbits

Replace the coin by a die. A die has six states, labelled $1, 2, \ldots, 6$. One possible law is the six-cycle

$$1 \mapsto 2 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6 \mapsto 1. \tag{1.7}$$

The physical tumbling needed to display these faces might look complicated, but the abstract law is simple: every state has a prescribed successor.

The numerical order is inessential. We could relabel the same cycle, or divide the states into several disjoint cycles. For example,

$$1 \mapsto 1, \qquad 2 \mapsto 3 \mapsto 4 \mapsto 2, \qquad 5 \mapsto 6 \mapsto 5. \tag{1.8}$$

This law contains a fixed point, a three-cycle, and a two-cycle. Wherever the system starts, the arrow diagram determines its future indefinitely. It also determines its past: every state has exactly one predecessor.

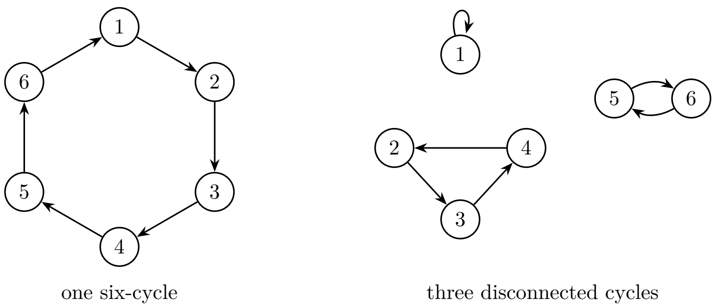

Figure 1.3: A finite reversible law is built from disjoint cycles, including cycles of length one.

In practice, a complicated system may be so sensitive to small errors that reconstructing its history is hopeless. That practical difficulty does not alter the ideal law. If the state were known exactly, its trajectory could be followed both forward and backward.

1.4 Forbidden Maps

Consider three states with the rule

$$1 \mapsto 2, \qquad 2 \mapsto 3, \qquad 3 \mapsto 2. \tag{1.9}$$

The future is unique. From any state there is one arrow to follow. The past is not unique, however. If the system is now at 2, it may have come from 1 or from 3. Two distinct histories have merged.

Reversing every arrow exchanges the defect. The past then becomes unique, but state 2 has two possible futures. Nothing in the state tells us whether to choose 1 or 3.

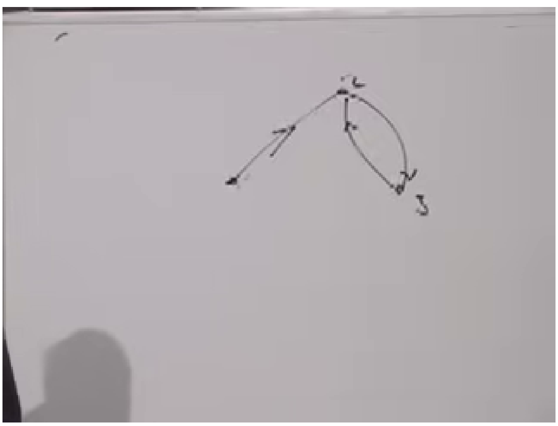

Figure 1.4: The board at 00:12:46, showing the merging three-state map. The two arrows entering state 2 make its past ambiguous.

Lecture frame: 00:12:46.

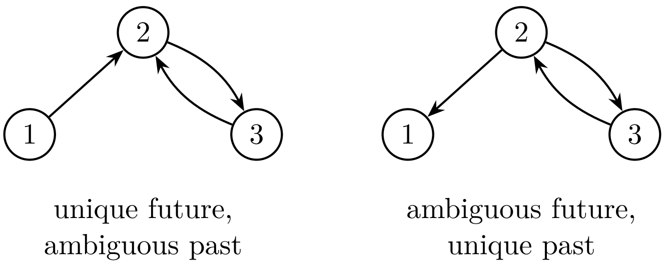

Figure 1.5: The two ways a state map can fail the classical one-in, one-out condition.

Question from the audience. Could we repair one of these diagrams by erasing an arrow, or by letting a chain terminate at a state?[a]

Response. Erasing an outgoing arrow leaves a state with no prescribed future. Turning the endpoint into a fixed point repairs the future but generally spoils the past: the fixed point then has both its self-loop and an incoming arrow from the chain. One way or another, a missing or dangling arrow violates uniqueness.

[a]Lecture timestamp: 00:14:46–00:15:34.

The criterion can now be stated exactly.

Proposition 1.2. *For a finite phase space* Γ*, a reversible deterministic update is a bijection*

$$\sigma : \Gamma \longrightarrow \Gamma. \tag{1.10}$$

Equivalently, every state has exactly one outgoing arrow and exactly one incoming arrow.

Proof. One outgoing arrow gives a unique future, and one incoming arrow gives a unique past. If either arrow is missing, evolution stops in one time direction. If either is duplicated, evolution is ambiguous in that direction. On a finite set, a map with one incoming and one outgoing arrow at every point is a permutation, hence a disjoint union of cycles. □

The inevitable cycling of a finite system may seem dull, but it is a consequence of finiteness, not of reversibility. An infinite phase space need not recur. If states are labelled by every integer and

$$n \mapsto n+1, \qquad n \in \mathbb{Z}, \tag{1.11}$$

then each state still has one predecessor and one successor, while the trajectory marches along the line forever.

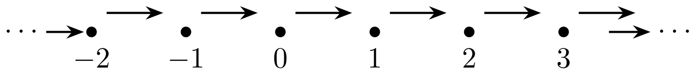

Figure 1.6: A reversible law on an infinite phase space need not return to its starting point.

1.5 Conservation as Memory

Phase space may split into disconnected components. Some initial states may lie on an infinite chain, while others lie on closed cycles. Once the system starts in one component, it can never reach another. That persistent distinction is the simplest model of a conservation law.

Assign a label C to each component, for example

$$C(q) = \begin{cases} +1, & q \in \Gamma_+, \\ -1, & q \in \Gamma_-. \end{cases} \tag{1.12}$$

Since the dynamics never crosses between Γ_+ and Γ_-,

$$C(q_{n+1}) = C(q_n). \tag{1.13}$$

The conserved quantity is a memory of where the motion began. By contrast, if the labels $+1$ and -1 are assigned to the alternating states H and T, the label changes on every beat and is not conserved.

Energy and momentum are much richer quantities, but the logical form is already present: a conserved function is constant along each allowed trajectory.

The same idea may be called *information conservation*. In the merging map of Eq. (1.9), arriving at state 2 erases the distinction between two possible pasts. In a reversible map, no such distinction disappears. Exact knowledge of the present determines both where the system came from and where it is going.

Why should fundamental laws have this form? At this level there is no deeper derivation. It is an empirical fact about the known fundamental dynamics. Perhaps a future "law of laws" would explain why information is conserved. Quantum mechanics changes the mathematical language, but its closed-system evolution is also information preserving. Our concern here is the classical version.

The qualification *closed system* matters. Friction, diffusion, and thermodynamic irreversibility often look many-to-one because unobserved environmental details have been discarded. Such effective descriptions do not overturn the reversible microscopic framework being isolated here.

1.6 When the Visible Symbol Is Not the State

An audience question exposes the most important subtlety in the definition of state.

Question from the audience. Suppose the next coin value depends not only on the present face but on the two most recent faces. Seeing H now would not determine what comes next. Has determinism been lost?[a]

Response. Not necessarily. The correct conclusion is that H or T alone was never a complete state. If two entries are needed to apply the law, the state must be the ordered pair of entries. What we call phase space must contain all the memory on which the update depends.

[a] Lecture timestamp: 00:22:09–00:23:18.

This observation leads directly to continuous mechanics. For a particle moving on a line, is its present position a complete state?

1.7 A Particle on a Line

Let x denote the particle's position. Knowing x alone is plainly insufficient. Two particles can occupy the same point while moving in opposite directions, or while moving at different speeds. To predict the next position we must also know the velocity

$$v = \dot{x} = \frac{dx}{dt}. \tag{1.14}$$

Equivalently, over a short discrete interval we need both the present position and the preceding one, since their difference determines the velocity. The continuous state is therefore the pair

$$(x, v). \tag{1.15}$$

The phase space of a particle on a line is two-dimensional, even though the particle moves in one-dimensional physical space. In canonical mechanics one often uses (x, p) with $p = mv$; for fixed mass this carries the same information as (x, v).

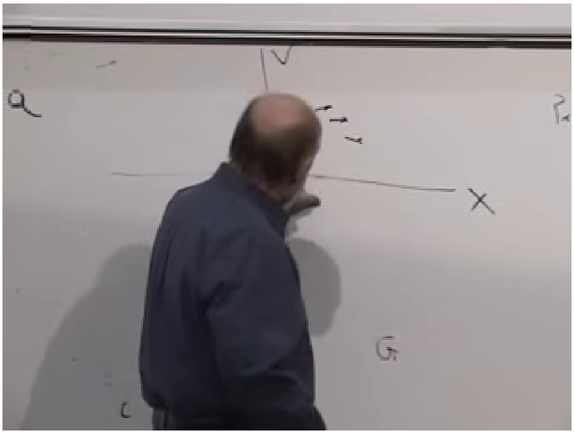

Figure 1.7: The board at 00:38:10 revisits the (x, v) phase-space plane. The horizontal x-axis and vertical v-axis are visible; the small arrows are qualitative, not a complete measured vector field.

Lecture frame: 00:38:10.

Imagine filling the (x, v) plane with short arrows. At a point with $v = 0$, the immediate horizontal displacement vanishes. At positive v, the particle moves to the right; a larger positive velocity produces a larger displacement during the same time interval. At negative v, it moves to the left. The arrow must also contain any change in velocity produced by the force, but even this elementary horizontal component shows why both coordinates are needed.

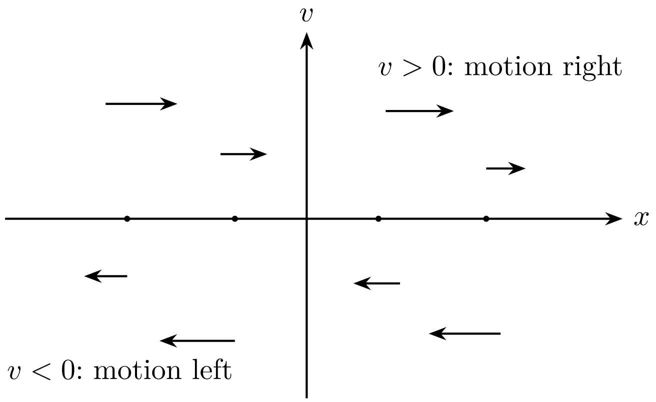

Figure 1.8: The kinematic part of the phase-space flow. The horizontal component of the arrow is set by $\dot{x} = v$.

Question from the audience. If continuous motion is approached by making the time steps arbitrarily small, why did velocity appear as an additional coordinate?[a]

Response. Because a rule may depend on more than one adjacent sample. In discrete language, the difference between the last two positions determines a velocity. We can either say that the rule remembers two positions or enlarge the state to include position and velocity. These are two descriptions of the same information. The phase space has one axis for the present position and another for the information carried by the previous step.

[a] Lecture timestamp: 00:27:09–00:28:18.

For a single particle in three-dimensional space, there are three position coordinates and three velocity coordinates, so phase space has six coordinates.

1.8 Determinism and Practical Prediction

Classical positions and velocities are real numbers, and no experiment specifies a real number with infinite precision. Therefore exact determinism of the equations does not imply unlimited practical prediction.

Imagine demanding the position of every molecule in the room thirty seconds from now. Some finite precision in the present data would, in principle, suffice for that finite task. The same data may fail at forty seconds because its errors have had longer to grow. To extend the forecast, we must improve the initial data. Classical mechanics promises that a sufficient precision exists for each fixed interval; it does not promise that any attainable measurement remains useful forever.

Suppose an initial uncertainty $\delta z(0)$ in phase space grows approximately as

$$\delta z(t) \sim \delta z(0)e^{\lambda t}, \qquad \lambda > 0. \tag{1.16}$$

Then predicting for a longer time requires exponentially more precise initial data. This is the characteristic difficulty of chaotic systems. The law still assigns one trajectory to each exact initial state, but nearby initial states separate rapidly.

Question from the audience. Can we call such a system deterministic even when it is not practically predictable?[a]

Response. The distinction is useful provided it is stated carefully. The equations are deterministic: infinitely precise initial data select a unique trajectory. Actual predictions are limited because initial data are never infinitely precise. For any fixed prediction interval, classical theory says that sufficiently accurate initial data exist; extending the interval generally demands still greater accuracy. Some speakers use "deterministic" and "predictable" interchangeably, but the equations-versus-measurement distinction removes the apparent contradiction.

[a]Lecture timestamp: 00:28:20–00:31:55.

This is not a quantum uncertainty argument. Quantum mechanics has been deliberately set aside here. The limitation comes from finite classical measurement precision and its possible amplification by the dynamics.

1.9 The Order of the Equation

The need for both position and velocity is encoded in the order of the equations of motion. A first-order differential equation contains a first time derivative. A second-order equation contains a second derivative,

$$\dot{x} = \frac{dx}{dt}, \qquad \ddot{x} = \frac{d^2x}{dt^2} = a. \tag{1.17}$$

The dot notation will be used throughout: one dot means one time derivative, two dots mean two.

To isolate the logic, consider the intentionally non-Newtonian first-order law used on the board,

$$F(x) = m\dot{x}. \tag{1.18}$$

It is a schematic law rather than a physically dimensioned Newtonian equation; a conventional first-order drag model would replace m by a coefficient with the proper units. Its pedagogical virtue is that position alone determines velocity:

$$\dot{x} = \frac{F(x)}{m}. \tag{1.19}$$

Once x is known, $F(x)$ is known, hence $\dot{x}$ is known.

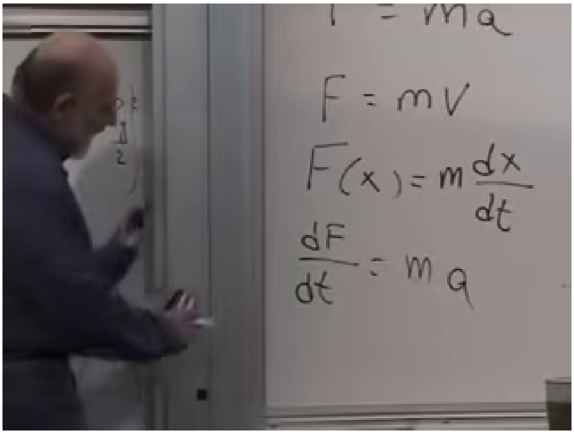

Figure 1.9: The board at 00:35:25. The deliberately phony law $F = mV$, its position-dependent form $F(x) = m\,dx/dt$, and the first differentiated relation are all legible.

Lecture frame: 00:35:25.

Differentiating Eq. (1.18) gives

$$\frac{dF}{dt} = m\ddot{x}, \tag{1.20}$$

$$\frac{dF}{dt} = \frac{dF}{dx}\frac{dx}{dt} = F'(x)\dot{x}. \tag{1.21}$$

Therefore

$$\ddot{x} = \frac{F'(x)}{m}\dot{x} = \frac{F'(x)F(x)}{m^2}. \tag{1.22}$$

Higher derivatives follow by differentiating again. Subject to the usual smoothness assumptions, specifying x at one time fixes the entire local Taylor expansion and hence the motion allowed by this first-order law.

Newton's actual equation is second order:

$$F(x) = m\ddot{x}. \tag{1.23}$$

Knowing x determines the force and therefore the acceleration, but it does not determine the velocity. The velocity must be supplied independently as part of the initial state. Once x

and v are known, Newton's equation can be rewritten as the first-order system on phase space

$$\dot{x} = v, \qquad \dot{v} = \frac{F(x)}{m}. \tag{1.24}$$

These two equations are the continuous version of the arrow leaving each phase-space point. They determine acceleration and, by further differentiation, all higher derivatives of the trajectory.

1.10 The Two-Step Coin Revisited

The discussion now returns to the coin because the discrete example makes the same point without calculus.

Question from the audience. Does the "bifurcated" heads-and-tails diagram require two pieces of information rather than one?[a]

Response. No. The diagram still describes one system. The number of arrows or clauses in its law is not the number of independent pieces of state information. In Eqs. (1.4) and (1.5), one symbol—H or T—is enough to determine the next symbol. A state may have many possible values without requiring several successive values as initial data.

[a]Lecture timestamp: 00:38:18–00:40:41.

If the rule depends on the previous two symbols, the four possible states are

$$\Gamma_2 = \{HH, HT, TH, TT\}. \tag{1.25}$$

The state update has the shift-register form

$$(a_n, a_{n+1}) \longmapsto (a_{n+1}, f(a_n, a_{n+1})). \tag{1.26}$$

The classroom builds a truth table in real time. Several first attempts fail: either the proposed output secretly depends on only one of the two entries, or two pair states flow into the same successor and information is lost. Those corrections are not incidental. They demonstrate how the one-in, one-out test detects an incomplete or irreversible rule.

One workable table is

previous pair	next symbol
HH	H
HT	T
TH	T
TT	H

(1.27)

and it induces

$$\begin{aligned} HH \mapsto HH, \quad HT \mapsto TT, \\ TT \mapsto TH, \quad TH \mapsto HT. \end{aligned} \tag{1.28}$$

Thus HH is a fixed point and the other three pair states form a cycle.

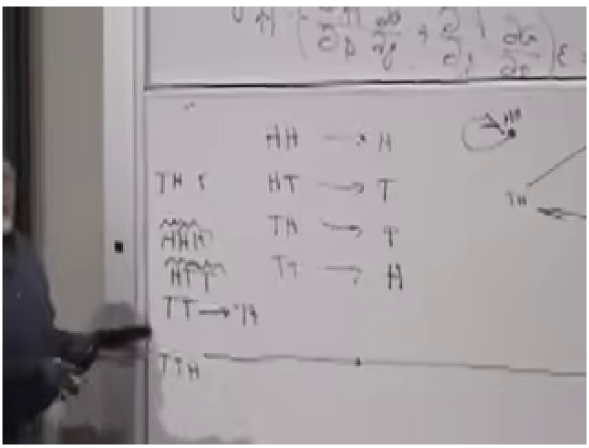

Figure 1.10: The board at 00:45:43 after the table has been corrected. The four pair states on the left generate a fixed point and a three-cycle on the right.

Lecture frame: 00:45:43.

A student recognizes the output rule as exclusive OR if $H = 0$ and $T = 1$:

$$a_{n+2} = a_n + a_{n+1} \pmod 2. \tag{1.29}$$

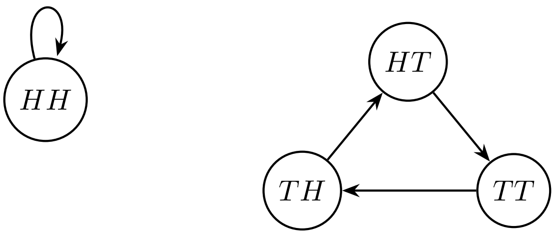

Figure 1.11: Clean reconstruction of the reversible four-state update in Eq. (1.28).

The pair update can then be written

$$\begin{pmatrix} a_{n+1} \\ a_{n+2} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n+1} \end{pmatrix} \pmod 2. \tag{1.30}$$

Its matrix has nonzero determinant modulo two, so the pair update is invertible. The new symbol alone does not preserve the past, but the shifted pair does. Once again, the cure is not to abandon determinism; it is to identify the complete state.

Question from the audience. What if still more history is needed—for example, three symbols rather than two?[a]

Response. Then the state must contain three symbols and the phase space contains 2^3 possible histories. The same principle applies in continuous mechanics. If position, velocity, and acceleration had to be specified independently, the fundamental equation would be third order and the state would be $(x, \dot{x}, \ddot{x})$. Experiment tells us that ordinary Newtonian mechanics is second order: position and velocity are enough.

[a]Lecture timestamp: 00:46:32–00:47:30.

1.11 What the Toy World Has Taught Us

The stroboscopic coin was not a diversion from mechanics. It isolated the structure that Newton's equations later realize. A state is all and only the information required to continue the motion. A classical closed-system law carries that information uniquely toward both the future and the past. On a finite phase space the law is a permutation; on a continuous phase space it is a flow.

For a particle, the state cannot be position alone. It must include velocity, and Newton's second-order equation is precisely what makes those two pieces of initial data necessary. Conservation laws, reversibility, the order of differential equations, and the geometry of phase space are therefore not separate topics. They are different expressions of one organizing principle: physical evolution must act on complete information.

CHAPTER 2

INITIAL DATA, CONSERVATION, AND THE ACTION

The laws of mechanics can be stated in two remarkably different ways. One description is local: specify the state now, use a differential equation, and advance the motion from one instant to the next. The other description treats an entire trajectory as a single object and selects the physical path by a stationary principle. Our purpose is to understand why both descriptions carry the same physics.

We shall approach the global principle slowly. First we return to initial data and the order of a differential equation. We then review energy and momentum, not because their elementary Newtonian derivations are the deepest proofs of those laws, but because they reveal structures that survive far beyond the derivations. Only after that preparation shall we pass from ordinary functions to functionals and uncover the action.

2.1 What the Order of a Law Demands

Begin in one dimension with a deliberately false law,

$$F = m\dot{x} = m\frac{dx}{dt}. \tag{2.1}$$

It says that a force is needed merely to sustain motion: remove the force and the body stops at once. This is not Newtonian physics. Its value is precisely that it lets us see what a first-order equation would demand of its initial data.

Suppose the force is prescribed by position,

$$F(x) = m\dot{x}. \tag{2.2}$$

If the position is known at one instant, then $F(x)$ is known, and the law immediately supplies the velocity,

$$\dot{x} = \frac{F(x)}{m}. \tag{2.3}$$

The same datum fixes the acceleration. Differentiating the force along the motion gives

$$\frac{dF}{dt} = \frac{dF}{dx}\dot{x}, \tag{2.4}$$

and differentiating Eq. (2.2) gives

$$m\ddot{x} = \frac{dF}{dx}\dot{x}. \tag{2.5}$$

Differentiate once more and use the product and chain rules:

$$\begin{aligned} m\,\dddot{x} &= \frac{d}{dt}\left(F'(x)\dot{x}\right) &(2.6)\\ &= F''(x)\dot{x}^{\,2} + F'(x)\ddot{x}. &(2.7)\end{aligned}$$

Thus position determines velocity, velocity and position determine acceleration, and the same chain continues through jerk and every higher time derivative. In this first-order world, one number $x(t_0)$ is enough to start the motion.

Throughout this calculation the mass is constant. A visible subsystem may seem to lose mass if a piece falls away, but then the discarded piece has simply been omitted from the description. For the complete closed system, the mass carried by the calculation has not disappeared.

Now replace the false law by Newton's law,

$$F(x) = m\ddot{x}. \tag{2.8}$$

Position still determines force and therefore acceleration, but it does not determine velocity. Two bodies can pass through the same point with different velocities and subsequently follow different motions. We must supply

$$x(t_0) = x_0, \qquad \dot{x}(t_0) = v_0. \tag{2.9}$$

Once these two data are known, Newton's equation supplies $\ddot{x}$, and successive differentiations supply the rest. The need for position and velocity is not an arbitrary convention: it is the signature of a second-order law.

For many particles, the statement simply expands. We must specify every particle coordinate, every corresponding velocity component, and the force law, which may depend on the relative positions of all the particles.

Question from the audience. Why does Newton's law require both position and velocity, whereas the false first-order law requires only position? [a]

Response. The distinction is the order of the equation. In $F(x) = m\dot{x}$, the position determines the first derivative directly. In $F(x) = m\ddot{x}$, it determines only the second derivative. The first derivative remains independent and must be included in the initial state.

[a] Lecture timestamp: 00:09:14–00:11:57.

2.2 Energy and Conservative Forces

Energy conservation is more general than the elementary argument we are about to give. It remains meaningful in electrodynamics, special and general relativity, and quantum mechanics. Here we ask the narrower question: under what conditions does ordinary Newtonian particle mechanics conserve $T + U$?

The qualification matters. A friction force appears to destroy mechanical energy if the resulting heat and the microscopic degrees of freedom of the environment are ignored. An even sharper counterexample is a tangential force that continually pushes a constrained particle around a circle. After one revolution the particle returns to the same position with a larger speed. Any single-valued potential has returned to its original value, while the kinetic energy has increased. Such a force cannot be written as the gradient of an ordinary potential.

For the class of forces considered here, there is a potential energy $U(\mathbf{x})$ such that

$$F_i = -\frac{\partial U}{\partial x_i}, \qquad \mathbf{F} = -\boldsymbol{\nabla} U. \tag{2.10}$$

The minus sign says that the force points toward decreasing potential energy.

Define

$$T = \frac{1}{2} m \mathbf{v}^2 = \frac{1}{2} m \sum_i v_i^2, \qquad E = T + U. \tag{2.11}$$

The board develops the proof in the useful order: write the two energies, differentiate them separately, and only then combine them.

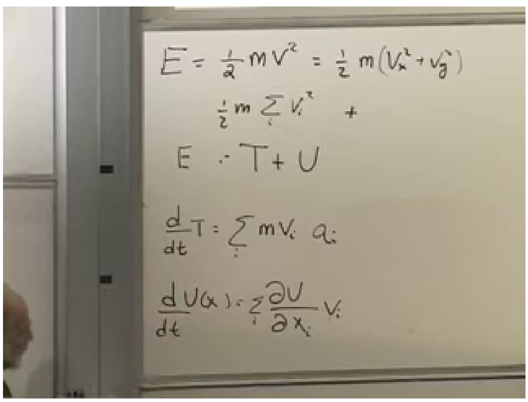

Figure 2.1: The energy calculation at 00:22:24. The upper handwritten label is read as T: the formula, transcript, and later identity $E = T + U$ fix that interpretation.

Lecture frame: 00:22:24.

For the kinetic energy,

$$\frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2} m \sum_i v_i^2 \right) \tag{2.12}$$

$$= \sum_i m v_i \frac{dv_i}{dt} = \sum_i m v_i a_i. \tag{2.13}$$

The potential changes because the particle moves through space. The multivariable chain rule gives

$$\frac{dU}{dt} = \sum_i \frac{\partial U}{\partial x_i}\frac{dx_i}{dt} \tag{2.14}$$

$$= \sum_i \frac{\partial U}{\partial x_i} v_i. \tag{2.15}$$

Adding the two derivatives,

$$\frac{dE}{dt} = \sum_i v_i \left(m a_i + \frac{\partial U}{\partial x_i} \right) \tag{2.16}$$

$$= \sum_i v_i (F_i - F_i) = 0. \tag{2.17}$$

The first F_i is Newton's ma_i; the second comes from $\partial_i U = -F_i$. The cancellation is exact.

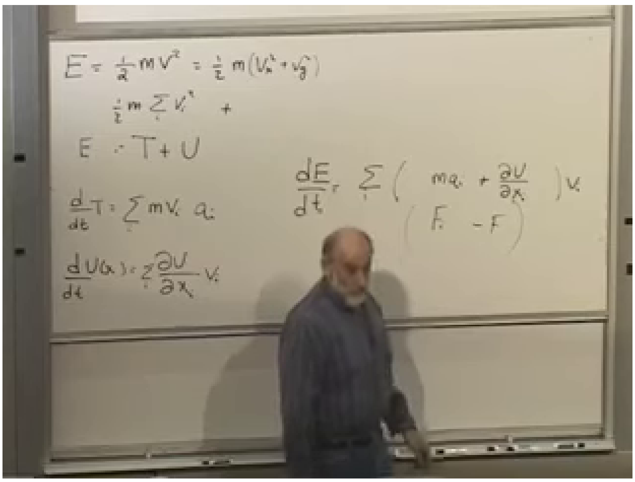

Figure 2.2: The completed energy derivative at 00:24:01. The right-hand bracket is the decisive $F_i - F_i$ cancellation.

Lecture frame: 00:24:01.

Nothing in the argument restricts i to the three directions of one particle. It may range over all coordinates of all particles. If each generalized force is the negative derivative of one common potential and the potential has no explicit time dependence, the total energy of the complete system is conserved.

Question from the audience. Why denote potential energy by U rather than the common letter V?[a]

Response. Either notation is possible. Here U avoids collisions: V is often used for electric voltage and v already denotes velocity. The symbol changes nothing in the physics.

[a] Lecture timestamp: 00:18:13–00:18:36.

2.3 Momentum and a Closed System

Momentum enters by rewriting the same Newtonian equation. With constant mass,

$$F_i = m\frac{dv_i}{dt} = \frac{d}{dt}(mv_i). \tag{2.18}$$

Define

$$\mathbf{p} = m\mathbf{v}. \tag{2.19}$$

Then Newton's law becomes

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}. \tag{2.20}$$

In the absence of force, $\dot{\mathbf{p}} = 0$, so a free particle carries a constant momentum. Intergalactic space is not perfectly empty, but it gives a good approximation to this ideal.

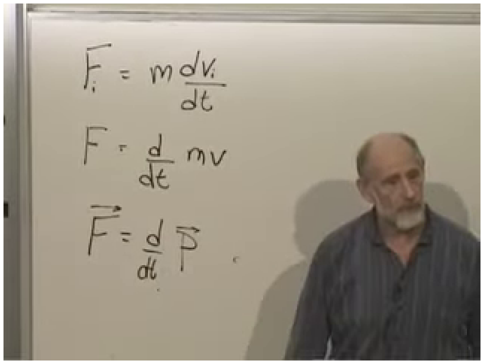

Figure 2.3: The component, scalar, and vector forms of Newton's momentum law at 00:29:00.

Lecture frame: 00:29:00.

This discussion uses a specifically Newtonian picture of time. Clocks can be synchronized everywhere, all observers agree on one universal time, and time flows uniformly. We are also considering force laws with no prescribed external time dependence. These assumptions will not survive unchanged in relativity, so they should be stated rather than hidden.

Now consider three interacting particles. Let $\mathbf{F}_{ab}$ mean the force on particle a due to particle b. Internal forces give

$$\dot{\mathbf{p}}_1 = \mathbf{F}_{12} + \mathbf{F}_{13}, \tag{2.21}$$

$$\dot{\mathbf{p}}_2 = \mathbf{F}_{21} + \mathbf{F}_{23}, \tag{2.22}$$

$$\dot{\mathbf{p}}_3 = \mathbf{F}_{31} + \mathbf{F}_{32}. \tag{2.23}$$

Newton's action–reaction law states

$$\mathbf{F}_{ab} = -\mathbf{F}_{ba}. \tag{2.24}$$

The equal-and-opposite forces lie along the line joining the pair in the elementary central-force picture, although only their cancellation is needed here.

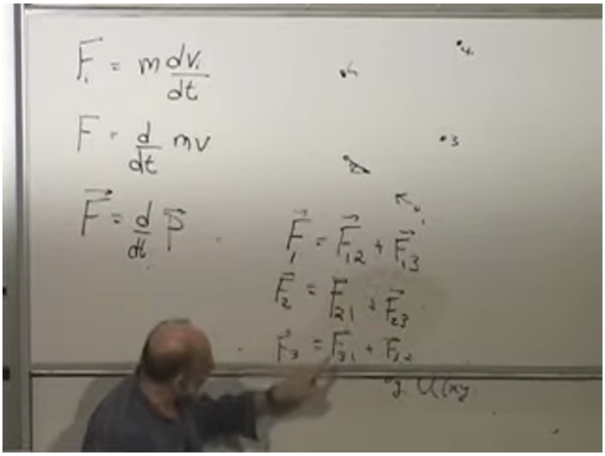

Figure 2.4: The board at 00:32:58, after the three force balances have been assembled. The frame is used for their layout; the exact subscripts are typeset in Eq. (2.23).

Lecture frame: 00:32:58.

Adding the three equations gives

$$\begin{aligned}\frac{d}{dt}(\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3) &= (\mathbf{F}_{12} + \mathbf{F}_{21}) + (\mathbf{F}_{13} + \mathbf{F}_{31}) \\ &\quad + (\mathbf{F}_{23} + \mathbf{F}_{32}) \\ &= 0. \end{aligned} \tag{2.25}$$

Therefore

$$\frac{d\mathbf{P}}{dt} = 0, \qquad \mathbf{P} = \sum_a \mathbf{p}_a. \tag{2.26}$$

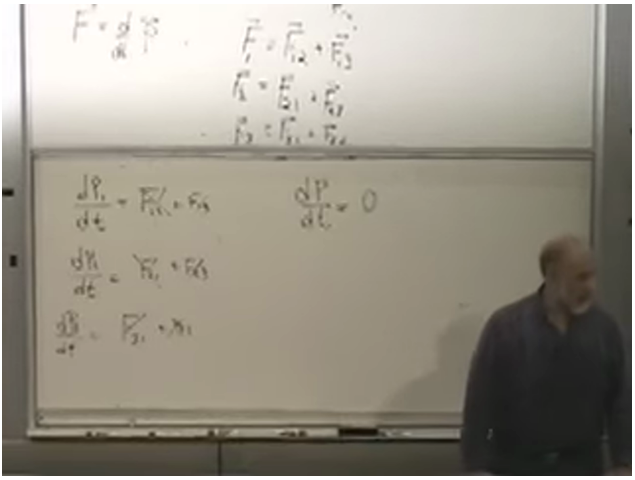

Figure 2.5: The completed cancellation board at 00:35:20. Each internal force appears once with each orientation, leaving $\dot{\mathbf{P}} = 0$.

Lecture frame: 00:35:20.

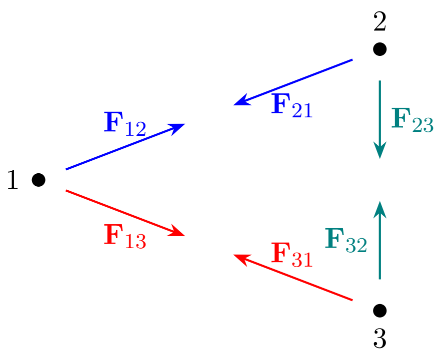

Figure 2.6: Pairwise internal forces cancel when the momentum equations for a closed system are summed.

A closed system is one for which every relevant force source has been included. If an external force is present, it does not acquire a partner inside the sum, and the result becomes $\dot{\mathbf{P}} = \mathbf{F}_{\text{ext}}$.

Question from the audience. Are action and reaction instantaneous? [a]

Response. They are instantaneous in Newtonian mechanics, where the speed at which influences propagate is effectively infinite. Real influences do not propagate instantaneously; special relativity forbids that. Yet momentum conservation survives. The Newtonian derivation is therefore less general than the conservation law it reveals.

[a] Lecture timestamp: 00:36:15–00:37:12.

Momentum conservation applies to radiation pressure, quantum systems, and relativistic physics, even though the simple pair-force proof cannot be carried unchanged into every one of those settings. There are three momentum laws, one for each spatial component, and one energy law. In special relativity these four quantities belong together as energy–momentum. This is a first hint that the conserved quantities have a deeper origin than the present calculations.

Question from the audience. Why does momentum seem to provide more conservation laws than energy?[a]

Response. Momentum has one conserved component for each independent direction of space, whereas energy is one scalar associated with evolution in time. In three spatial dimensions that gives three momentum components and one energy. Relativity later unifies the four rather than treating them as an accidental list.

[a] Lecture timestamp: 00:37:37–00:38:24.

2.4 Stationarity in Ordinary Calculus

We now prepare for what is really the central law of the course: the principle of least action. The required mathematics begins with an ordinary function of one variable, $f(y)$. At a smooth local minimum,

$$\frac{df}{dy} = 0. \tag{2.27}$$

The derivative also vanishes at a smooth maximum. It can even vanish at a flat inflection point, such as the origin of $f(y) = y^3$. The first-derivative condition therefore identifies a *stationary point*; the second derivative and higher information distinguish the possibilities.

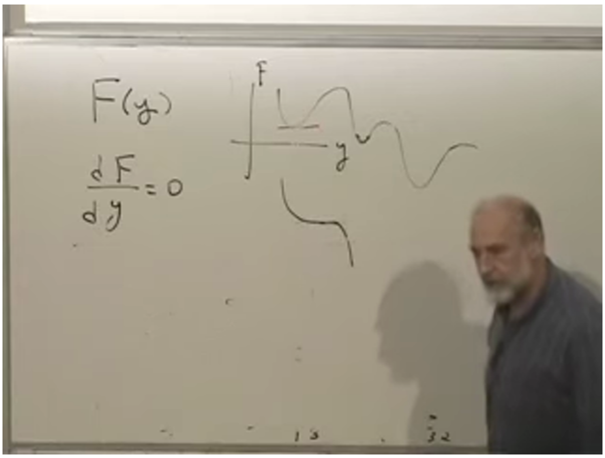

Figure 2.7: Minimum, maximum, and flat inflection behavior on the board at 00:43:20. All are stationary to first order.

Lecture frame: 00:43:20.

For a function of many variables, stationarity must hold along every coordinate direction:

$$\frac{\partial f}{\partial x_i} = 0 \qquad \text{for every } i. \tag{2.28}$$

Geometrically, at the bottom of a bowl there is no downhill first-order direction. The same equations also find maxima and saddles, so “stationary” remains the accurate word.

As an example, take

$$f(x, y) = x^2 + y^2 + xy + 3x + 2y + 7. \tag{2.29}$$

The stationarity equations are

$$\frac{\partial f}{\partial x} = 2x + y + 3 = 0, \tag{2.30}$$
$$\frac{\partial f}{\partial y} = x + 2y + 2 = 0. \tag{2.31}$$

Solving the linear system gives

$$x = -\frac{4}{3}, \qquad y = -\frac{1}{3}. \tag{2.32}$$

The constant 7 never enters the equations because a vertical shift changes the value of f, not the location of its stationary point.

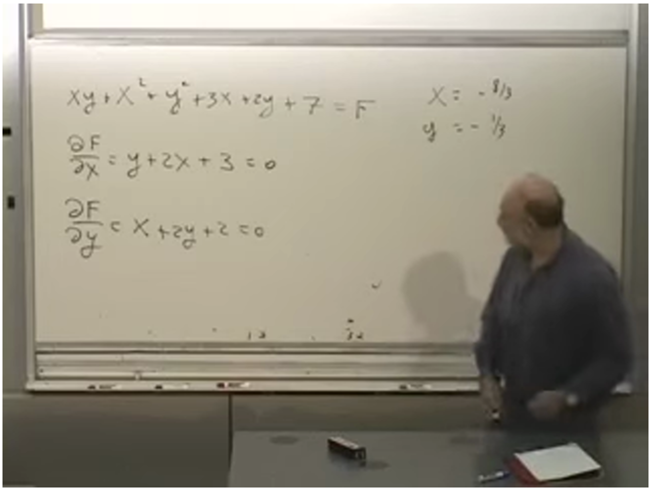

Figure 2.8: The two-variable example at 00:51:10. The displayed derivative equations are correct; the board's $x = -8/3$ is an arithmetic slip. Solving those equations gives Eq. (2.32).

Lecture frame: 00:51:10.

For n variables there are n first-order equations. They may have no solution, one solution, or several solutions. A monotone function need not have a stationary point, while a landscape with several valleys can have many.

Question from the audience. Is the physical principle really one of least action, or of stationary action?[a]

Response. Stationary action is the precise statement. The first-order change of the action vanishes; that does not by itself prove a minimum. In many elementary mechanical problems the physical path is locally minimizing, so "least action" is useful language, but the distinction should not be forgotten. Velocity-dependent interactions are one reason not to build the theory around a naive minimum test.

[a] Lecture timestamp: 00:42:05–00:46:39.

2.5 Generalized Trajectories

The basic mathematical problem of mechanics is to determine the trajectory of a system from suitable data. For 10^{23} particles, no one attempts to write every trajectory in practice; statistical or approximate descriptions are used instead. Nevertheless, the exact classical question remains well defined.

Describe a system by generalized coordinates

$$q_1, q_2, \ldots, q_N. \tag{2.33}$$

They may be Cartesian positions, angles, orientations, latitude and longitude, or any other independent variables that locate the system in configuration space. A single particle in three-dimensional space has three such coordinates; two particles have six; n unconstrained particles have $3n$.

The motion is the whole collection

$$\mathbf{q}(t) = \big(q_1(t), q_2(t), \ldots, q_N(t)\big). \tag{2.34}$$

This curve through configuration space is a generalized trajectory. To begin a Newtonian prediction, we specify both $\mathbf{q}(t_0)$ and $\dot{\mathbf{q}}(t_0)$. Newton's equations then act locally: knowing where the system is and how it is moving tells us its acceleration, and hence how to advance to the neighboring point of the path.

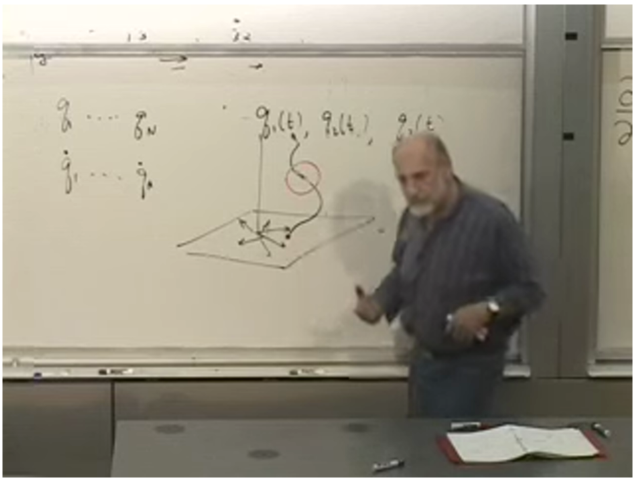

Figure 2.9: The generalized trajectory shown at 01:00:20. The horizontal plane is configuration space, with all its generalized coordinates; the vertical axis is time.

Lecture frame: 01:00:20.

There is another way to pose the problem. Instead of giving position and velocity at one time, hold the configurations fixed at two times,

$$\mathbf{q}(t_0) = \mathbf{q}_0, \qquad \mathbf{q}(t_1) = \mathbf{q}_1, \tag{2.35}$$

and ask which trajectory connects them. In the elementary well-posed cases being considered, this supplies the same amount of boundary data as position and velocity at one time. More complicated systems or long time intervals can admit several classical paths, but each stationary path still satisfies the same equations of motion.

The contrast is conceptual. Newton's differential equation gives a local rule along the curve. A variational principle assigns one number to the whole curve and asks which admissible curve makes that number stationary.

2.6 From Functions to Functionals

Take two fixed points in the xy-plane and describe a connecting curve by $y(x)$. The shortest connection is plainly a straight line, but we want the mathematical question that produces that answer.

For a small segment,

$$ds^2 = dx^2 + dy^2, \tag{2.36}$$

and therefore

$$ds = dx\sqrt{1 + \left(\frac{dy}{dx}\right)^2}. \tag{2.37}$$

Adding all the little segments gives the length

$$\mathcal{S}[y] = \int_{x_1}^{x_2} \sqrt{1 + (y'(x))^2}\, dx. \tag{2.38}$$

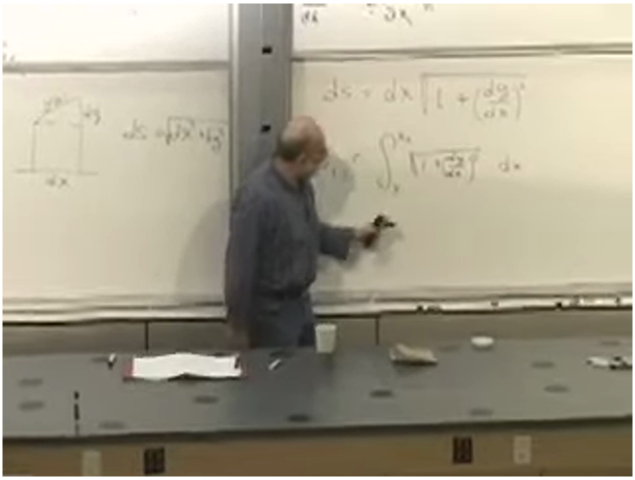

Figure 2.10: The curve, its dx-dy triangle, and the arc-length integral at 01:12:00.

Lecture frame: 01:12:00.

The input to $\mathcal{S}$ is not one number or a finite list of numbers. It is the entire function $y(x)$, and the output is a number. Such an object is a *functional.*

There are complementary global and local views. Globally, compare every curve between the endpoints and select a shortest one. Locally, break the curve into tiny pieces and focus on a point between two nearby pinned neighbors. Moving that point until the two small segments line up says simply: continue straight ahead. Applying the same local condition everywhere gives the straight line. The global extremization and the local differential equation encode the same geometry.

One tempting shortcut fails. Since the integrand in Eq. (2.38) is smallest when $y' = 0$, why not set $y' = 0$ at every point? Because a horizontal line will not generally meet the fixed endpoints. The admissible curves are constrained globally. Minimizing an integrand point by point while ignoring its boundary conditions solves a different problem.

2.6.1 Fermat's Principle

Optics supplies a more revealing functional. Suppose a layered isotropic medium has a light speed $c(y)$ that varies with height but not with direction at a fixed point. Fermat's principle says that the physical ray makes the travel time stationary. Locally,

$$c(y) = \frac{ds}{dt}, \qquad dt = \frac{ds}{c(y)}. \tag{2.39}$$

Using Eq. (2.37),

$$dt = \frac{\sqrt{1 + (y')^2}}{c(y)}\, dx, \tag{2.40}$$

so the total time is the functional

$$\mathcal{T}[y] = \int_{x_1}^{x_2} \frac{\sqrt{1 + \left(y'(x)\right)^2}}{c(y)}\, dx. \tag{2.41}$$

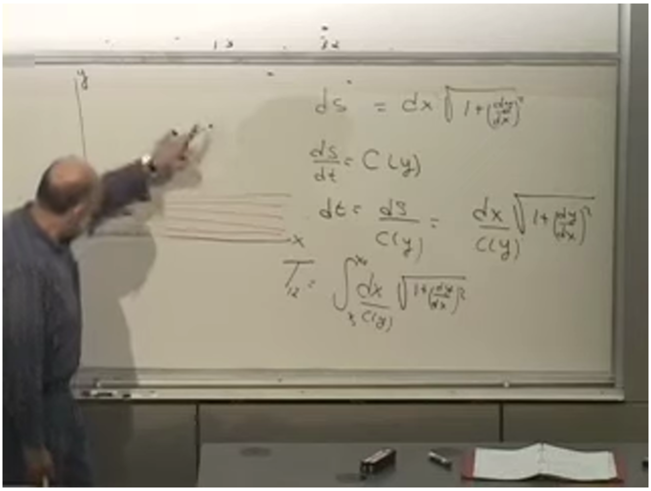

Figure 2.11: The layered-medium ray and completed least-time functional at 01:23:10. The ray bends toward the faster region while still advancing toward its endpoint.

Lecture frame: 01:23:10.

If c is constant, minimizing time is the same as minimizing distance, and the ray is straight. If light is faster higher in the medium, the best path can curve upward to spend more of its journey in the faster region. It cannot gain indefinitely by going straight up and then straight across: the vertical detour costs time without advancing toward the destination. The physical path is a compromise selected by the whole integral.

2.6.2 The Hanging Chain

A chain of fixed length suspended from two points supplies another variational problem. Gravity drives each small mass element downward, while the fixed length prevents every element from falling independently. The equilibrium curve minimizes the total gravitational potential energy subject to the length constraint.

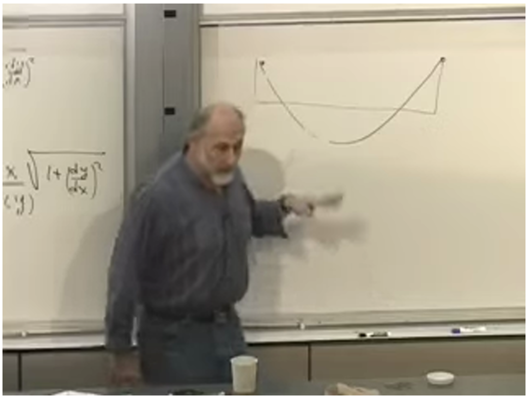

Figure 2.12: The hanging-chain sketch at 01:25:20. The fixed endpoints and fixed length constrain the chain while gravity lowers its center of mass.

Lecture frame: 01:25:20.

The resulting curve is a catenary, which may be written

$$y(x) = a\cosh\left(\frac{x - x_0}{a}\right) + y_0 \tag{2.42}$$

after the constants are chosen to fit the endpoints and length. Its derivation is not needed yet; its role is to show that minimizing a functional is a common physical task, not an artificial game invented for mechanics.

Question from the audience. From what height is the chain's potential energy measured? [a]

Response. Any fixed reference height will do. Shifting the zero adds the same constant to the energy of every allowed configuration, so it cannot change which configuration minimizes the energy.

[a] Lecture timestamp: 01:25:50–01:26:16.

These examples define the calculus of variations: find a function that makes a functional stationary. Mechanics will ask the same question for all the functions $q_i(t)$ that make up a trajectory.

2.7 The Action and the Lagrangian

We can now state the global formulation of mechanics. For every mechanical system there is a quantity called the action. Hold the endpoint configurations fixed and compare neighboring trajectories between them. The physical trajectory makes the action stationary.

Question from the audience. Is this really equivalent to Newton's formulation, or merely an approximation to it?[a]

Response. For systems governed by Newtonian mechanics, it is an equivalent formulation: varying the correct action yields Newton's equations. Which language is more efficient depends on the problem. The action principle also extends naturally to relativistic particles, fields, waves, and electromagnetism, where a literal catalogue of pairwise Newtonian forces is no longer the right description.

[a]Lecture timestamp: 01:28:21–01:30:13.

The practical advantage can be dramatic. Imagine describing a rigid eraser by writing force equations for every microscopic constituent. The local equations exist in principle, but they hide the small set of collective variables we actually care about: translation and rotation. A suitable action written in those variables reaches the same physics far more directly. Likewise, in a frictionless block-on-wedge problem, conservation laws or a Lagrangian can avoid pages of constraint forces. A more general principle is often not only deeper but faster.

For one particle moving in one dimension, the unknown path is $x(t)$. Since kinetic and potential energy have dominated the discussion, a natural first guess is

$$\int_{t_0}^{t_1} (T + U)\, dt. \tag{2.43}$$

That is the total energy integrated over time, and it is the wrong guess. The correct action is

$$A[x] = \int_{t_0}^{t_1} (T - U)\, dt \tag{2.44}$$

$$= \int_{t_0}^{t_1} \left[\frac{1}{2} m\dot{x}^2 - U(x)\right] dt. \tag{2.45}$$

The difference

$$L(x, \dot{x}) = T - U \tag{2.46}$$

is the Lagrangian, and the action is its integral over time.

The minus sign is not a decorative convention. If U is defined by $F = -dU/dx$, varying $T - U$ produces the correct sign in Newton's equation. One could rename $-U$ as a new quantity and write a plus sign, but that new quantity would then be the negative of potential energy. Renaming the symbol does not change the relation.

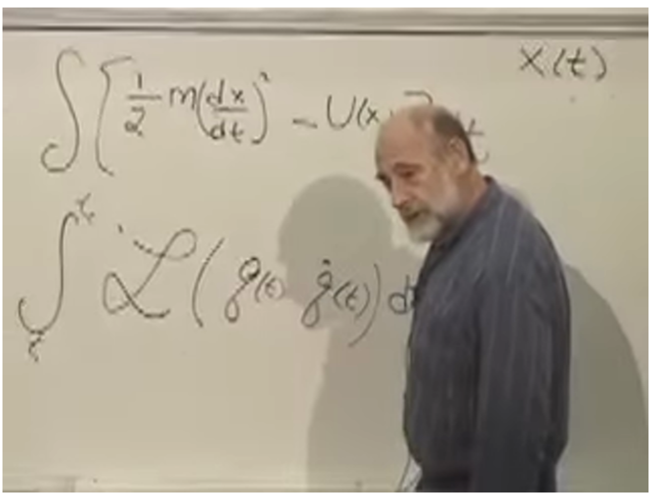

Figure 2.13: The action and its general Lagrangian form at 01:43:20. The independent variable is now time, and the path is the collection of functions $q_i(t)$.

Lecture frame: 01:43:20.

For generalized coordinates, write

$$L = L(\mathbf{q}, \dot{\mathbf{q}}), \qquad A[\mathbf{q}] = \int_{t_0}^{t_1} L(\mathbf{q}, \dot{\mathbf{q}})\, dt. \tag{2.47}$$

The analogy with the earlier examples is exact at the structural level. The arc-length integrand depends on y and y'; the optical integrand depends on y, y', and $c(y)$; the mechanical integrand depends on $\mathbf{q}$ and $\dot{\mathbf{q}}$. In every case an integral assigns a number to a whole path with fixed endpoints.

For ordinary Newtonian particles, $L = T - U$. More general classical systems can have less familiar Lagrangians, especially when velocity-dependent interactions or fields are present. What survives is the architecture: construct the action, vary the path while holding its endpoints fixed, and set the first-order change to zero.

Question from the audience. Why kinetic energy minus potential energy, rather than the total energy?[a]

Response. At this stage the honest test is operational. Use the calculus of variations and see which choice reproduces Newton's equations. $T-U$ does; $T+U$ gives the wrong force sign. The derivation, rather than an analogy with conserved energy, will explain why this is the right combination.

[a]Lecture timestamp: 01:33:30–01:36:38.

That derivation is the next task. Given a Lagrangian, we must learn how to turn stationary action into local equations of motion, and how to move back and forth between those equations and Newton's law. The same technique will solve the optical least-time problem. Mechanics and geometrical optics have arrived at the same mathematical threshold from different directions.

CHAPTER 3

LEAST ACTION, LOCAL EQUATIONS, AND SYMMETRY

The laws of physics have many subjects but a remarkably common form. Newtonian particles, electromagnetic fields, gravitation, and even the fields of modern physics can all be described by an action principle. Thermodynamics looks different because it is statistical, but its microscopic degrees of freedom still obey dynamical laws of this kind.

The phrase laws of physics is being used specifically here. Mathematical identities such as $1 + 1 = 2$, and the internal rules of probability theory, are not dynamical laws of a physical system. The fact that probability describes nature may itself be a physical fact, but the laws at issue now are those that govern motion and fields.

This is a stronger claim than saying that least action is a clever alternative to $F = ma$. The action is the organizing language; Newton's equations are one local expression of it. Our first task is therefore to understand how a global statement about an entire history can become a differential equation at each instant of time. Two elementary facts from calculus will do nearly all of the work.

3.1 Two Small Theorems

3.1.1 Integration by parts

Begin with the fundamental theorem of calculus,

$$\int_{t_1}^{t_2} \dot{F}(t)\, dt = F(t_2) - F(t_1). \tag{3.1}$$

There is a useful elementary picture behind this formula. Divide the interval into short steps. Each contribution to the integral is approximately a difference,

$$F(t_{k+1}) - F(t_k).$$

On adding the differences, every intermediate value cancels:

$$(F_2 - F_1) + (F_3 - F_2) + \cdots + (F_N - F_{N-1}) = F_N - F_1.$$

Only the two endpoints survive.

Now take $F = fg$. The product rule gives

$$\frac{d}{dt}(fg) = \dot{f}\, g + f\, \dot{g}. \tag{3.2}$$

Substitution into Eq. (3.1) yields

$$\int_{t_1}^{t_2} \dot{f}\, g\, dt = [fg]_{t_1}^{t_2} - \int_{t_1}^{t_2} f\, \dot{g}\, dt. \tag{3.3}$$

Thus a derivative may be moved from one factor to the other, at the cost of a minus sign and a boundary term.

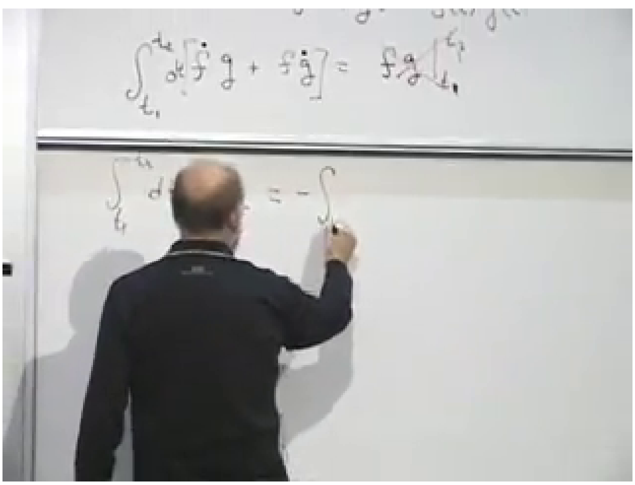

Figure 3.1: Integration by parts on the board at 00:10:20. The upper line retains the endpoint term; the lower line is being reduced to the endpoint-vanishing case used in the variation.

Lecture frame: 00:10:20.

Question from the audience. Why must the integral contain dt?

Response. An integral is always taken with respect to a variable. Here time is the independent variable, so the complete expression is $\int \dot{f}(t)\,dt$. Writing only $\int \dot{f}(t)$ leaves the integration measure unstated. Small notational questions are worth stopping for because they force every ingredient of the calculation into view.

We shall use a special case repeatedly. If f vanishes at t_1 and t_2, then $[fg]_{t_1}^{t_2} = 0$, and

$$\int_{t_1}^{t_2} \dot{f}\,g\,dt = -\int_{t_1}^{t_2} f\,\dot{g}\,dt. \tag{3.4}$$

It is enough that the product fg have the same value at both endpoints, but fixed-endpoint variations will make f vanish separately at each end.

3.1.2 An arbitrary test function

The second fact is almost obvious, but it is the step that turns an integral condition into a local law.

Proposition 3.1. *If*

$$\int_{t_1}^{t_2} a(t)f(t)\,dt = 0 \tag{3.5}$$

for every sufficiently smooth function $f(t)$ that vanishes at the endpoints, then $a(t) = 0$ throughout the interval.

Proof. Choose f to be a narrow smooth blip around an arbitrary point t_* and zero away from that neighborhood. If $a(t_*) \neq 0$, choose the sign of the blip to agree with the sign of a; for a sufficiently narrow neighborhood the integral cannot vanish. Hence $a(t_*) = 0$. Since the point was arbitrary, a vanishes everywhere. $\square$

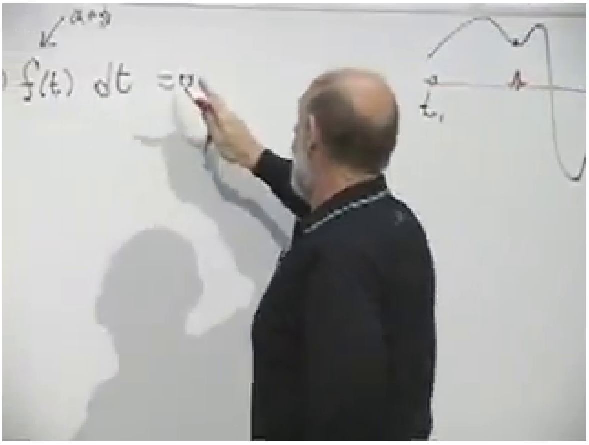

Figure 3.2: The arbitrary-function argument at 00:13:30. A narrow test function isolates the value of $a(t)$ near any chosen point of the interval.

Lecture frame: 00:13:30.

One does not verify Eq. (3.5) by trying functions one after another; that would take forever. In an application we prove the integral vanishes for an arbitrary f, and the proposition then gives the pointwise conclusion in one step.

3.2 Histories: Local and Global Descriptions

Let a system have generalized coordinates

$$q_1(t), q_2(t), \ldots, q_n(t). \tag{3.6}$$

The complete collection of these functions is a *history*, or a trajectory in configuration space. For a system of N unconstrained particles in three dimensions, $n = 3N$.

Whether time is drawn horizontally, as the independent variable often is in elementary mathematics, or vertically, as it often is in physics diagrams, is only a convention. The object represented is the same set of functions $q_i(t)$.

There are two apparently different ways to state a law of motion. A local law uses information near one instant to continue the trajectory. In Newtonian mechanics, knowledge of q_i and $\dot{q}_i$, together with equations of the form $F = ma$, determines the acceleration and advances the solution to the next instant.

A global law considers the whole path between two endpoint configurations. It assigns a number, the action, to every candidate history and says that the physical history makes that number stationary. Instead of giving q_i and $\dot{q}_i$ at one time, this boundary-value formulation fixes q_i at an initial and a final time.

The two formulations must be equivalent when they describe the same physics. There is already a simple reason to expect locality to emerge. If a path is stationary on a long interval, its segment between any two neighboring points must also be stationary with those points held fixed. Otherwise that segment could be replaced and the action of the whole path would change to first order.

3.3 Varying a Path

Write the physical history as $\hat{q}_i(t)$. A nearby history can be parameterized by

$$q_i(t, \alpha) = \hat{q}_i(t) + \alpha f_i(t), \qquad f_i(t_1) = f_i(t_2) = 0. \tag{3.7}$$

The number α controls the size and sign of the deformation; it does not depend on time. The functions $f_i(t)$ describe its shape and are otherwise arbitrary. The endpoint conditions nail the beginning and end of every candidate path in place.

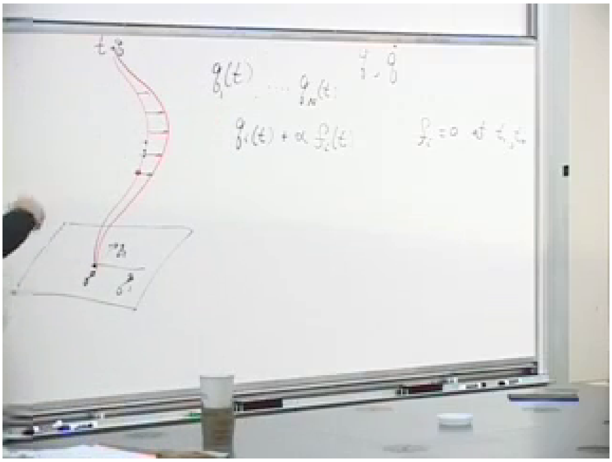

Figure 3.3: A fixed-endpoint deformation at 00:22:55. The red curve is displaced in the interior while the endpoint conditions $f_i(t_1) = f_i(t_2) = 0$ remain visible on the board.

Lecture frame: 00:22:55.

For fixed $\hat{q}_i$ and fixed deformation functions f_i, the action is now an ordinary function $A(\alpha)$. The physical path corresponds to $\alpha = 0$. If it is a minimum, then in particular it is stationary:

$$\left.\frac{dA}{d\alpha}\right|_{\alpha=0} = 0. \tag{3.8}$$

The first derivative is the condition we need; whether the stationary path is a minimum, maximum, or saddle requires additional information.

Question from the audience. What is the action being varied?

Response. For a history between t_1 and t_2, the action is

$$A[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt. \tag{3.9}$$

The Lagrangian L is evaluated at every point of the path. In the Newtonian examples below it is $L = T - U$, but the variational calculation needs only that L be a function of the coordinates, velocities, and possibly time.

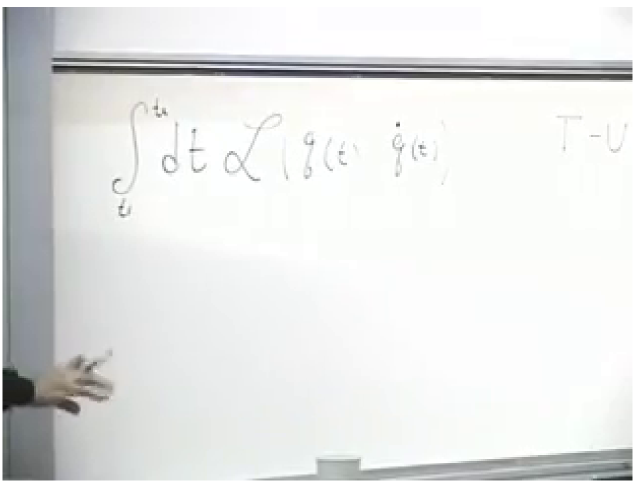

Figure 3.4: The action integral at 00:28:42, written before any particular mechanical form of the Lagrangian is needed.

Lecture frame: 00:28:42.

Equation (3.7) immediately gives

$$\frac{\partial q_i}{\partial \alpha} = f_i, \qquad \frac{\partial \dot{q}_i}{\partial \alpha} = \dot{f}_i. \tag{3.10}$$

Applying the chain rule inside the action,

$$\frac{dA}{d\alpha} = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} \frac{\partial q_i}{\partial \alpha} + \frac{\partial L}{\partial \dot{q}_i} \frac{\partial \dot{q}_i}{\partial \alpha} \right) dt \tag{3.11}$$

$$= \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} f_i + \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i \right) dt. \tag{3.12}$$

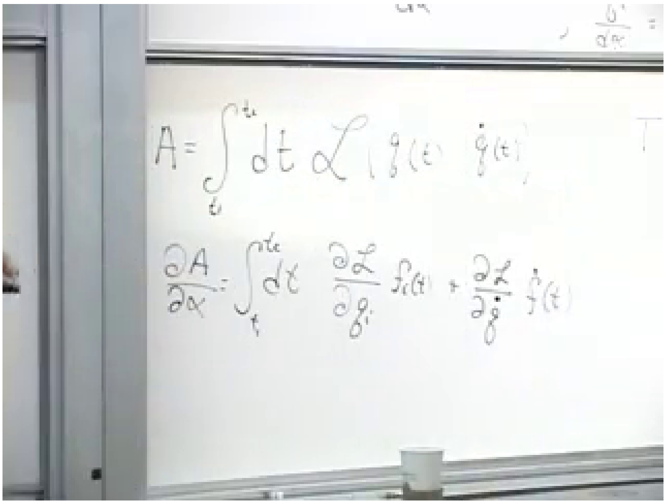

Figure 3.5: The first variation at 00:34:10. Changes of both q_i and $\dot{q}_i$ contribute, and the audience supplies the required sum over coordinates.

Lecture frame: 00:34:10.

Question from the audience. Should there be a sum over i?

Response. Yes. Each coordinate and its velocity are independent arguments of L, so the chain rule contains one pair of terms for every i. For a one-coordinate problem the sum is invisible; for a general system it is essential.

The unwanted $\dot{f}_i$ in the second term is precisely why integration by parts was prepared. Applying Eq. (3.3),

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i \, dt = \left[f_i \frac{\partial L}{\partial \dot{q}_i} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} f_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) dt. \tag{3.13}$$

In the earlier notation, f_i plays the role of f, while $\partial L/\partial \dot{q}_i$ plays the role of g. The operation removes the dot from f_i

and differentiates the momentum-like factor instead. The boundary term vanishes because $f_i(t_1) = f_i(t_2) = 0$. Consequently,

$$\left.\frac{dA}{d\alpha}\right|_{\alpha=0} = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\right]_{\hat{q}} f_i(t)\, dt. \tag{3.14}$$

Question from the audience. What happened to the endpoint contribution?

Response. It is exactly $[f_i\, \partial L/\partial \dot{q}_i]_{t_1}^{t_2}$. The varied and physical paths have the same endpoints, so every f_i vanishes there and the contribution is zero. The endpoint condition was chosen at the start for this reason.

Stationarity requires Eq. (3.14) to vanish for every collection of functions $f_i(t)$. The arbitrary-function proposition applies separately to every i, giving

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}} \qquad (i = 1, \ldots, n). \tag{3.15}$$

These are the Euler–Lagrange equations. The global condition $\delta A = 0$ has become one local differential equation for each generalized coordinate.

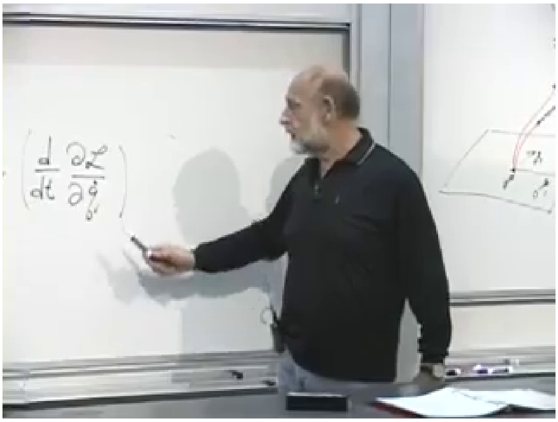

Figure 3.6: The Euler–Lagrange time-derivative term at 00:45:02, beside the trajectory picture used to connect the global path with its local equation.

Lecture frame: 00:45:02.

The result is not confined to point particles. Field theory, general relativity, and string theory use the same variational structure with more elaborate coordinates. Quantum mechanics also retains it in another form: Feynman's path integral assigns amplitudes to histories through their action.

3.4 Momentum and Generalized Force

The quantity differentiated with respect to time in Eq. (3.15) has a name:

$$\pi_i \equiv \frac{\partial L}{\partial \dot{q}_i}. \tag{3.16}$$

It is the canonical momentum conjugate to q_i. The corresponding $\partial L/\partial q_i$ is the generalized force. The equation of motion therefore reads

$$\dot{\pi}_i = \frac{\partial L}{\partial q_i}. \tag{3.17}$$

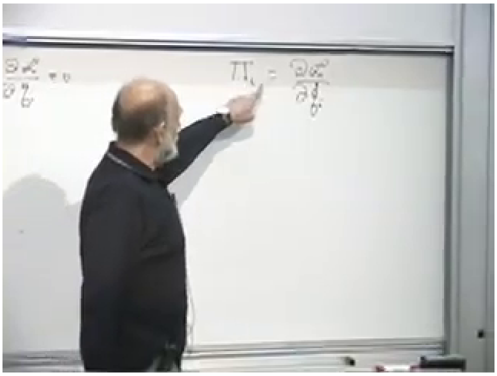

Figure 3.7: Canonical momentum introduced on the board at 00:46:32. The standard lowercase notation is $\pi_i = \partial L/\partial \dot{q}_i$.

Lecture frame: 00:46:32.

Question from the audience. What does the Euler–Lagrange equation say in ordinary language?

Response. First, it says that a global statement about the entire trajectory is equivalent to a local differential law. Second, after naming the two sides, it says that the time derivative of momentum equals generalized force. The concrete examples now show that this is $F = ma$, written in a form that survives arbitrary choices of coordinates.

One may try to picture the left side as a time change of the velocity-derivative of L, and the right side as the coordinate-derivative at the present point. That intuition is useful, but the names momentum and force become convincing only after the first calculation.

3.5 Recovering Newton's Equation

For one particle of mass m moving on a line,

$$L(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 - U(x). \tag{3.18}$$

The generalized coordinate q is simply x, and its canonical momentum is the ordinary momentum:

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} = p. \tag{3.19}$$

The coordinate derivative is

$$\frac{\partial L}{\partial x} = -\frac{dU}{dx}. \tag{3.20}$$

Hence

$$m\ddot{x} = -\frac{dU}{dx}, \tag{3.21}$$

which is Newton's law for a conservative force.

The minus sign is not optional. Had we chosen $L = T + U$, the force would have come out with the wrong sign. The Lagrangian $T - U$ is not the energy and is not generally conserved along a trajectory. For a time-independent conservative system the familiar conserved energy is $T + U$; its derivation from symmetry is deliberately postponed.

For many particles, combine every particle label and Cartesian direction into one index i. Then

$$L = \sum_i \frac{1}{2} m_i \dot{x}_i^2 - U(x_1, \ldots, x_n), \tag{3.22}$$

and, coordinate by coordinate,

$$p_i = m_i \dot{x}_i, \qquad \dot{p}_i = -\frac{\partial U}{\partial x_i}. \tag{3.23}$$

For a physical particle the same mass occurs in its three Cartesian components, although the notation is general enough to attach a coefficient to every coordinate. A rigid body, an eraser, or a person can ultimately be viewed as a large collection of particles held together by internal forces, so this construction already covers an enormous class of mechanical systems.

3.6 Translation Symmetry and Momentum

Consider two interacting particles on a line:

$$L = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 - U(x_1 - x_2). \tag{3.24}$$

The potential depends only on their separation. If both coordinates are shifted by the same constant a,

$$x_1 \longmapsto x_1 + a, \qquad x_2 \longmapsto x_2 + a, \tag{3.25}$$

their velocities and the difference $x_1 - x_2$ are unchanged. The Lagrangian, and therefore the action, is invariant. This is translation symmetry.

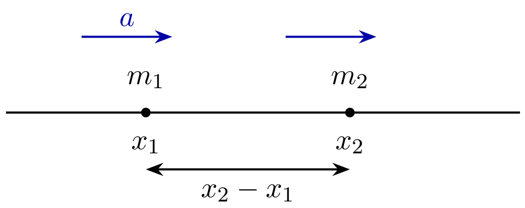

Figure 3.8: A rigid translation moves both particles by the same amount without changing their separation or velocities.

This form would fail if a third, external lump of matter exerted a position-dependent force. The potential would then depend on the distance from that external object as well as on the separation between the two particles. The two-particle subsystem would no longer be closed or translation invariant.

Set $d = x_1 - x_2$. The chain rule gives

$$\frac{\partial U}{\partial x_1} = \frac{dU}{dd}, \qquad \frac{\partial U}{\partial x_2} = -\frac{dU}{dd}. \tag{3.26}$$

Therefore

$$\dot{p}_1 = -\frac{\partial U}{\partial x_1}, \qquad \dot{p}_2 = -\frac{\partial U}{\partial x_2} = +\frac{\partial U}{\partial x_1}. \tag{3.27}$$

The forces are equal and opposite, and adding the equations gives

$$\frac{d}{dt}(p_1 + p_2) = 0. \tag{3.28}$$

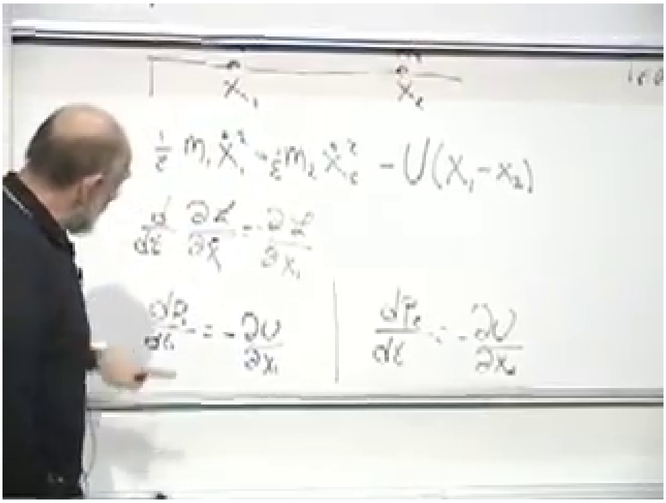

Figure 3.9: The translation-invariant Lagrangian and its two force equations at 01:08:50. Dependence on $x_1 - x_2$ makes the derivatives equal and opposite.

Lecture frame: 01:08:50.

This is the first instance of the central pattern:

translation symmetry
$\Downarrow$
conservation of linear momentum.

The symmetry is an active operation on the system that leaves the action unchanged. Here the operation can be made by an arbitrarily small amount and built up continuously, so it is a continuous symmetry.

Question from the audience. If U depends on $|x_1 - x_2|$, is inversion another symmetry?

Response. Yes. Interchanging the orientation of the separation leaves such a potential unchanged. Unlike translation, inversion is discrete: there is no continuously adjustable parameter connecting the two alternatives. Continuous symmetries generate the classical conservation laws being studied here. Discrete symmetries become especially consequential in quantum mechanics, which we leave for later.

3.7 One Translation Survives

Place a particle near Earth's surface and retain horizontal x and vertical y motion. With y measured upward,

$$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 - mgy, \tag{3.29}$$

where g is the approximately constant local gravitational acceleration. Numerically it is about $9.8\,\mathrm{m\,s^{-2}}$, commonly rounded to $10\,\mathrm{m\,s^{-2}}$ for this discussion.

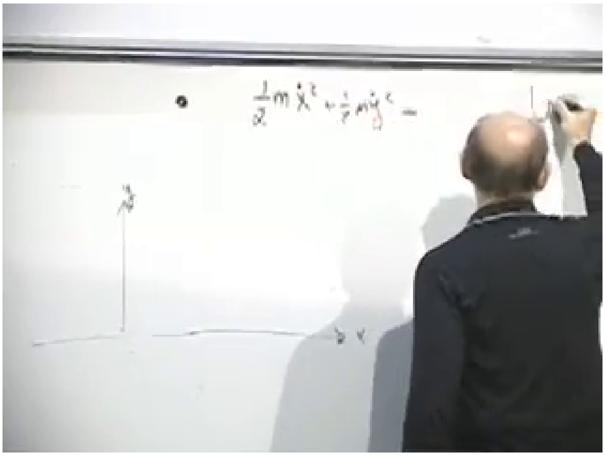

Figure 3.10: The two-coordinate kinetic energy and near-Earth setup at 01:15:15. The stable board content is $T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2$; the potential is completed immediately afterward.

Lecture frame: 01:15:15.

The Lagrangian contains no x, so horizontal translation is a symmetry:

$$\frac{d}{dt}(m\dot{x}) = 0. \tag{3.30}$$

It does contain y, so vertical translation is not a symmetry:

$$\frac{d}{dt}(m\dot{y}) = -mg, \qquad \ddot{y} = -g. \tag{3.31}$$

There is one translation symmetry and one corresponding conserved momentum, not two.

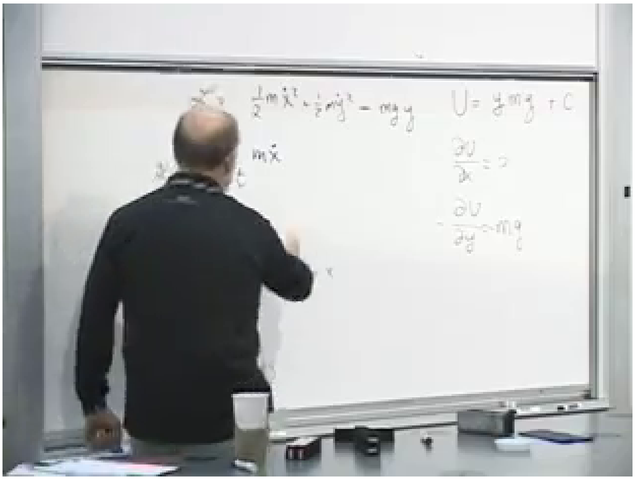

Figure 3.11: The completed near-Earth Lagrangian and coordinate derivatives at 01:19:15. The absence of x gives horizontal momentum conservation, while $\partial U/\partial y = mg$ gives downward acceleration.

Lecture frame: 01:19:15.

Question from the audience. For a very small vertical displacement, is there an approximate y-translation symmetry?

Response. No exact symmetry appears merely because the displacement is small. In the near-Earth approximation $U = mgy$, any nonzero change of y changes the potential by $mg\,\Delta y$. We may add one fixed constant to U and thereby choose a different zero of potential energy, but that does not remove the y-dependence or the vertical force.

Adding C to the potential changes neither force nor trajectory:

$$U(y) \longmapsto U(y) + C, \qquad -\frac{d}{dy}(U + C) = -\frac{dU}{dy}. \tag{3.32}$$

Sea level, the floor, or any other fixed height may therefore be chosen as the zero of potential energy.

Question from the audience. Is this just the familiar falling-body problem and its energy conservation in another form?

Response. It is the same mechanics, but so far we have derived the equations of motion and identified momentum conservation. The conserved energy is $T + U$, not the Lagrangian $T - U$. Its origin will be traced to another symmetry, time translation, in a later step.

3.8 Changing Coordinates

The action does not depend on how we label the configurations. Rectangular, polar, parabolic, or curved coordinates may make the formulas look different, but a correctly transformed action assigns the same number to the same physical history. Stationarity of that action is therefore coordinate-independent.

For planar motion, replace x, y by polar coordinates r, θ. At every instant the velocity has a radial component and a perpendicular component:

$$v_r = \dot{r}, \qquad v_\perp = r\dot{\theta}. \tag{3.33}$$

The factor of r has a simple geometric origin. A fixed angular change $d\theta$ sweeps the arc length $r\,d\theta$, so the same angular speed corresponds to a larger linear speed farther from the origin. The fingertips of a rotating arm move faster than the elbow.

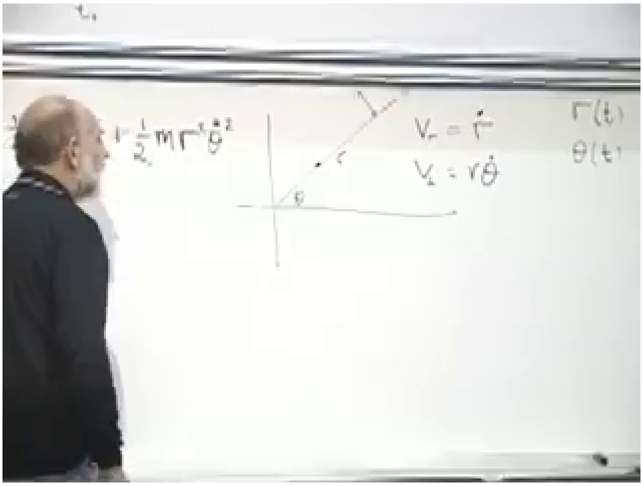

Figure 3.12: Polar-coordinate velocity components at 01:27:50. The board records $v_r = \dot{r}$, $v_\perp = r\dot{\theta}$, and the resulting kinetic-energy terms.

Lecture frame: 01:27:50.

Hence

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right). \tag{3.34}$$

For a central potential $U(r)$,

$$L = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 - U(r). \tag{3.35}$$

It is useful to write both Euler–Lagrange equations. The radial one is

$$m\ddot{r} = mr\dot{\theta}^2 - \frac{dU}{dr}. \tag{3.36}$$

The $mr\dot{\theta}^2$ term comes from differentiating the angular kinetic energy with respect to r. Radial momentum is not conserved: the central force and the changing radial direction both enter its evolution.

The angular coordinate is different. The Lagrangian depends on $\dot{\theta}$ but not on θ itself. Thus θ is a cyclic coordinate and

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}. \tag{3.37}$$

Its equation of motion is

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = \frac{\partial L}{\partial \theta} = 0. \tag{3.38}$$

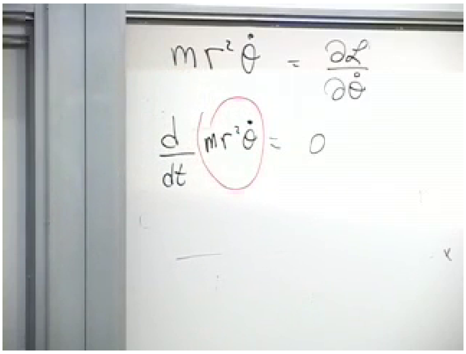

Figure 3.13: Angular momentum conservation completed on the board at 01:33:35: $\frac{d}{dt}(mr^2\dot{\theta}) = 0$.
Lecture frame: 01:33:35.

The conserved quantity $mr^2\dot{\theta}$ is the angular momentum, usually denoted by ℓ when L is already reserved for the Lagrangian. Solving for the angular speed,

$$\dot{\theta} = \frac{\ell}{mr^2}. \tag{3.39}$$

As r decreases, $\dot{\theta}$ increases. This is the spinning-skater effect in its simplest form.

We have now seen the same structure twice:

$$\begin{gathered}\text{translation invariance}\\ \Downarrow\\ \text{conserved linear momentum,}\end{gathered}$$

$$\begin{gathered}\text{rotational invariance}\\ \Downarrow\\ \text{conserved angular momentum.}\end{gathered}$$

The action principle makes the pattern visible. A coordinate that does not appear in the Lagrangian has a conserved conjugate momentum, and the local equation expressing that fact is itself the consequence of stationarity of the entire history.

CHAPTER 4

SYMMETRY, NOETHER'S THEOREM, AND ENERGY

Our destination is energy conservation, but first there is a correction to make. In the previous discussion of motion in polar coordinates, the outward term in the radial equation was called Coriolis. The correct name is *centrifugal.* Rather than repair only the word, we shall reconstruct the whole example. It provides the simplest route from the Euler–Lagrange equations to symmetry, conservation laws, and finally the Hamiltonian.

4.1 A Particle in Polar Coordinates

The working procedure is straightforward. Choose coordinates that describe the system, write the kinetic and potential energies, form the Lagrangian, and apply the Euler–Lagrange equations. The difficult part is usually choosing a good description; after that, the calculation is largely systematic.

For a particle moving in a plane, Cartesian coordinates x, y are possible, but polar coordinates r, θ are better adapted to a force directed toward or away from an origin. The radial velocity is

$$v_r = \dot{r}, \tag{4.1}$$

whereas the transverse velocity is

$$v_\theta = r\dot{\theta}. \tag{4.2}$$

The factor of r matters. At the same angular velocity, a point farther from the origin sweeps out a larger distance in the same time.

The two velocity components are perpendicular, so

$$v^2 = \dot{r}^{\,2} + r^2\dot{\theta}^{\,2}, \tag{4.3}$$

and the kinetic energy is

$$T = \frac{m}{2}\left(\dot{r}^{\,2} + r^2\dot{\theta}^{\,2}\right). \tag{4.4}$$

We now choose a central potential,

$$U = U(r), \tag{4.5}$$

and therefore

$$\mathcal{L} = \frac{m}{2}\left(\dot{r}^{\,2} + r^2\dot{\theta}^{\,2}\right) - U(r). \tag{4.6}$$

This restriction is not compulsory. A potential may depend on θ, but then the system distinguishes one direction from another. Independence from θ is precisely rotational symmetry.

Question from the audience. Does $U(r)$ also depend on θ?

Response. Not in the central-force problem being studied. Gravity due to a fixed central body is the standard example: the potential depends on the distance from the body, not on the direction from which it is approached. A more general potential $U(r,\theta)$ is possible, but it would not possess this rotational symmetry.

4.2 The Radial Equation

For any generalized coordinate q_i, define its canonical momentum by

$$\pi_i = \frac{\partial\mathcal{L}}{\partial\dot{q}_i}. \tag{4.7}$$

The Euler–Lagrange equation may then be written

$$\frac{d\pi_i}{dt} = \frac{\partial\mathcal{L}}{\partial q_i}. \tag{4.8}$$

For the radial coordinate,

$$\pi_r = \frac{\partial\mathcal{L}}{\partial\dot{r}} = m\dot{r}, \tag{4.9}$$

and hence $d\pi_r/dt = m\ddot{r}$. There are two contributions to $\partial\mathcal{L}/\partial r$: one from the potential and one from the factor r^2 in the angular kinetic energy. Thus

$$\frac{\partial \mathcal{L}}{\partial r} = \frac{\partial}{\partial r}\left(\frac{m}{2}r^2\dot{\theta}^2 - U(r)\right) \tag{4.10}$$

$$= mr\dot{\theta}^2 - \frac{dU}{dr}. \tag{4.11}$$

The radial equation is therefore

$$m\ddot{r} = mr\dot{\theta}^2 - \frac{dU}{dr}. \tag{4.12}$$

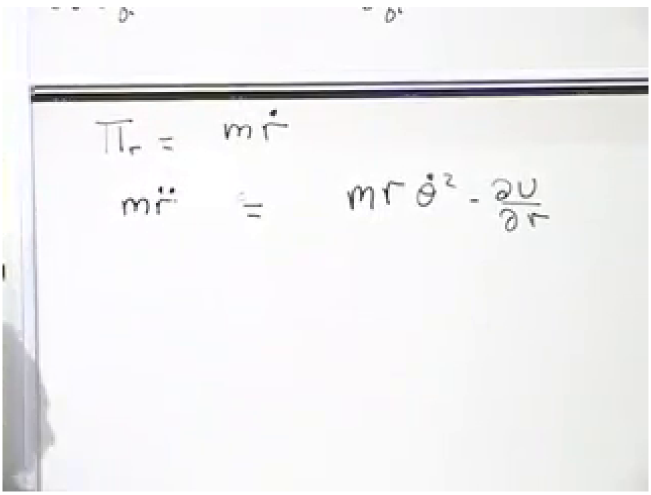

Figure 4.1: The completed radial equation on the board at 00:09:55. The term $mr\dot{\theta}^2$ comes from differentiating the angular part of the kinetic energy, while $-dU/dr$ comes from the potential.

Lecture frame: 00:09:55.

The first term on the right is positive in the outward radial direction. It is the centrifugal contribution. Nothing was added by hand: it emerged from the r-dependence of the kinetic energy. This is why the Lagrangian method is so efficient. Equations of motion in curvilinear coordinates may look complicated, but they follow from the same mechanical steps as the Cartesian equations.

Question from the audience. What does it mean to call the centrifugal contribution *not a real force*?

Response. The word *real* adds nothing and should be withdrawn. The useful statement is only that $mr\dot{\theta}^2$ is not obtained from the derivative of the potential U. It arises from the coordinate dependence of the kinetic term. Nevertheless it is present on the right-hand side of the radial equation and is part of the generalized radial force. A change of coordinates changes the appearance of the equation, not the motion.

4.3 Rotational Symmetry and Angular Momentum

The equation for θ is even more revealing. Its canonical momentum is

$$\pi_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2\dot{\theta}. \tag{4.13}$$

Since θ does not occur in Eq. (4.6),

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0, \tag{4.14}$$

and the angular Euler–Lagrange equation becomes

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = 0. \tag{4.15}$$

Thus

$$mr^2\dot{\theta} = L, \qquad L = \text{constant}. \tag{4.16}$$

The momentum conjugate to the angle is the angular momentum. Its value is not fixed by the law itself; it is fixed by the initial position and velocity. By contrast, $\pi_r = m\dot{r}$ generally changes because the radial force is not zero.

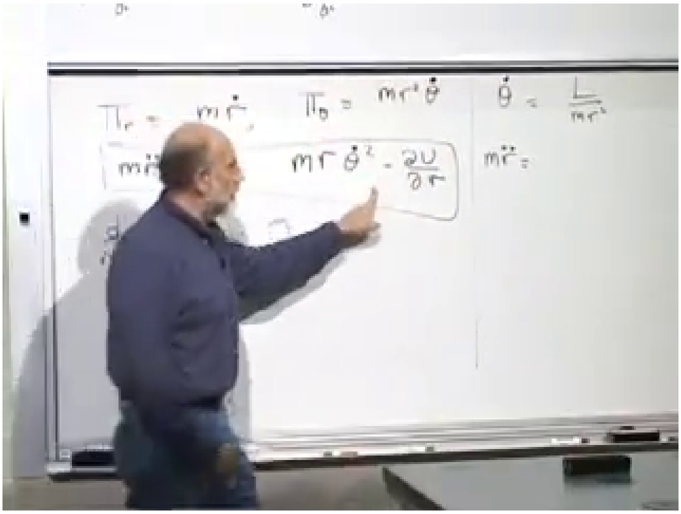

Figure 4.2: The polar equations assembled at 00:18:05. Rotational symmetry makes $mr^2\dot{\theta}$ constant, and the same board retains the radial equation that will receive this constant as an input.

Lecture frame: 00:18:05.

Solving Eq. (4.16) for the angular velocity gives

$$\dot{\theta} = \frac{L}{mr^2}. \tag{4.17}$$

Substitution into Eq. (4.12) yields

$$m\ddot{r} = -\frac{dU}{dr} + mr\left(\frac{L}{mr^2}\right)^2 \tag{4.18}$$

$$= -\frac{dU}{dr} + \frac{L^2}{mr^3}. \tag{4.19}$$

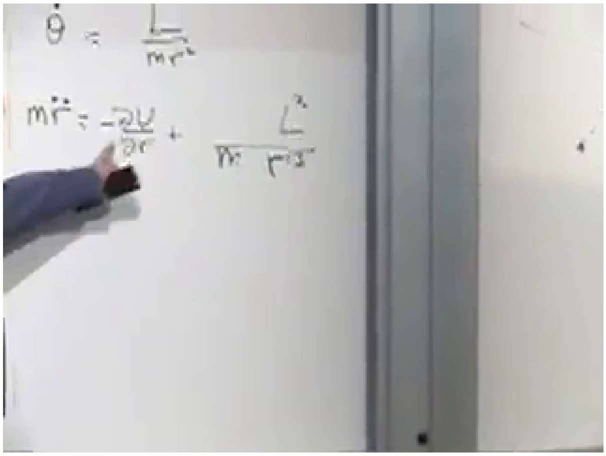

Figure 4.3: The effective radial equation at 00:21:10. Once the angular momentum is fixed, angular motion contributes the outward term $L^2/(mr^3)$.

Lecture frame: 00:21:10.

The whole influence of the angular coordinate has now become an outward force proportional to $1/r^3$. Far from the origin it is weak; close to the origin it becomes very strong. It may also be written as the derivative of an effective potential,

$$m\ddot{r} = -\frac{dU_{\text{eff}}}{dr}, \qquad U_{\text{eff}}(r) = U(r) + \frac{L^2}{2mr^2}. \tag{4.20}$$

The second term is the centrifugal barrier.

For gravity or the Coulomb force, the attractive force varies as $1/r^2$, whereas the centrifugal contribution varies as $1/r^3$. At sufficiently small r, nonzero angular momentum therefore wins. This is why an orbiting planet does not simply crash into the central body. If the motion is exactly radial, however, then $L = 0$, the barrier disappears, and a collision may occur.

Question from the audience. Is centrifugal force just inertia trying to carry the particle along a straight line?

Response. Yes. In Cartesian coordinates the free particle continues along a straight line. In polar coordinates that same tendency appears as an outward term in the radial equation. With nonzero angular momentum a straight line misses the origin; with zero angular momentum it may pass through it. The coordinate descriptions differ, but the trajectory is the same.

4.4 Variations and Stationarity

We need one piece of notation before turning to the general theorem. Let F be a function of many variables α_i. Under small changes $\delta\alpha_i$, its first-order change is

$$\delta F = \sum_i \frac{\partial F}{\partial \alpha_i}\, \delta\alpha_i + O(\delta\alpha^2). \tag{4.21}$$

If $\delta F = 0$ for every independent choice of $\delta\alpha_i$, then

$$\frac{\partial F}{\partial \alpha_i} = 0 \qquad \text{for every } i. \tag{4.22}$$

This is the meaning of stationarity to first order. A minimum is stationary, but stationarity is the precise property needed in the action principle.

For a trajectory the same shorthand becomes

$$A[q] = \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}, t)\, dt, \qquad \delta A = 0. \tag{4.23}$$

The variation of the action is

$$\delta A = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial \mathcal{L}}{\partial q_i}\, \delta q_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\, \delta \dot{q}_i \right) dt. \tag{4.24}$$

This notation is not an additional physical principle. It is a compact way of collecting all first-order changes.

4.5 Continuous Symmetries

A symmetry is a transformation that leaves the action unchanged. Translating an isolated system is a symmetry when the action depends only on relative positions. Rotating a rotationally invariant system is another. In each case the transformation acts on the entire history, not merely on one particle at one instant.

We shall study continuous symmetries through infinitesimal transformations,

$$q_i \longrightarrow q_i + \epsilon f_i(q), \qquad \delta q_i = \epsilon f_i(q), \tag{4.25}$$

where ϵ is small. The displacement may depend on the point in configuration space. A finite translation or rotation can be built from many small ones, so first-order information is sufficient.

For an infinitesimal counterclockwise rotation in the x-y plane,

$$\delta x = -\epsilon y, \qquad \delta y = \epsilon x, \tag{4.26}$$

and therefore

$$f_x = -y, \qquad f_y = x. \tag{4.27}$$

The signs can be checked on the axes. At a point on the positive x-axis, $y = 0$, so x does not change to first order while y increases. At a point on the positive y-axis, x decreases while y does not change to first order.

Question from the audience. Must the small rotation angle be measured in radians?

Response. Radians make Eq. (4.26) take this simple form. Another angular unit merely rescales ϵ and hence rescales the generator by a constant; the conservation law is unchanged.

Question from the audience. Does a cube have rotational symmetry?

Response. It has a discrete rotational symmetry: certain rotations, such as ninety degrees about suitable axes, return it to itself. It is not invariant under an arbitrarily small rotation. A sphere is. The argument below concerns continuous symmetries that can be built from infinitesimal transformations, not isolated discrete transformations.

4.6 Noether's Theorem

Begin with a trajectory $q_i(t)$ that satisfies the equations of motion. An ordinary least-action variation keeps its endpoints fixed. A symmetry variation is different: it transforms every point of the trajectory, including the endpoints. The action still does not change, but now it vanishes for a different reason. It vanishes because the transformation is a symmetry, not because it belongs to the fixed-endpoint family used to derive the equations of motion.

Integrating the second term of Eq. (4.24) by parts gives

$$\delta A = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \right] \delta q_i \, dt \tag{4.28}$$

$$+ \left[\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \, \delta q_i \right]_{t_1}^{t_2}. \tag{4.29}$$

The bulk integral vanishes because the original path obeys the Euler–Lagrange equations. The boundary term does not vanish by an endpoint condition, because the symmetry moved those endpoints. Instead symmetry tells us that the full δA vanishes. Consequently,

$$\left[\sum_i \pi_i \, \delta q_i \right]_{t_1}^{t_2} = 0. \tag{4.30}$$

Substitute Eq. (4.25) and remove the common constant ϵ:

$$[Q]_{t_1}^{t_2} = 0, \qquad Q = \sum_i \pi_i f_i(q). \tag{4.31}$$

Since the endpoints may be any two times,

$$\frac{dQ}{dt} = 0. \tag{4.32}$$

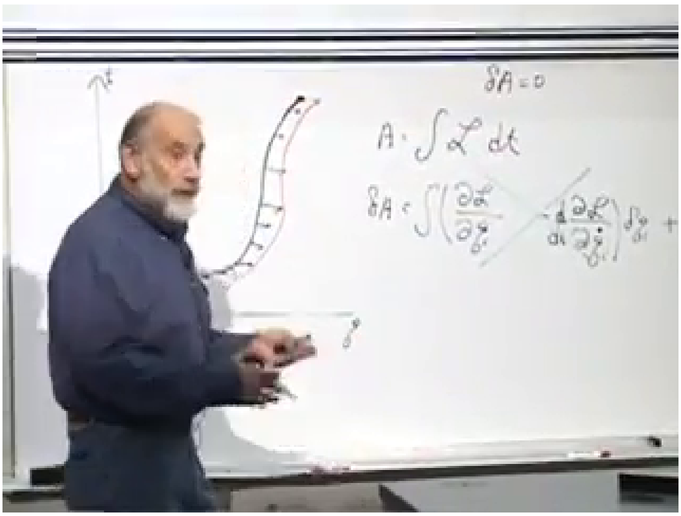

Figure 4.4: The endpoint form of the variational argument at 00:47:50. The bulk Euler–Lagrange term is crossed out because the original path obeys the equations of motion, leaving the boundary contribution whose equality at two times is the conservation law.

Lecture frame: 00:47:50.

This is Noether's theorem in the form needed here. The quantity Q is called the Noether charge. The word *charge* is used broadly: it means a conserved quantity, not necessarily electric charge. Momentum, angular momentum, and energy are all Noether charges. The theorem also uses two facts together: the equations arise from an action principle, and the action has a continuous symmetry. Invariance of an arbitrary equation, without an action principle, is not by itself this theorem.

Question from the audience. Is the rotation in this argument a parallel transport?

Response. No. Here a rotation is simply a transformation of the coordinates and of the entire trajectory. Parallel transport is a different geometric construction and is not needed for this derivation.

4.7 Translation and Rotation Charges

Consider a closed system of many particles. Under a common translation in the x-direction,

$$\delta x_a = \epsilon, \qquad \delta y_a = \delta z_a = 0 \tag{4.33}$$

for every particle label a. Thus $f_{x_a} = 1$, and Eq. (4.31) gives

$$Q_x = \sum_a \pi_{x_a} = \sum_a m_a \dot{x}_a = \sum_a p_{x_a}. \tag{4.34}$$

The conserved Noether charge is total momentum in the translation direction. The detailed internal forces were never needed. We required only that moving the whole isolated system leave its potential energy unchanged. A fixed external body would spoil that symmetry unless it too were included in the dynamical system.

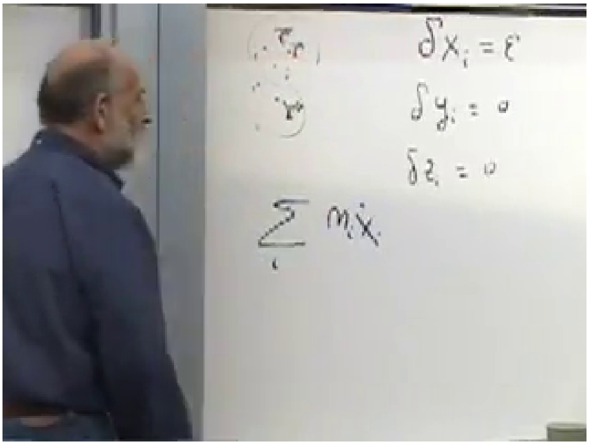

Figure 4.5: Translation along x at 00:57:30. Every particle has $\delta x_a = \epsilon$, while the conserved charge is the summed x-momentum $\sum_a m_a \dot{x}_a$.

Lecture frame: 00:57:30.

For a rotation in the x-y plane, use Eq. (4.26). For one particle,

$$Q_{\text{rot}} = p_x(-y) + p_y x \tag{4.35}$$

$$= x p_y - y p_x. \tag{4.36}$$

This is the component of $\mathbf{r} \times \mathbf{p}$ perpendicular to the plane. For many particles,

$$L_z = \sum_a \left(x_a p_{y_a} - y_a p_{x_a} \right). \tag{4.37}$$

Rotations about the other axes give the other components of the angular momentum vector.

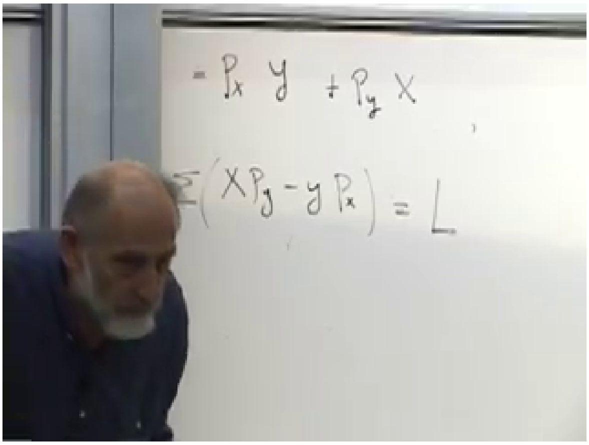

Figure 4.6: The rotational Noether charge at 01:04:10. The board identifies $\sum_a(x_a p_{y_a} - y_a p_{x_a})$ with angular momentum.
Lecture frame: 01:04:10.

Once a candidate symmetry is known, constructing its charge is mechanical: find the infinitesimal generator f_i, multiply by the corresponding canonical momenta, and sum. Finding all the symmetries of a complicated system may be difficult; using Noether's formula after finding one is not.

4.8 Time Translation

Suppose the laws and external conditions have no explicit dependence on time. If one solution is begun a little later, the displaced history is again a solution. An orbit does not care whether we label a particular state as eleven o'clock or one minute later; the whole orbit is merely shifted in time. This is time-translation invariance.

The qualification about external conditions is essential. If we study only part of a system while a massive external body moves according to a prescribed schedule, the forces on our subsystem depend explicitly on time. Translating only the subsystem's history is then not a symmetry.

Let the original trajectory be $q_i(t)$. Pushing it forward in time by ϵ gives the new function

$$q_i^{(\epsilon)}(t) = q_i(t - \epsilon). \tag{4.38}$$

At a fixed value of t, the induced displacement in configuration space is

$$\delta q_i(t) = q_i(t - \epsilon) - q_i(t) \tag{4.39}$$

$$= -\epsilon \dot{q}_i(t) + O(\epsilon^2). \tag{4.40}$$

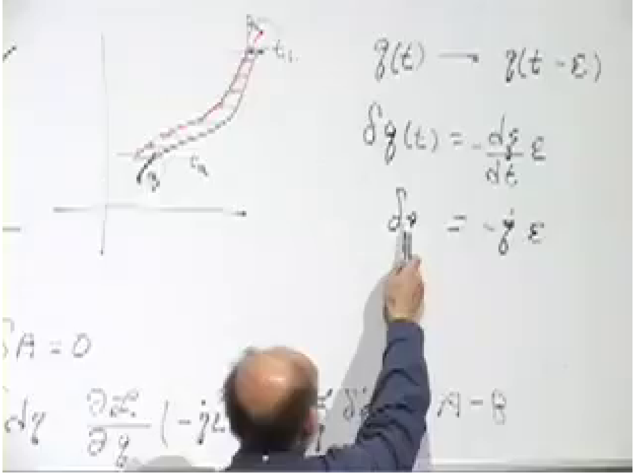

Figure 4.7: The time-shifted trajectory at 01:23:30. The board keeps both views of the transformation together: $q(t) \mapsto q(t - \epsilon)$ and the induced fixed-time displacement $\delta q = -\epsilon \dot{q}$.

Lecture frame: 01:23:30.

Question from the audience. Why does $\delta q_i = -\epsilon \dot{q}_i$ contain a minus sign?

Response. The whole curve is moved forward in time, but at a fixed time t the new curve samples the old curve at the earlier time $t - \epsilon$. If q was increasing, moving the graph upward shifts its intersection with a fixed horizontal time slice toward smaller q. Drawing the trajectory with time vertical makes the sign immediate. If the slope is reversed, then $\dot{q} < 0$ and the same formula reverses the displacement.

4.9 The Endpoint Overhangs

A time translation is not quite an ordinary symmetry of configuration space: it also shifts the interval over which the action is evaluated. Draw the original trajectory and its translate with time running vertically. Through most of the interval they overlap, but the translated path leaves one short extra segment at the upper endpoint while the original leaves one at the lower endpoint. These short segments are the overhangs.

On the common interval, the same integration by parts as before gives the endpoint contribution

$$\left[\sum_i \pi_i \delta q_i\right]_{t_1}^{t_2} = -\epsilon \left[\sum_i \pi_i \dot{q}_i\right]_{t_1}^{t_2}. \tag{4.41}$$

The bulk term again vanishes on the equations of motion.

The upper overhang lasts for the small time ϵ, so its action is, to first order,

$$A_{\text{upper}} = \epsilon \mathcal{L}(t_2) + O(\epsilon^2). \tag{4.42}$$

Similarly, the lower overhang contributes

$$A_{\text{lower}} = \epsilon \mathcal{L}(t_1) + O(\epsilon^2). \tag{4.43}$$

This approximation assumes that $\mathcal{L}$ is regular and changes little over a sufficiently short interval. No singular behavior is being allowed at the endpoints.

Time-translation symmetry says that the total action difference is zero. Combining the ordinary boundary term with the two overhangs gives

$$0 = \epsilon \left[\mathcal{L} - \sum_i \pi_i \dot{q}_i \right]_{t_1}^{t_2}. \tag{4.44}$$

Therefore

$$\mathcal{L} - \sum_i \pi_i \dot{q}_i = \text{constant}. \tag{4.45}$$

Question from the audience. Why did an extra $\mathcal{L}$ appear here but not in the spatial translation or rotation examples?

Response. A spatial transformation moved the path in configuration space while the times t_1, t_2 remained fixed. A time translation moves those temporal endpoints as well. The short overhangs contribute $\mathcal{L}\epsilon$ at the two ends, and their difference supplies the new term.

4.10 The Hamiltonian Is the Energy

The conserved expression has first appeared with the opposite sign from the usual convention. The negative of a conserved quantity is also conserved, so we define

$$H = \sum_i \pi_i \dot{q}_i - \mathcal{L}. \tag{4.46}$$

This is the Hamiltonian, and it is the Noether charge associated with time translation.

For a one-dimensional particle with

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 - U(x), \tag{4.47}$$

the canonical momentum is $\pi_x = m\dot{x}$, and therefore

$$H = \pi_x \dot{x} - \mathcal{L} \tag{4.48}$$
$$= m\dot{x}^2 - \left(\frac{1}{2}m\dot{x}^2 - U(x)\right) \tag{4.49}$$
$$= \frac{1}{2}m\dot{x}^2 + U(x). \tag{4.50}$$

Thus, for the familiar Lagrangian $T - U$, the Hamiltonian is the familiar energy $T + U$.

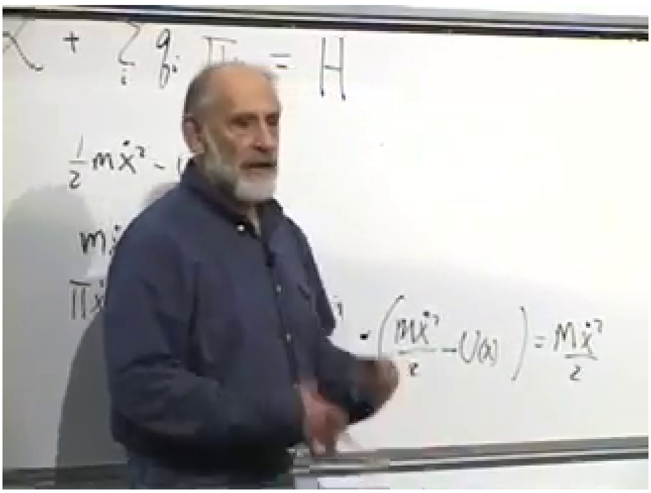

Figure 4.8: The Hamiltonian definition and its one-dimensional check at 01:33:20. The upper line shows $H = \sum_i \dot{q}_i \pi_i - \mathcal{L}$; the lower calculation reduces it to kinetic plus potential energy.

Lecture frame: 01:33:20.

The deeper definition is not simply *kinetic plus potential.* Energy is the Noether charge generated by time translation. In ordinary mechanics that charge reduces to $T + U$, but Eq. (4.46) also applies when the Lagrangian has a less familiar form.

Question from the audience. Is this circular? Was the Lagrangian not defined using energy in the first place?

Response. The general definition of the Lagrangian is the integrand of the action. For the elementary system we happened to know the convenient form $\mathcal{L} = T - U$, but that is not the universal definition. Once a Lagrangian is known, Eq. (4.46) determines the Hamiltonian. For the conventional form it reproduces $T + U$, which is a check rather than an assumption.

4.10.1 Regularity and explicit time dependence

Not every arbitrary function is equally useful as a Lagrangian. At this stage, two conditions are worth recording. It should depend locally on the coordinates and velocities, and the momenta

$$\pi_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \tag{4.51}$$

should normally determine the velocities. Locally, the standard regularity condition is

$$\det\left(\frac{\partial^2 \mathcal{L}}{\partial \dot{q}_i\, \partial \dot{q}_j}\right) \neq 0. \tag{4.52}$$

This is what permits the passage from Lagrangian to Hamiltonian variables. A full treatment belongs to Hamiltonian mechanics.

If the Lagrangian depends explicitly on time,

$$\mathcal{L} = \mathcal{L}(q, \dot{q}, t), \tag{4.53}$$

then shifting a trajectory in time need not leave the action invariant. The derivation above no longer gives a conservation law, and energy need not be conserved. The precise evolution law for that case will be developed after the Hamiltonian formalism is in place.

Question from the audience. How is a Lagrangian obtained from observation rather than guessed?

Response. Historically, classical equations of motion were often found first and only later recognized as Euler–Lagrange equations. The Lorentz force and Maxwell's equations are examples of laws whose action formulations were identified after the physical equations had been assembled from experiment. In modern particle physics, measured scattering processes are tied more directly to interaction terms in a Lagrangian. In either case the Lagrangian is constrained by observation and symmetry; it is not pulled from the air.

The compactness of the action is one reason this language became fundamental. The polar-coordinate equations contain centrifugal terms that would be awkward to guess directly, while the Lagrangian in Eq. (4.6) is easy to remember. The same economy persists in field theory, electromagnetism, relativity, and quantum theory. Lagrange's formulation came first; Hamilton's reformulation followed in the nineteenth century and became the natural bridge to quantum mechanics.

The next practical example will be the harmonic oscillator. At the same time, the formal development will pass from $(q_i, \dot{q}_i)$ to (q_i, π_i), where the Hamiltonian becomes the central dynamical object.

4.11 Symmetry and Dynamics in One View

The polar-coordinate correction already contains the pattern of the whole chapter. Rotational invariance removes θ from the Lagrangian, so its conjugate momentum

$$L = mr^2\dot{\theta} \tag{4.54}$$

is conserved. Substituting that constant into the radial equation exposes the centrifugal barrier $L^2/(mr^3)$.

The general pattern is Noether's theorem. An infinitesimal symmetry $\delta q_i = \epsilon f_i(q)$ has the conserved charge

$$Q = \sum_i \pi_i f_i(q). \tag{4.55}$$

Translations give momentum and rotations give angular momentum. Time translation requires one extra piece of bookkeeping because the integration interval moves. Its endpoint overhangs combine with the ordinary variational boundary term to give

$$H = \sum_i \pi_i \dot{q}_i - \mathcal{L}. \tag{4.56}$$

The Hamiltonian is therefore not merely a convenient formula. It is the conserved charge that expresses the independence of the laws of physics from when an isolated experiment is performed.

CHAPTER 5

PENDULUMS, OSCILLATORS, AND PHASE SPACE

We shall work through three examples. The first is simple, the second is much more complicated, and the third is simpler again. There is a reason for this order. The simple and double pendulums show why Lagrange's equations are often far easier to use than direct force resolution. The harmonic oscillator then introduces the variables and geometry of Hamiltonian mechanics.

The advantage is elementary but decisive. Newton's equation asks for accelerations, which are second derivatives. In curvilinear or angular coordinates, their Cartesian components can become unpleasant. A Lagrangian calculation ordinarily asks first for velocities, which are only first derivatives. Once the kinetic and potential energies are known,

$$\mathcal{L} = T - U,$$

the Euler–Lagrange equations turn the remaining work into a fixed procedure. If the resulting differential equations resist analytic solution, they can be integrated numerically. The difficulty of solving a trajectory is separate from the difficulty of formulating the correct equations.

5.1 The Simple Pendulum

Take a mass m at the end of a rigid, massless rod of fixed length r. The rod pivots in a vertical plane. Its only changing coordinate is the angle θ, measured from the downward vertical. With horizontal coordinate x and downward-positive vertical coordinate y,

$$x = r\sin\theta, \qquad y = r\cos\theta. \tag{5.1}$$

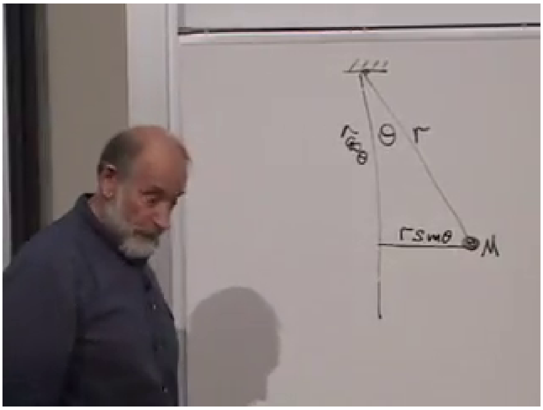

Figure 5.1: The pendulum geometry at 00:05:32. The rod length r, angle θ, and horizontal projection $r\sin\theta$ are complete and unobstructed.

Differentiation gives

$$v_x = r\cos\theta\,\dot{\theta}, \qquad v_y = -r\sin\theta\,\dot{\theta}. \tag{5.2}$$

The two trigonometric factors disappear when the speed is squared:

$$\begin{aligned} v^2 &= v_x^2 + v_y^2 \\ &= r^2\dot{\theta}^2\left(\cos^2\theta + \sin^2\theta\right) = r^2\dot{\theta}^2. \end{aligned} \tag{5.3}$$

Hence

$$T = \frac{1}{2}mr^2\dot{\theta}^2. \tag{5.4}$$

Only differences of gravitational potential matter. Taking zero potential when the rod is horizontal, the bob's height relative to the pivot is $-r\cos\theta$, so

$$U = -mgr\cos\theta. \tag{5.5}$$

At $\theta = 0$ the bob is at the bottom and $U = -mgr$; at $\theta = \pi$ it is upright and $U = +mgr$.

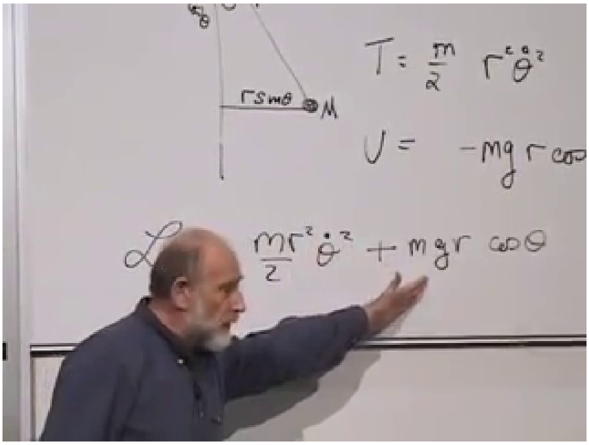

Figure 5.2: The completed kinetic energy, potential energy, and pendulum Lagrangian at 00:10:50. The gesture emphasizes the plus sign created by subtracting the negative gravitational potential.

The Lagrangian is therefore

$$\boxed{\mathcal{L} = \frac{1}{2}mr^2\dot{\theta}^2 + mgr\cos\theta.} \tag{5.6}$$

Question from the audience. Why does the Lagrangian contain $+mgr\cos\theta$ when gravity pulls downward?

Response. The potential itself is $U = -mgr\cos\theta$ in the chosen convention, and the Lagrangian is $\mathcal{L} = T - U$. Subtracting this negative potential produces the plus sign. The force will acquire the expected restoring sign when $\mathcal{L}$ is differentiated with respect to θ.

5.1.1 Equation of motion

The momentum conjugate to θ is

$$\pi_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2\dot{\theta}. \tag{5.7}$$

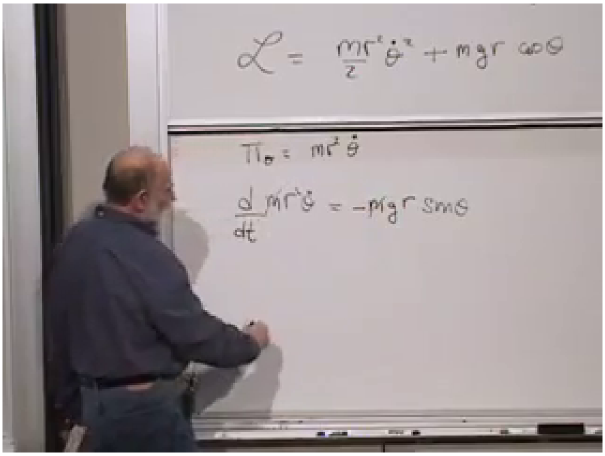

Figure 5.3: The completed pendulum Euler–Lagrange equation at 00:12:27. The upper board retains the Lagrangian from which the lower equation follows.

It is the angular momentum about the pivot, although the derivation needs only its definition. The Euler–Lagrange equation gives

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = -mgr\sin\theta, \tag{5.8}$$

or

$$\boxed{\ddot{\theta} + \frac{g}{r}\sin\theta = 0.} \tag{5.9}$$

This nonlinear equation can be integrated directly on a computer by advancing θ and $\dot{\theta}$ over short time steps.

5.1.2 Hamiltonian and energy

There is one generalized coordinate, so the Hamiltonian is

$$\begin{aligned} H &= \pi_\theta\dot{\theta} - \mathcal{L} \\ &= mr^2\dot{\theta}^2 - \left(\frac{1}{2}mr^2\dot{\theta}^2 + mgr\cos\theta\right) \\ &= \frac{1}{2}mr^2\dot{\theta}^2 - mgr\cos\theta. \end{aligned} \tag{5.10}$$

It is exactly $T + U$, as expected.

Question from the audience. Should $\sum_i \pi_i \dot{q}_i$ always equal $2T$?

Response. It does for the ordinary kinetic energies used here. If T is homogeneous of degree two in the generalized velocities, Euler's homogeneous-function theorem gives $\sum_i (\partial T/\partial \dot{q}_i)\dot{q}_i = 2T$. It is not a universal identity for every possible Lagrangian, so it should be checked rather than assumed.

5.2 The Pendulum Energy Landscape

We do not need the exact nonlinear solution to classify the motion. The potential changes by

$$U(\pi) - U(0) = mgr - (-mgr) = 2mgr \tag{5.11}$$

between the bottom and the upright position.

Suppose the pendulum starts at the bottom with a push. If its initial kinetic energy is less than $2mgr$, it spends that energy climbing, reaches a turning point with $\dot{\theta} = 0$, and returns. It oscillates between two turning points. If the kinetic energy exceeds $2mgr$, some remains when the bob passes the upright point, and the pendulum continues to rotate. The rotation may be clockwise or counterclockwise. Exactly at the threshold, the ideal trajectory approaches the unstable upright configuration with zero speed. This separatrix motion is possible but requires exact tuning.

Question from the audience. What is meant by the *top* of the orbit? Is it the ceiling?

Response. The pivot is imagined to permit a complete 360° rotation. The top is the configuration $\theta = \pi$, with the bob vertically above the pivot. The phrase does not refer to a physical ceiling obstructing the rod.

Thus the familiar swinging pendulum and the continuously rotating pendulum are different families of motion separated

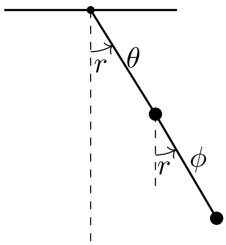

Figure 5.4: Both angles are measured from the same vertical direction. This choice makes a common rotation appear as equal shifts of θ and ϕ.

by the threshold energy. This qualitative conclusion follows immediately from conservation of Eq. (5.10).

5.3 The Double Pendulum

The single pendulum is simple enough that direct force resolution remains possible. The double pendulum is the real test. Its motion can become chaotic, and resolving all rod tensions and accelerations by $F = ma$ is treacherous. The Lagrangian method, by contrast, changes only the amount of algebra.

Take two equal point masses m joined by two massless rods of equal length r, moving in one vertical plane. Let θ be the angle of the first rod from the downward vertical and let ϕ be the angle of the second rod from the same vertical.

Question from the audience. Must the double pendulum remain in one plane?

Response. No. A spatial double pendulum is a legitimate problem, but it needs more coordinates. We choose planar motion here so that the two angles θ and ϕ form a minimal complete set.

The first mass has kinetic energy

$$T_1 = \frac{1}{2} m r^2 \dot{\theta}^2. \tag{5.12}$$

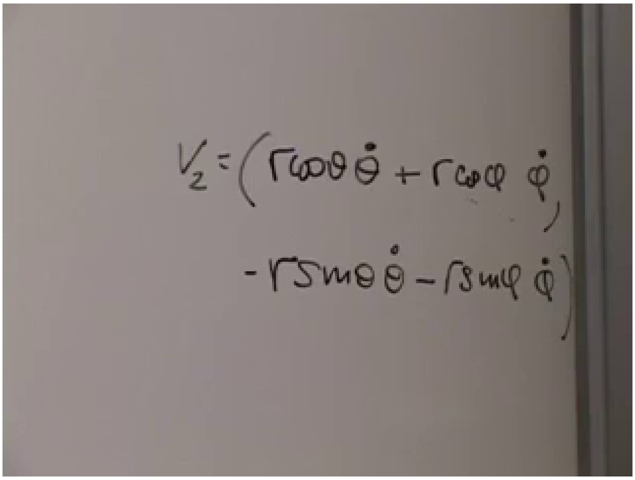

Figure 5.5: The two contributions to the lower bob's velocity at 00:29:02. Each Cartesian component contains motion inherited from the upper bob and motion due to the second rod.

The second mass moves both because the first mass moves and because the second rod rotates relative to the first mass. Its velocity components are

$$v_{2x} = r\cos\theta\,\dot\theta + r\cos\phi\,\dot\phi, \tag{5.13}$$

$$v_{2y} = -r\sin\theta\,\dot\theta - r\sin\phi\,\dot\phi. \tag{5.14}$$

Squaring and adding produces two diagonal terms and one mixed term:

$$\begin{aligned} v_2^2 &= r^2\dot\theta^2 + r^2\dot\phi^2 \\ &\quad + 2r^2\dot\theta\dot\phi\left(\cos\theta\cos\phi + \sin\theta\sin\phi\right). \end{aligned} \tag{5.15}$$

The trigonometric identity

$$\cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta - \phi) \tag{5.16}$$

then gives

$$T_2 = \frac{1}{2}mr^2\dot\theta^2 + \frac{1}{2}mr^2\dot\phi^2 + mr^2\dot\theta\dot\phi\cos(\theta - \phi). \tag{5.17}$$

Adding the first bob,

$$\boxed{T = mr^2\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2 + mr^2\dot{\theta}\dot{\phi}\cos(\theta - \phi).} \tag{5.18}$$

Question from the audience. Why did the factor 1/2 disappear from the $\dot{\theta}^2$ term?

Response. Both masses carry the motion generated by the first rod. The upper bob contributes $\frac{1}{2}mr^2\dot{\theta}^2$, and the lower bob contributes another identical term. Their sum is $mr^2\dot{\theta}^2$. The simple coefficient depends on our choice of equal masses and equal rod lengths.

The gravitational energy must also count the first rod twice, because both masses rise when θ changes:

$$\begin{aligned} U &= -mgr\cos\theta - mg(r\cos\theta + r\cos\phi) \\ &= -2mgr\cos\theta - mgr\cos\phi. \end{aligned} \tag{5.19}$$

Question from the audience. Are the two gravitational energies measured from the same reference point?

Response. Yes. The lower bob's vertical coordinate is the sum of the two rod projections. Its energy therefore includes both $-mgr\cos\theta$ and $-mgr\cos\phi$, while the upper bob supplies a second $-mgr\cos\theta$. A common additive constant would still be irrelevant.

The full Lagrangian is

$$\boxed{\begin{aligned} \mathcal{L} = mr^2\dot{\theta}^2 &+ \frac{1}{2}mr^2\dot{\phi}^2 \\ &+ mr^2\dot{\theta}\dot{\phi}\cos(\theta - \phi) \\ &+ mgr(2\cos\theta + \cos\phi). \end{aligned}} \tag{5.20}$$

Nothing was guessed. We chose coordinates, differentiated positions, formed T and U, and subtracted.

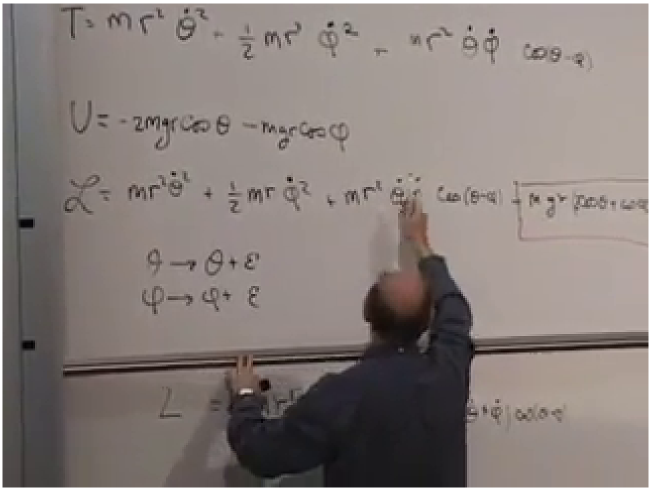

Figure 5.6: The double-pendulum kinetic energy, potential energy, Lagrangian, and common angle shift at 00:52:47. The board preserves the $\cos(\theta - \phi)$ coupling that controls the later derivatives.

5.4 Rotation Symmetry Without Gravity

Gravity singles out the vertical direction, so the full Lagrangian Eq. (5.20) is not invariant under a common planar rotation. Now imagine the apparatus far from a gravitational field. Removing the last two potential terms leaves

$$\mathcal{L}_0 = mr^2\dot\theta^2 + \frac{1}{2}mr^2\dot\phi^2 + mr^2\dot\theta\dot\phi\cos(\theta - \phi). \tag{5.21}$$

It is invariant under

$$\theta \longmapsto \theta + \epsilon, \qquad \phi \longmapsto \phi + \epsilon. \tag{5.22}$$

Adding a constant changes neither angular velocity, and the difference $\theta - \phi$ remains fixed.

Question from the audience. Why does the coupling contain only $\theta - \phi$, rather than $\theta + \phi$?

Response. Without gravity there is no absolute direction in the plane. A common rotation must leave the kinetic energy unchanged. $\theta - \phi$ is invariant under that rotation, whereas $\theta + \phi$ changes by 2ϵ. The trigonometric identity in Eq. (5.16) is therefore also a visible statement of rotational symmetry.

For this symmetry the two generators are both 1. Noether's charge is

$$Q = \pi_\theta + \pi_\phi. \tag{5.23}$$

The mixed kinetic term makes each canonical momentum depend on both angular velocities:

$$\pi_\theta = 2mr^2\dot{\theta} + mr^2\dot{\phi}\cos(\theta - \phi), \tag{5.24}$$

$$\pi_\phi = mr^2\dot{\phi} + mr^2\dot{\theta}\cos(\theta - \phi). \tag{5.25}$$

Thus

$$\boxed{\begin{aligned} Q = mr^2\Big[&2\dot{\theta} + \dot{\phi} \\ &+ (\dot{\theta} + \dot{\phi})\cos(\theta - \phi)\Big] \\ = \text{constant}&. \end{aligned}} \tag{5.26}$$

This is the total angular momentum. Once its initial value is known, one relation among the velocities remains fixed throughout the motion.

5.4.1 The coupled equations

The equations may be unpleasant, but their construction is unambiguous:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{\partial \mathcal{L}}{\partial \theta}, \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{\partial \mathcal{L}}{\partial \phi}. \tag{5.27}$$

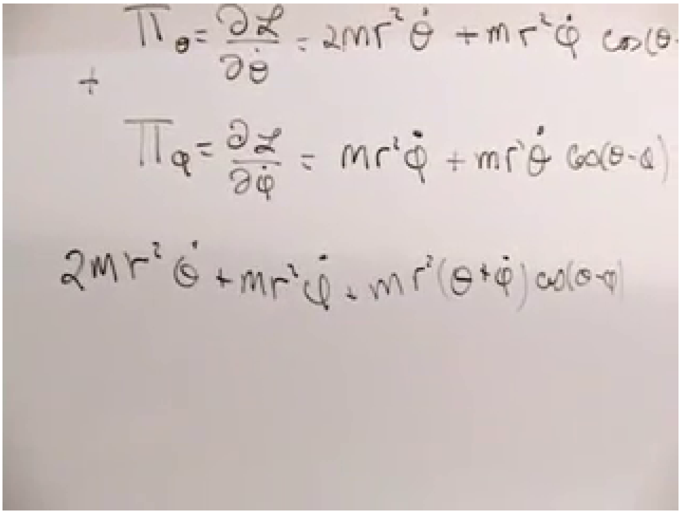

Figure 5.7: The two canonical momenta and their conserved sum at 00:46:15. The mixed velocity dependence is complete and unobstructed.

Expanding Eq. (5.20) carefully, with $\Delta = \theta - \phi$, gives the useful implicit pair

$$2\ddot\theta + \ddot\phi\cos\Delta + \dot\phi^2\sin\Delta + 2\frac{g}{r}\sin\theta = 0, \tag{5.28}$$

$$\ddot\phi + \ddot\theta\cos\Delta - \dot\theta^2\sin\Delta + \frac{g}{r}\sin\phi = 0. \tag{5.29}$$

Setting $g = 0$ recovers the rotation-invariant case.

Question from the audience. When differentiating $\cos(\theta - \phi)$, where do the opposite signs enter?

Response. A partial derivative with respect to θ gives $-\sin(\theta - \phi)$, while a partial derivative with respect to ϕ gives $+\sin(\theta - \phi)$. A time derivative also differentiates both angles:

$$\frac{d}{dt}\cos(\theta - \phi) = -\sin(\theta - \phi)(\dot{\theta} - \dot{\phi}).$$

Keeping partial and total derivatives distinct removes the apparent ambiguity.

The energy is obtained by reversing the sign of the potential contribution:

$$\begin{aligned} E = T + U = mr^2\dot{\theta}^2 &+ \frac{1}{2}mr^2\dot{\phi}^2 \\ &+ mr^2\dot{\theta}\dot{\phi}\cos\Delta \\ &- mgr(2\cos\theta + \cos\phi). \end{aligned} \tag{5.30}$$

Without gravity, energy and Q provide two conserved relations. They do not make the trajectories visually simple, but they substantially reduce the problem. This is the desired lesson: difficult mechanics can still be reduced to a reliable algorithm.

5.5 Why Harmonic Motion Is Universal

We now turn to the most basic nontrivial problem in theoretical physics. A free particle is simpler, but the harmonic oscillator is the structure that reappears whenever a stable system is disturbed slightly from equilibrium. The pendulum already contains it.

Near $\theta = 0$,

$$\begin{aligned} U(\theta) &= -mgr\cos\theta \\ &= -mgr + \frac{1}{2}mgr\,\theta^2 \\ &\quad - \frac{1}{24}mgr\,\theta^4 + \cdots . \end{aligned} \tag{5.31}$$

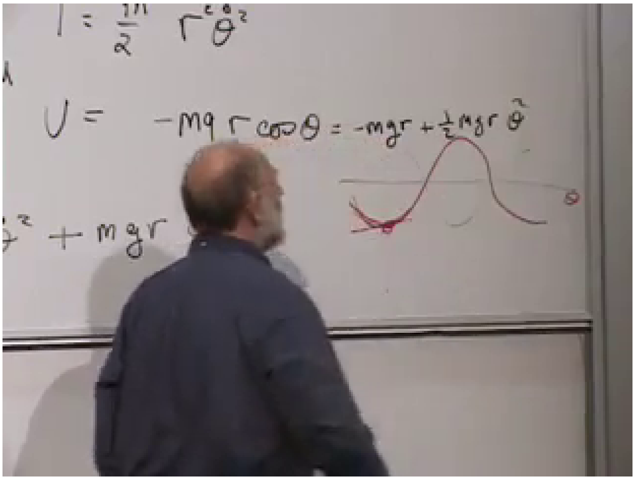

Figure 5.8: The exact pendulum potential and its quadratic approximation at 00:59:15. The parabola agrees with the cosine potential near the stable minimum and departs from it for larger angles.

The constant $-mgr$ has no effect on the equations of motion. To quadratic order,

$$\mathcal{L} \simeq \frac{1}{2}mr^2\dot{\theta}^2 - \frac{1}{2}mgr\,\theta^2, \tag{5.32}$$

which is the Lagrangian of a harmonic oscillator.

Question from the audience. What happened to the constant $-mgr$?

Response. It remains in the numerical value assigned to the energy, but it has zero derivative. Adding or subtracting a constant from $\mathcal{L}$ changes neither Euler–Lagrange equation. We may choose the zero of potential for convenience.

The canonical realization is a mass on a spring. Let $x = 0$ be its equilibrium position and write

$$U(x) = \frac{1}{2}kx^2, \qquad \mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2. \tag{5.33}$$

The force is

$$F = -\frac{dU}{dx} = -kx, \tag{5.34}$$

which is Hooke's law. Positive displacement produces a negative force and negative displacement produces a positive force: both point back toward equilibrium.

This law is not an exact global description of a real spring. Stretch far enough and nonlinearities appear; farther still, the spring breaks. Its generality is local. For a smooth potential expanded about $x = 0$,

$$U(x) = U_0 + ax + \frac{1}{2}kx^2 + c_3x^3 + c_4x^4 + \cdots . \tag{5.35}$$

At an equilibrium $a = U'(0) = 0$. At a stable nondegenerate minimum $k = U''(0) > 0$, so the quadratic term dominates all higher powers for sufficiently small x.

Question from the audience. What if a linear term appears in the expansion?

Response. Then $x = 0$ was not chosen at the equilibrium. Shift the origin to the point where $U'(x) = 0$; the linear term disappears. The constant term fixes only the zero of energy, while the quadratic term controls the leading restoring force.

Question from the audience. Is Hooke's law exact, and can it be derived without experiment?

Response. Its coefficient k and its range of validity must be measured. Mathematics explains why a quadratic approximation is generic near a nondegenerate minimum, not why a particular material has a particular k. The approximation improves as the displacement shrinks, and its error is estimated by comparing the higher terms with $kx^2/2$.

There is one refinement worth making. A cubic correction may accompany a positive quadratic term. But if the quadratic coefficient were tuned to zero and $x = 0$ were still a genuine smooth minimum, a cubic term could not be the leading nonzero term, because an odd power changes sign across the origin. The next stable leading possibility is typically quartic. The absence of the ordinary quadratic term is therefore exceptional.

This same small-oscillation logic applies to mechanical vibration, sound, water waves, molecular motion, and fields. It fails for sufficiently large amplitude, where the higher terms can no longer be neglected.

Question from the audience. If the small-oscillation approximation is initially valid, what prevents the motion from growing until it leaves that regime?

Response. Energy conservation. For the stable oscillator,

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2,$$

and both terms are nonnegative. Small initial energy bounds both $|x|$ and $|\dot{x}|$. An inverted potential has the opposite behavior: near the top of a hill the quadratic potential is negative, and a small displacement can grow rather than oscillate.

5.6 Equation and General Solution

Applying the Euler–Lagrange equation to Eq. (5.33),

$$m\ddot{x} = -kx. \tag{5.36}$$

Define

$$\omega^2 = \frac{k}{m}. \tag{5.37}$$

Then

$$\boxed{\ddot{x} + \omega^2 x = 0.} \tag{5.38}$$

Only the ratio k/m controls the time dependence. Increasing k and m by the same factor leaves the equation unchanged.

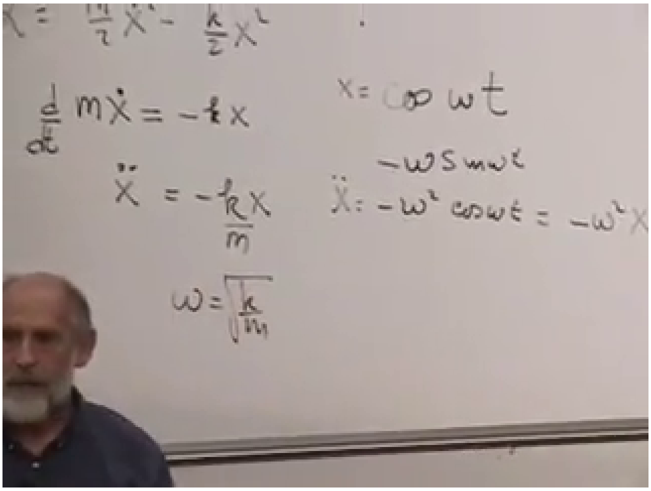

Figure 5.9: The oscillator equation, cosine trial solution, and $\omega = \sqrt{k/m}$ at 01:20:42.

The reason sines and cosines solve the equation is immediate:

$$\frac{d^2}{dt^2}\cos\omega t = -\omega^2\cos\omega t,$$
$$\frac{d^2}{dt^2}\sin\omega t = -\omega^2\sin\omega t.$$

The general real solution is

$$x(t) = A\cos\omega t + B\sin\omega t. \tag{5.39}$$

Two constants are required because the equation is second order: initial position and initial velocity must both be specified. The same family may be written

$$x(t) = C\cos[\omega(t - t_0)]\,, \tag{5.40}$$

where C is the amplitude and t_0 specifies the phase, here chosen as a time at which x is maximal.

Question from the audience. Is the phase $t - t_0$, t_0, or $\omega(t - t_0)$?

Response. The dimensionless phase appearing inside the cosine is $\omega(t - t_0)$. The parameter t_0 is a time shift, often loosely called the phase parameter. Equivalently one may write $C\cos(\omega t - \delta)$ with dimensionless phase $\delta = \omega t_0$.

5.7 From Velocities to Canonical Momenta

For the oscillator,

$$p = \pi_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}. \tag{5.41}$$

The Hamiltonian is

$$\begin{aligned} H &= p\dot{x} - \mathcal{L} \\ &= m\dot{x}^2 - \left(\frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2\right) \\ &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2. \end{aligned} \tag{5.42}$$

Hamiltonian mechanics takes x and p, rather than x and $\dot{x}$, as its fundamental variables. Here the conversion is immediate:

$$\dot{x} = \frac{p}{m}. \tag{5.43}$$

Substitution gives

$$\boxed{H(x,p) = \frac{p^2}{2m} + \frac{1}{2}kx^2.} \tag{5.44}$$

Notice the useful change: m multiplies $\dot{x}^2/2$, but it divides $p^2/2$. This is a mnemonic for ordinary quadratic kinetic energy, not a substitute for carrying out the Legendre transform.

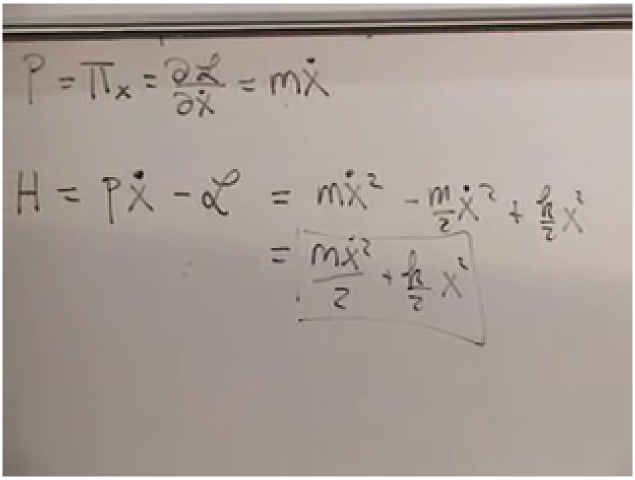

Figure 5.10: The canonical momentum and Hamiltonian calculation at 01:26:53. The completed board state is visible without the earlier hand occlusion.

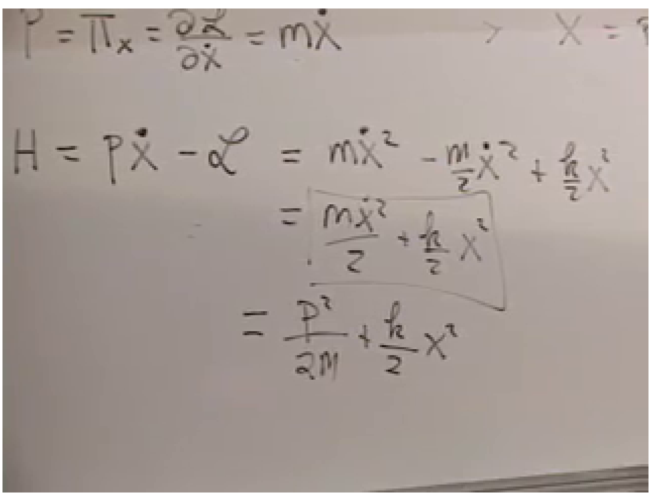

Figure 5.11: The oscillator Hamiltonian rewritten in canonical variables at 01:33:32. The final line contains $p^2/(2m) + kx^2/2$.

Question from the audience. Why reorganize mechanics around q and p instead of q and $\dot{q}$?

Response. The full motivation is not yet visible. Hamilton's equations will place coordinates and momenta in a nearly symmetric first-order form, and this structure becomes indispensable in quantum mechanics. For now the oscillator provides the first hint: after a rescaling, H treats the quadratic coordinate and momentum terms symmetrically.

Question from the audience. Does the oscillator have a special role in quantum mechanics too?

Response. Yes. Small disturbances about equilibrium lead to oscillators in quantum theory for the same local reason. Quantum mechanics adds zero-point energy, so the energy and fluctuation amplitude cannot both be reduced arbitrarily to zero. That belongs to the quantum course; here we retain the classical limit.

5.8 Motion in Phase Space

The plane with horizontal coordinate x and vertical coordinate p is called phase space. A point (x, p) specifies the instantaneous state of this one-dimensional system. As time passes, that point traces a curve.

Energy conservation confines the curve to

$$\frac{p^2}{2m} + \frac{1}{2}kx^2 = E. \tag{5.45}$$

This is an ellipse. Its intercepts are

$$x_{\max} = \sqrt{\frac{2E}{k}}, \qquad p_{\max} = \sqrt{2mE}. \tag{5.46}$$

Changing E changes the size but not the eccentricity of the ellipses. Rescaling the axes by

$$X = \sqrt{k}\,x, \qquad P = \frac{p}{\sqrt{m}}$$

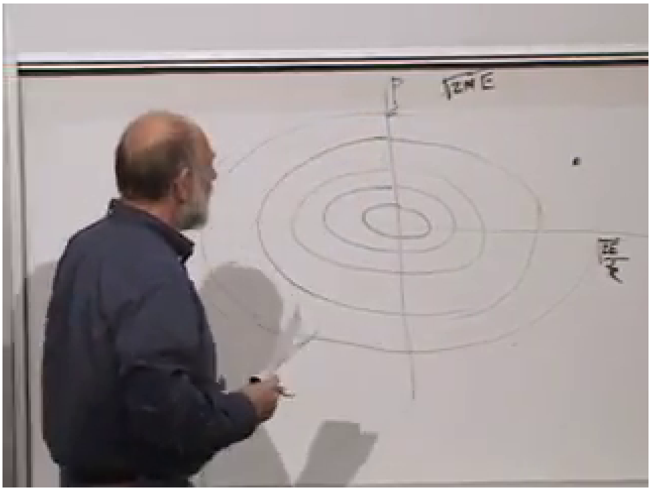

Figure 5.12: Nested constant-energy ellipses in the x-p plane at 01:40:34. The board also marks the momentum and position intercepts.

turns them into circles $X^2 + P^2 = 2E$.

The phase point goes once around its ellipse in the oscillator period

$$\tau = \frac{2\pi}{\omega}. \tag{5.47}$$

Every ellipse has the same period because the ideal oscillator frequency is independent of amplitude. In the rescaled plane the entire family rotates uniformly with angular frequency ω.

Ordinary oscillation is a projection of this motion. Projection onto the x-axis gives the back-and-forth displacement; projection onto the p-axis gives the momentum oscillation, shifted by a quarter period.

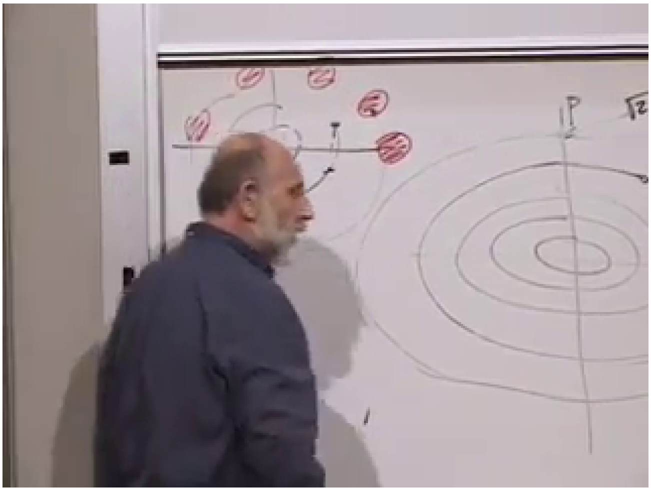

Figure 5.13: A patch of neighboring initial conditions transported around the phase portrait at 01:45:51. Its position changes while its phase-space area is preserved.

Question from the audience. Is the oscillator moving fastest when it is farthest from the origin?

Response. No. At $x = \pm x_{\max}$, the phase ellipse meets the x-axis, so $p = 0$ and the mass is momentarily at rest. At $x = 0$, the ellipse reaches $p = \pm p_{\max}$, so the speed is greatest.

The ellipse is special to the harmonic oscillator, but the larger statement is general: for a time-independent Hamiltonian, trajectories remain on constant-energy sets in phase space.

5.8.1 Area and invertibility

Imagine a small patch of initial conditions rather than one perfectly known point. Oscillator evolution carries the patch around phase space without changing its area. The corresponding general theorem is Liouville's theorem: Hamiltonian flow preserves phase-space volume. Its proof belongs to the Hamiltonian development that follows.

One technical condition is already unavoidable. To replace velocities by momenta, the relations

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \tag{5.48}$$

must be invertible for $\dot{q}_i(q, p)$. For an ordinary unconstrained system, local invertibility requires

$$\det\left(\frac{\partial^2 \mathcal{L}}{\partial \dot{q}_i\, \partial \dot{q}_j}\right) \neq 0. \tag{5.49}$$

Question from the audience. What happens if one momentum corresponds to several velocities, or the relation cannot be inverted?

Response. Then the ordinary Legendre transform does not define a single Hamiltonian in the variables q, p. For elementary unconstrained mechanics that signals a defective formulation. More generally, singular Lagrangians can describe constrained or gauge systems, but they require a separate constrained-Hamiltonian treatment.

We have therefore arrived at the threshold of Hamiltonian mechanics. The examples began as a practical argument for calculating velocities instead of accelerations. They end with a new state space, a conserved function $H(q, p)$, motion along its level sets, and an area-preserving flow. Those structures, rather than any one pendulum trajectory, are what we shall develop next.

CHAPTER 6

HAMILTONIAN MECHANICS AND PHASE-SPACE FLOW

Tonight we turn to Hamiltonian mechanics. This is still classical mechanics, not quantum mechanics, but the change of language will matter enormously when we reach quantum theory. The immediate purpose is more elementary: we want a description in which a complete state is a point in phase space and the law of motion carries that point along a reversible flow.

The route has three stages. First we ask what information a state must contain. Then a Legendre transformation replaces velocities by canonical momenta. Finally Hamilton's equations turn one function, the Hamiltonian, into the velocity field of the entire phase space.

6.1 Reversible Laws and Complete States

Begin with a coin whose state is either heads or tails. Imagine that time advances in discrete steps. One possible rule leaves the coin unchanged: heads goes to heads and tails goes to tails. Another rule flips it at every step. In either case the present determines not only the next state but also the preceding one. The evolution can be run forward or backward without ambiguity.

For a system with many possible states, represent each state by a point and draw an arrow from every point to its successor. A reversible rule permits exactly one arrow into and one arrow out of every point. For a finite state space, the arrows must therefore decompose into cycles. If the cycles are marked by different labels, such as $+1$ and -1, the label is conserved: a trajectory cannot jump from one cycle to another.

Now contrast this with a many-to-one rule. Several states

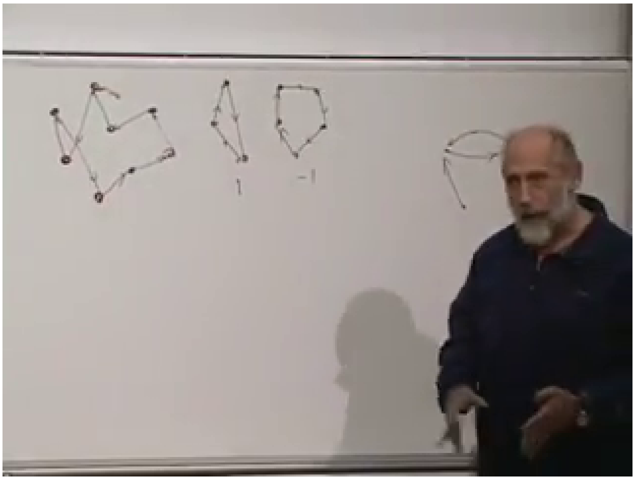

Figure 6.1: Discrete evolution at 00:08:05. The central cycles belong to separate sectors, while the map at the left illustrates how several distinct histories can merge when backward evolution is not unique.

may have the same successor. Forward prediction remains possible, but retrodiction does not: after reaching the common image, we cannot infer which predecessor was occupied. Distinguishable histories have been erased.

The same idea can be pictured by covering the state space with red dots and letting every dot move. Under a reversible rule no dot disappears, splits, or merges with another. In a continuous Hamiltonian system something stronger is true: phase-space volume is preserved. The points move as an incompressible fluid. Reversibility by itself does not prove incompressibility for every conceivable continuous flow; the canonical structure of Hamilton's equations does. We shall see the local reason for that structure in the crossed derivatives of those equations.

The phrase *phase space* will mean the space of complete instantaneous states. A point in it must carry enough information for the equations of motion to determine a unique trajectory. If the microscopic law is reversible, the same point also determines the past.

6.2 Why Position Is Not a State

Suppose a particle is observed at a position q. Where will it be one second later? Position alone does not answer the question. The particle may be moving in any direction and at any speed. A complete state must include a second set of data. It may be written as

$$(q_i, \dot{q}_i) \qquad \text{or} \qquad (q_i, p_i). \tag{6.1}$$

The second form follows after a change of variables. The coordinates q_i form configuration space. The pairs (q_i, p_i) form phase space.

This distinction is already visible in Newton's law. For N Cartesian coordinates,

$$m\ddot{x}_i = F_i, \qquad i = 1, \ldots, N, \tag{6.2}$$

is a set of N second-order equations. Define

$$p_i = m\dot{x}_i. \tag{6.3}$$

The same physics becomes $2N$ first-order equations,

$$\boxed{\dot{x}_i = \frac{p_i}{m}, \qquad \dot{p}_i = F_i.} \tag{6.4}$$

This rewriting is not yet the full Hamiltonian construction. Any ordinary second-order equation can be made first order by introducing an extra variable. Hamiltonian mechanics demands the more special pairing of coordinates and canonical momenta, with both first-order equations generated by derivatives of a single function.

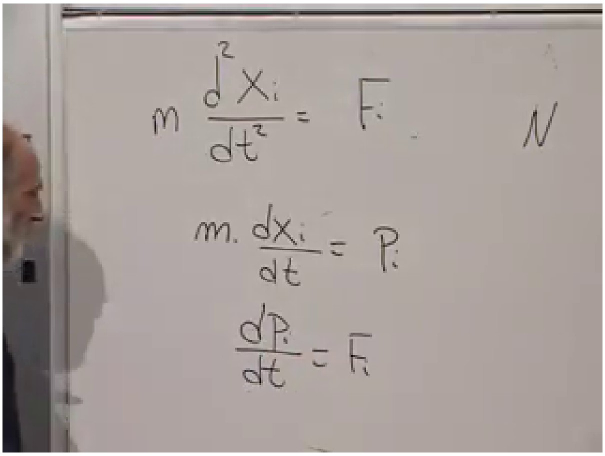

Figure 6.2: The Newton equation rewritten as two first-order equations at 00:18:48. No new physics has been added; the complete state has been made explicit.

Question from the audience. Why reformulate $F = ma$ if it already predicts the motion?

Response. The Lagrangian formulation freed us from Cartesian coordinates. Motion on a plane may be described by x and y, while a pendulum or a particle on a sphere is more naturally described by angles. Hamilton's formulation reveals an additional structure: mechanics becomes a flow on a space of canonical pairs. The reformulation is useful in classical mechanics and becomes indispensable in quantum mechanics. Historical motives are harder to reconstruct than the mathematics, so the important test is what the construction accomplishes.

Imagine placing one identical system at every phase-space point. Each copy has an instantaneous velocity $(\dot{q}_i, \dot{p}_i)$, so the equations define a vector field on phase space. Following that field carries the whole cloud of systems forward. Our task is to find the function that generates this flow.

6.3 The Legendre Transformation

The required mathematical device is a Legendre transformation. Start with one variable V and a differentiable function $L(V)$. Define the conjugate variable

$$P = \frac{dL}{dV}. \tag{6.5}$$

Assume, at least in the region of interest, that this relation can be inverted to give $V = V(P)$. Geometrically the graph of $P(V)$ must not double back in a way that assigns several relevant values of V to the same P.

With convenient additive constants one may write

$$L(V) = \int_0^V P(V')\,dV', \qquad H(P) = \int_0^P V(P')\,dP'. \tag{6.6}$$

For a monotone graph through the chosen origin, these are complementary areas inside a rectangle of area PV:

$$H(P) + L(V) = PV. \tag{6.7}$$

The invariant definition, independent of the choice of additive constants, is

$$\boxed{H(P) = PV - L(V),} \qquad P = L'(V), \tag{6.8}$$

where V on the right is to be expressed as a function of P.

The derivative of the transformed function follows by varying Eq. (6.8):

$$\begin{aligned} \delta H &= P\,\delta V + V\,\delta P - \frac{dL}{dV}\,\delta V \\ &= P\,\delta V + V\,\delta P - P\,\delta V = V\,\delta P. \end{aligned} \tag{6.9}$$

Therefore

$$\boxed{\frac{dH}{dP} = V.} \tag{6.10}$$

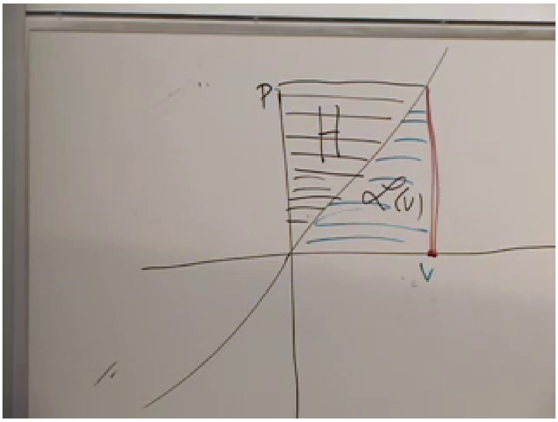

Figure 6.3: The complementary-area construction at 00:32:05. The area labeled $L(V)$ lies under $P(V)$; the remaining part of the PV rectangle is the transformed function $H(P)$.

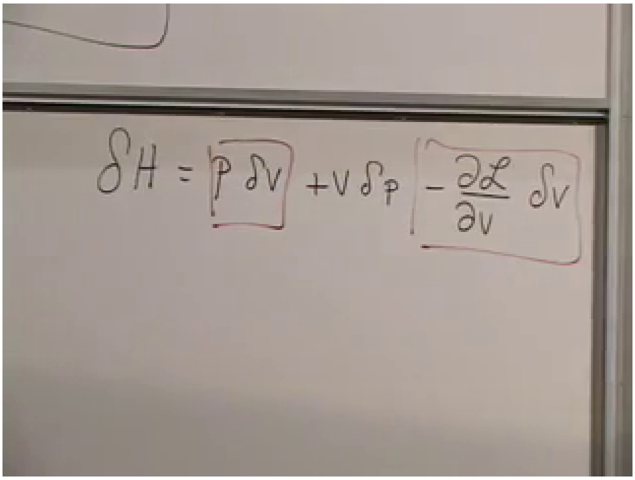

Figure 6.4: The cancellation in the variation of $H = PV - L$ at 00:36:30. The boxed terms cancel because $dL/dV = P$, leaving $\delta H = V\,\delta P$.

Differentiation created P from $L(V)$; after the transform, differentiation creates V from $H(P)$. This reciprocity is the point of the construction.

For several variables, let $L = L(v_1, \ldots, v_N)$ and define

$$p_i = \frac{\partial L}{\partial v_i}. \tag{6.11}$$

If these equations can be inverted for all the v_i, the transform is

$$H(p_1, \ldots, p_N) = \sum_i p_i v_i - L, \tag{6.12}$$

and the same cancellation gives

$$\frac{\partial H}{\partial p_i} = v_i. \tag{6.13}$$

Legendre transformations appear repeatedly in thermodynamics, where changing which variables are held fixed produces the familiar thermodynamic potentials. Here we use the same mathematics to exchange velocities for momenta.

6.3.1 Coordinates Go Along for the Ride

In mechanics the Lagrangian is not merely $L(v)$, but

$$L = L(q, v, t), \qquad v_i = \dot{q}_i. \tag{6.14}$$

The transformation is performed only in the velocity variables. During it, the coordinates q_i and time t are held fixed as parameters:

$$p_i = \frac{\partial L}{\partial v_i}, \qquad H(q, p, t) = \sum_i p_i v_i - L(q, v, t). \tag{6.15}$$

The inverted velocities $v_i = v_i(q, p, t)$ are then substituted on the right, so H is genuinely a function of q, p, t.

Question from the audience. May the velocity determined by the momentum also depend on position?

Response. Yes. The Legendre transformation holds q fixed, but the relation $p_i = \partial L/\partial v_i$ may contain q. A particle in curvilinear coordinates is a standard example: the coefficients in the kinetic energy depend on position, so solving for v_i generally produces $v_i(q, p)$.

Local invertibility is controlled by the velocity Hessian

$$W_{ij} = \frac{\partial^2 L}{\partial v_i \partial v_j}. \tag{6.16}$$

For ordinary unconstrained mechanics, $\det W \neq 0$ allows the velocities to be solved locally in terms of the momenta. A singular Hessian does not mean that nature is inconsistent; it signals a constrained system requiring a more careful Hamiltonian treatment.

6.4 Hamilton's Equations

Now vary the mechanical transform while allowing both q_i and p_i to change. At fixed time,

$$\begin{aligned} \delta H &= \sum_i (p_i\, \delta v_i + v_i\, \delta p_i) - \sum_i \frac{\partial L}{\partial q_i}\, \delta q_i - \sum_i \frac{\partial L}{\partial v_i}\, \delta v_i \\ &= \sum_i v_i\, \delta p_i - \sum_i \frac{\partial L}{\partial q_i}\, \delta q_i. \end{aligned} \tag{6.17}$$

The two terms proportional to δv_i cancel because $p_i = \partial L/\partial v_i$. Reading off the coefficients gives

$$\frac{\partial H}{\partial p_i} = v_i = \dot{q}_i, \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i}. \tag{6.18}$$

The Euler–Lagrange equations say

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i}. \tag{6.19}$$

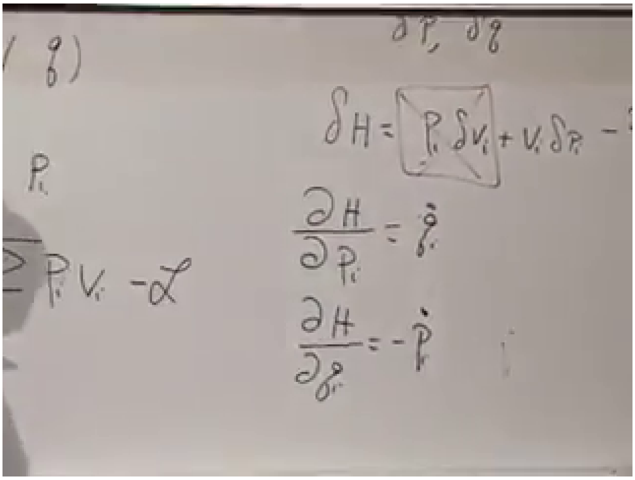

Figure 6.5: The completed pair of Hamilton equations at 00:52:40. The crossed derivatives and the single minus sign encode the canonical phase-space structure.

Since the derivative on the left is $\dot{p}_i$, we arrive at

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}.} \tag{6.20}$$

The equations are first order and nearly symmetric in q_i and p_i. Their derivatives are crossed: the rate of change of q_i comes from a momentum derivative, while the rate of change of p_i comes from a coordinate derivative. The minus sign is essential. It makes the phase-space velocity field

$$\boldsymbol{u}_H = \left(\frac{\partial H}{\partial p_1}, \ldots, \frac{\partial H}{\partial p_N}, -\frac{\partial H}{\partial q_1}, \ldots, -\frac{\partial H}{\partial q_N}\right) \tag{6.21}$$

divergence-free:

$$\nabla_{q,p} \cdot \boldsymbol{u}_H = \sum_i \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i}\right) = 0. \tag{6.22}$$

Thus the red-dot fluid is locally incompressible. This is the elementary coordinate form of Liouville's theorem.

Question from the audience. Is Hamiltonian mechanics merely the trick of replacing each second-order equation by two first-order equations?

Response. No. Doubling the variables is always possible, but an arbitrary doubled system need not have the crossed-gradient form in Eq. (6.20). The canonical form, with one Hamiltonian generating both equations, is the stronger statement.

Question from the audience. What happens if the relation between velocity and momentum cannot be inverted?

Response. For the regular systems being discussed, the ordinary Legendre transform then fails and $H(q, p)$ cannot be constructed by this recipe. Such a model may be pathological, but singular Lagrangians also arise in legitimate constrained and gauge systems. They require additional constraint equations rather than the naive transform.

Question from the audience. Does friction, including terminal velocity, violate the Hamiltonian picture?

Response. A friction law for a reduced set of variables is generally not Hamiltonian: phase-space volumes contract and mechanical information is discarded. At the microscopic level, however, the object and its environment may form a larger Hamiltonian system. Energy and information then flow into unobserved molecular, acoustic, or radiative degrees of freedom.

Question from the audience. Where, exactly, did the Legendre transformation enter the mechanics?

Response. It is the passage from $L(q,v)$ to $H(q,p) = \sum_i p_i v_i - L$, with $p_i = \partial L/\partial v_i$ and the velocities eliminated in favor of the momenta. The geometric area picture explains the reciprocal derivatives; the cancellation in Eq. (6.17) is what turns that reciprocity into Hamilton's equations.

Question from the audience. Can a Legendre transform fail because the velocity becomes singular or several velocities correspond to one momentum?

Response. Yes. Global one-to-one behavior can fail even if a local branch exists, and the Hessian can become singular at special points. Before using the ordinary Hamiltonian construction one must specify a region where the momentum–velocity relation is regular, or adopt the constrained formalism appropriate to the system.

Question from the audience. Do infinitely many degrees of freedom, as in a field, automatically produce friction?

Response. No. A classical field can have infinitely many canonical pairs and still evolve Hamiltonianly. Apparent damping arises when energy is carried away into modes that are omitted from the reduced description. An accelerating charge, for example, may lose mechanical energy to radiation, while the combined particle-plus-field system retains the full accounting.

6.5 The Ordinary Particle

For one particle in a potential $U(x)$,

$$L(x,\dot{x}) = \frac{1}{2}m\dot{x}^2 - U(x). \tag{6.23}$$

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \dot{x} = \frac{p}{m}. \tag{6.24}$$

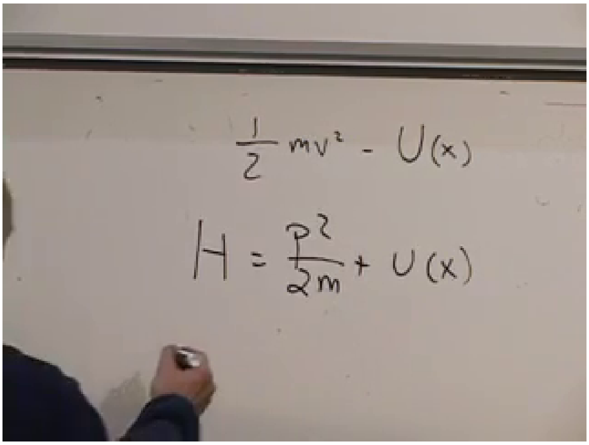

Figure 6.6: The ordinary-particle Hamiltonian at 01:12:35. The velocity has been eliminated, leaving $H = p^2/(2m)+U(x)$.

Its Legendre transform is

$$\begin{aligned} H &= p\dot{x} - L \\ &= p\frac{p}{m} - \left(\frac{1}{2}m\frac{p^2}{m^2} - U(x)\right) \\ &= \boxed{\frac{p^2}{2m} + U(x)}. \end{aligned} \tag{6.25}$$

For this familiar Lagrangian, the Hamiltonian is the total energy $T + U$.

The first Hamilton equation gives

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}, \tag{6.26}$$

which is the momentum–velocity relation. The second gives

$$\dot{p} = -\frac{\partial H}{\partial x} = -\frac{dU}{dx}. \tag{6.27}$$

Since $p = m\dot{x}$, this is Newton's equation

$$m\ddot{x} = -\frac{dU}{dx}. \tag{6.28}$$

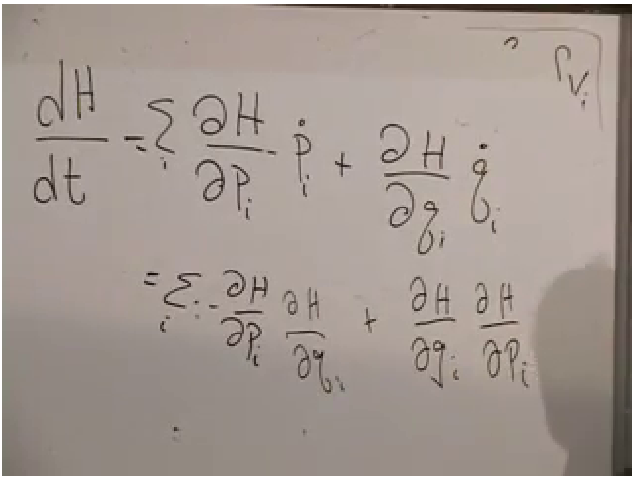

Figure 6.7: Energy conservation at 01:16:10. Substituting Hamilton's equations into dH/dt makes the two products cancel term by term.

The reformulation has changed the organization of the equations, not their physical predictions.

6.6 Energy and the Geometry of Its Contours

Suppose $H(q,p)$ has no explicit time dependence. Along a trajectory,

$$\begin{aligned}\frac{dH}{dt} &= \sum_i \left(\frac{\partial H}{\partial q_i}\dot{q}_i + \frac{\partial H}{\partial p_i}\dot{p}_i \right) \\ &= \sum_i \left(\frac{\partial H}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i}\frac{\partial H}{\partial q_i} \right) = 0. \end{aligned} \tag{6.29}$$

The cancellation is pairwise, so

$$\boxed{H = \text{constant along the motion.}} \tag{6.30}$$

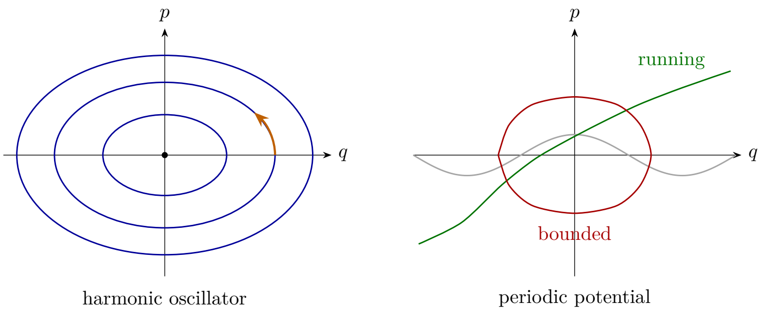

Figure 6.8: Typeset phase portraits. Small energy in a stable well produces a closed orbit. Above a periodic potential barrier, the coordinate can keep running and the constant-energy curve is open.

Question from the audience. Is the converse true? Does energy conservation by itself imply Hamiltonian dynamics?

Response. No. A time-independent Hamiltonian evolving by Hamilton's equations is conserved, but a dynamical rule may preserve some function called energy without possessing canonical Hamiltonian structure. Energy conservation is one consequence, not a complete characterization, of Hamiltonian flow.

For one degree of freedom, constant values of $H(q, p)$ are curves in the q-p plane. The phase point must move along the curve on which it starts. For a harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 q^2, \tag{6.31}$$

the curves are ellipses, or circles after rescaling the axes. The origin is a fixed point. Every nonzero bounded orbit closes around it.

Closed and open curves can coexist in one phase space. A pendulum below the separatrix oscillates and traces a closed curve. Above the barrier it rotates continuously, producing an open branch when the angle is unwrapped. The same distinction appears for a particle moving through a periodic washboard potential.

Question from the audience. If the phase-space fluid is incompressible, how can neighboring energy contours bunch together and spread apart?

Response. Think of an incompressible fluid passing through a narrow channel. Where the channel narrows, the fluid moves faster; where it widens, the fluid moves more slowly. Closely spaced contours mean that the gradient of H is large, and Hamilton's equations therefore give a large phase-space speed. The density need not change.

Question from the audience. May closed and open constant-energy curves occur in the same system?

Response. Yes. A periodic potential provides the simplest example. Below a barrier, the particle is trapped in one well and the orbit is closed. Above it, the particle passes from well to well and the corresponding branch is open. The pendulum has the same oscillating and rotating regimes.

6.6.1 Momentum Need Not Be Mass Times Velocity

The relation $p = m\dot{q}$ belongs to the ordinary quadratic kinetic energy; it is not the definition of canonical momentum in general. Hamilton's first equation always says

$$\dot{q} = \frac{\partial H}{\partial p}. \tag{6.32}$$

For the illustrative Hamiltonian written in the lecture,

$$H = p^3 + q^3, \tag{6.33}$$

one finds

$$\dot{q} = 3p^2, \qquad \dot{p} = -3q^2. \tag{6.34}$$

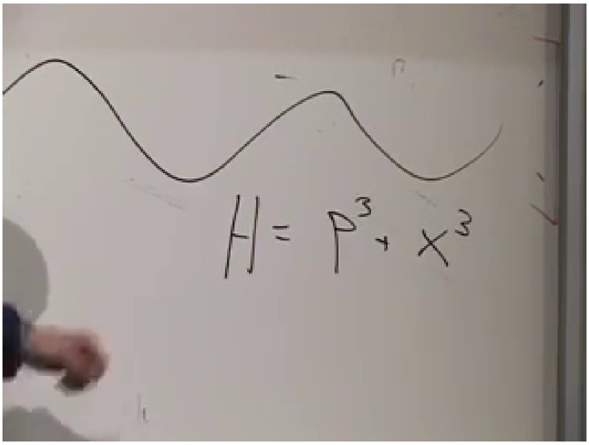

Figure 6.9: A deliberately nonstandard Hamiltonian at 01:25:50. It shows directly that the canonical momentum need not be proportional to velocity.

Question from the audience. Is canonical momentum always a constant multiple of velocity?

Response. No. It is defined by $p_i = \partial L/\partial \dot{q}_i$, or equivalently related to velocity by $\dot{q}_i = \partial H/\partial p_i$. That relation may be nonlinear and may depend on the coordinates. The cubic example makes the nonlinearity obvious, although its momentum–velocity map is not globally one-to-one and therefore does not come from one globally regular ordinary Lagrangian.

Question from the audience. If microscopic Hamiltonian motion conserves energy, does that make every machine a perpetual-motion machine?

Response. No. Total energy conservation does not mean that all energy remains available as organized mechanical work. Energy can spread among enormous numbers of environmental degrees of freedom and become heat. Understanding that irreversibility and the distinction between total and usable energy requires statistical mechanics, entropy, and often chaotic dynamics.

6.7 Poisson Brackets and General Conservation Laws

Energy is not the only quantity that may be conserved. Let $A(q,p)$ be any function on phase space. Angular momentum in a plane is one example:

$$L_z = xp_y - yp_x. \tag{6.35}$$

For an observable with no explicit time dependence, the chain rule and Hamilton's equations give

$$\begin{aligned} \frac{dA}{dt} &= \sum_i \left(\frac{\partial A}{\partial q_i}\dot{q}_i + \frac{\partial A}{\partial p_i}\dot{p}_i \right) \\ &= \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial H}{\partial q_i} \right). \end{aligned} \tag{6.36}$$

This combination is the Poisson bracket. With the convention used here,

$$\boxed{\{A,B\} = \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i} \right).} \tag{6.37}$$

Consequently,

$$\boxed{\dot{A} = \{A,H\}.} \tag{6.38}$$

The Hamiltonian is the generator of time evolution.

Taking $A = q_i$ or $A = p_i$ recovers Hamilton's equations:

$$\{q_i,H\} = \frac{\partial H}{\partial p_i} = \dot{q}_i, \qquad \{p_i,H\} = -\frac{\partial H}{\partial q_i} = \dot{p}_i. \tag{6.39}$$

The fundamental brackets are

$$\{q_i,q_j\} = 0, \qquad \{p_i,p_j\} = 0, \qquad \{q_i,p_j\} = \delta_{ij}. \tag{6.40}$$

A quantity $A(q,p)$ is conserved precisely when

$$\boxed{\{A,H\} = 0.} \tag{6.41}$$

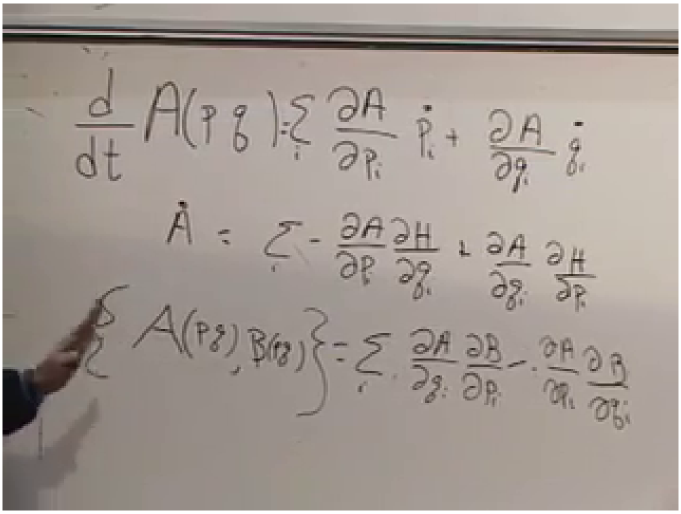

Figure 6.10: The chain rule and Poisson-bracket definition at 01:35:20. The bracket packages the crossed derivatives that already appeared in Hamilton's equations.

For a rotationally invariant Hamiltonian, direct differentiation verifies $\{L_z, H\} = 0$. Later we will connect this calculation to symmetry; for now the bracket gives the mathematical test.

Question from the audience. How could Poisson have discovered such a combination of derivatives?

Response. Historical reconstruction is uncertain, but the combination is not arbitrary once Hamilton's equations are written. Apply the chain rule to an arbitrary $A(q, p)$, substitute the two canonical equations, and the bracket appears. Its importance is then confirmed by the algebraic properties and conservation laws it organizes.

6.8 Explicit Time Dependence

So far time entered only implicitly through the moving variables $q(t)$ and $p(t)$. An external apparatus can also make the Hamiltonian depend on time explicitly. A potential $U(q, t)$,

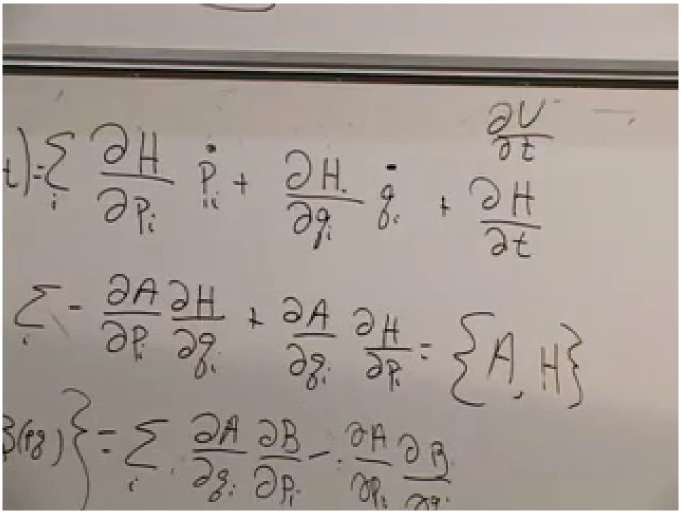

Figure 6.11: Explicit and implicit time dependence separated at 01:45:55. Hamilton's equations cancel the terms involving $\dot{q}_i$ and $\dot{p}_i$, leaving $dH/dt = \partial H/\partial t$.

for example, may be produced by a source whose prescribed position changes while the source itself is not included among the dynamical variables.

For a general observable $A(q, p, t)$,

$$\frac{dA}{dt} = \{A, H\} + \frac{\partial A}{\partial t}. \tag{6.42}$$

Setting $A = H$ gives

$$\frac{dH}{dt} = \{H, H\} + \frac{\partial H}{\partial t} = \boxed{\frac{\partial H}{\partial t}}, \tag{6.43}$$

because every function has zero Poisson bracket with itself. Thus a time-independent Hamiltonian is conserved, while explicit time dependence measures the energy supplied to or removed from the chosen subsystem.

The same conclusion follows directly from the time-dependent

Legendre transform:

$$dH = \sum_i v_i \, dp_i - \sum_i \dot{p}_i \, dq_i - \frac{\partial L}{\partial t} \, dt, \tag{6.44}$$

so

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \tag{6.45}$$

when the canonical variables are held fixed on their respective sides of the transform.

Question from the audience. What is the difference between implicit and explicit time dependence?

Response. The value $H(q(t), p(t))$ changes its arguments as the system moves; that is implicit dependence. Explicit dependence remains even when q and p are held fixed, as in $U(q, t)$ produced by a moving external source. The canonical terms cancel in dH/dt; only the explicit partial derivative remains.

Question from the audience. Why do some books reverse the order or sign of the Poisson bracket? Which slot represents momentum?

Response. The names A and B label whole phase-space functions; neither argument slot is the momentum slot. The canonical variables q_i, p_i occur inside the definition. Since $\{A, B\} = -\{B, A\}$, reversing the order reverses the sign. Either convention is consistent if Hamilton's evolution equation is adjusted with it. Here we use $\dot{A} = \{A, H\}$.

Question from the audience. Is it the Poisson bracket that appears everywhere outside mechanics?

Response. The broad remark at this point concerns the Legendre transformation, which recurs throughout thermodynamics and other variational theories. Poisson brackets are fundamental in Hamiltonian mechanics and lead directly toward quantum commutators, but they are a different piece of the construction.

We began with a reversible map of abstract states and ended with a precise continuous law. The Legendre transformation changes variables from $(q, \dot{q})$ to (q, p); Hamilton's equations generate an incompressible phase-space flow; constant H confines that flow to energy surfaces; and the Poisson bracket tells how every observable changes. These are not separate tricks. They are four views of the same canonical structure.

CHAPTER 7
LIOUVILLE'S THEOREM AND MOTION IN A MAGNETIC FIELD

Liouville's theorem is one of the central facts of Hamiltonian mechanics. It says that Hamiltonian evolution preserves volume in phase space. In quantum mechanics the corresponding idea is unitarity: distinguishable states do not quietly collapse into one another under the fundamental law of motion. In both theories the underlying issue is the conservation of information.

We shall first make this geometric statement precise and prove it directly from Hamilton's equations. We shall then use it to understand canonical area, chaotic stretching, coarse-graining, and entropy. In the second half we turn to a different problem, the motion of a charged particle in a magnetic field. That example introduces a velocity-dependent force while preserving the Lagrangian and Hamiltonian structures developed so far.

7.1 A Flow That Does Not Lose Information

A state of a classical system is a point in phase space. If we observe the system at equally spaced instants, the law of motion draws an arrow from each point to its successor. Determinism requires one outgoing arrow: the present state must determine what happens next. Reversibility also requires one incoming arrow: the present state must determine where it came from.

This already rules out two kinds of evolution. A trajectory cannot split into two futures, and two distinct trajectories cannot merge into a single state. Either event would destroy the one-to-one character of the microscopic law. For a finite set of states, a reversible update rule is therefore a permutation, decomposing the state space into disjoint cycles.

Hamiltonian mechanics makes a stronger statement. Fill a continuous phase space uniformly with imaginary dust. Each dust grain follows the local equations of motion. A small cloud of grains may stretch, bend, and shear, but its phase-space volume remains unchanged. The flow is incompressible.

Question from the audience. Could a distribution of states converge or diverge even when each individual trajectory is deterministic?[a]

Response. For a general deterministic rule that possibility must be examined separately. Uniqueness of the future only forbids splitting. Liouville's theorem is stronger: for a Hamiltonian system, an entire region cannot be compressed into a smaller phase-space volume or expanded into a larger one.

[a]Lecture timestamp: 00:05:05–00:06:15.

There are several equivalent ways to picture the theorem. We may follow a small volume surrounding each point; we may follow an arbitrarily shaped region and compare its initial and final volumes; or we may begin with uniform dust and demand that its density remain uniform. These are local, regional, and density versions of the same incompressibility statement.

Volume preservation does not mean rigidity. A square can shear into a parallelogram, or a nearly round cloud can become a long, thin filament. The amount of phase space is fixed even though its shape and every ordinary distance inside it may change.

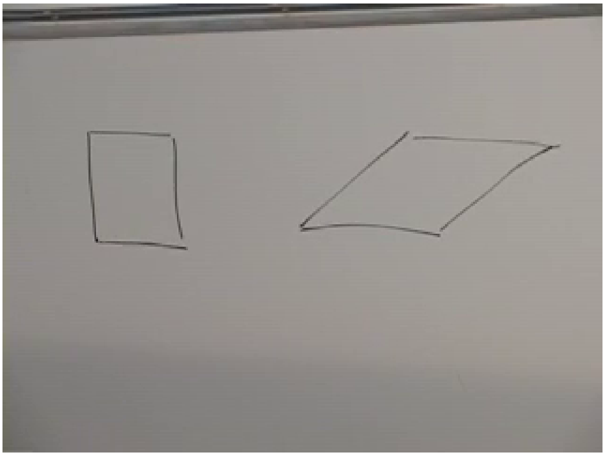

Figure 7.1: A square and its sheared image at 00:13:16. Hamiltonian flow need not preserve lengths or angles; the relevant invariant is phase-space area.

Question from the audience. If the initial region is connected, can the flow tear it apart or create a hole?[a]

Response. Not under a smooth Hamiltonian flow for which the solution exists uniquely. The time-evolution map and its inverse are continuous, so connectedness and topology are preserved. The region may become extremely distorted, but it is not cut and reattached.

[a]Lecture timestamp: 00:10:08–00:11:11.

Question from the audience. Can two points pass one another, and can the orientation of a little square change?[a]

Response. Their projections onto one coordinate axis may pass. What cannot happen is that two complete phase-space trajectories intersect at the same time and state. A little square may rotate or shear into a parallelogram; those changes are compatible with preserving its area.

[a]Lecture timestamp: 00:12:20–00:13:53.

Question from the audience. Does a chaotic double pendulum contradict this picture? Nearby initial conditions can separate rapidly.[a]

Response. No. Chaos concerns the growth of separations, not the growth of exact phase-space volume. Expansion along unstable directions is compensated by compression along other directions. A compact patch can become a complicated filament while retaining precisely the same volume.

[a]Lecture timestamp: 00:13:54–00:16:12.

For one canonical pair (q, p), volume means area. For N pairs it means the $2N$-dimensional volume element

$$d\Gamma = \prod_{i=1}^{N} dq_i \, dp_i. \tag{7.1}$$

Two points are close only when their complete states are close: their coordinates and their conjugate momenta must both be nearly equal.

Question from the audience. What exactly is the volume being preserved, and what does "nearby" mean?[a]

Response. For a single degree of freedom it is an area in the q-p plane; with more degrees of freedom it is the product of all the canonical coordinate and momentum intervals. Nearby states are close in this full space, not merely close in ordinary position.

[a]Lecture timestamp: 00:17:03–00:18:42.

Ordinary friction seems to violate the theorem. Many sliding states approach the same visible state of rest, and many throws of a die can settle on the same face. The apparent contraction arises because the description omits the table, the air, and the internal degrees of freedom. Mechanical energy becomes heat, sound, and microscopic motion. Information is transferred into those unobserved variables. A reduced phase space may contract even though the larger closed system remains Hamiltonian.

Question from the audience. When a die or a sliding block comes to rest, have many states not genuinely merged?[a]

Response. They have merged only in a coarse description that records the visible object and ignores its environment. Include the molecular state of the table, the air, and the emitted sound, and the final microscopic states remain distinct. The situation is analogous to energy: the block loses kinetic energy while the table warms by a correspondingly tiny amount.

[a] Lecture timestamp: 00:18:52–00:22:18.

7.2 Incompressible Flow and the Divergence

Before proving the theorem in phase space, consider an ordinary fluid of points. In one dimension, let $v_x(x)$ be the instantaneous velocity. A small interval keeps a uniform density only if the flux entering one end equals the flux leaving the other. Thus

$$\frac{\partial v_x}{\partial x} = 0. \tag{7.2}$$

In one dimension an incompressible instantaneous flow is locally a rigid translation.

In two dimensions take a small rectangle with sides Δx and Δy. The difference between outward fluxes through the right and left faces is, to first order,

$$\frac{\partial v_x}{\partial x} \Delta x \, \Delta y. \tag{7.3}$$

The corresponding difference through the upper and lower faces is

$$\frac{\partial v_y}{\partial y} \Delta x \, \Delta y. \tag{7.4}$$

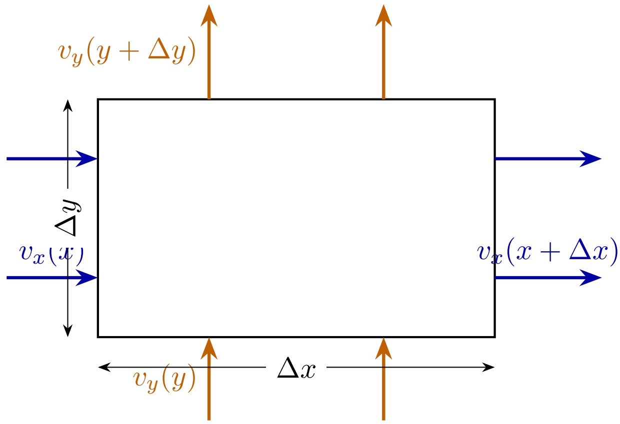

Figure 7.2: Instantaneous flux through a small rectangle. Incompressibility requires the sum of the two first-order outward-flux differences to vanish.

After dividing by the area, incompressibility becomes

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0, \qquad \text{or} \qquad \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0. \tag{7.5}$$

This does not require the velocity to be the same everywhere. An increase of the x-velocity along x can be balanced by a decrease of the y-velocity along y. The rectangle may stretch in one direction and contract in the other.

Question from the audience. A point entering through one side may turn and later leave through another. Does the flux argument overlook that motion? [a]

Response. The argument is instantaneous. Take a snapshot and ask for the present rate of change of the number of points inside the rectangle. A later turn belongs to a later snapshot. Summing the flux through all faces at the same instant gives the local density change.

[a]Lecture timestamp: 00:37:09–00:37:49.

The same reasoning works in any number of dimensions:

$$\sum_{a=1}^{D} \frac{\partial v_a}{\partial x_a} = 0. \tag{7.6}$$

For a nonuniform density ρ, the general conservation law is the continuity equation

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{v}) = 0. \tag{7.7}$$

Uniform dust reduces this to zero divergence.

Question from the audience. Does the argument depend on two dimensions or on a uniform density?[a]

Response. The divergence formula extends to any number of dimensions. Uniform density is the convenient case used here. For a varying density one conserves the flux $\rho\boldsymbol{v}$, as in Eq. (7.7).

[a] Lecture timestamp: 00:37:50–00:40:07.

7.3 Proof of Liouville's Theorem

Return now to a phase space with canonical coordinates $(q_1, \ldots, q_N, p_1, \ldots, p_N)$. Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{7.8}$$

They define a phase-space velocity field

$$\boldsymbol{u}_H = (\dot{q}_1, \ldots, \dot{q}_N, \dot{p}_1, \ldots, \dot{p}_N). \tag{7.9}$$

Its divergence is

$$\begin{aligned} \boldsymbol{\nabla}_{q,p} \cdot \boldsymbol{u}_H &= \sum_{i=1}^{N} \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) \\ &= \sum_{i=1}^{N} \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0. \end{aligned} \tag{7.10}$$

Each pair of mixed derivatives cancels. This local identity is the whole proof.

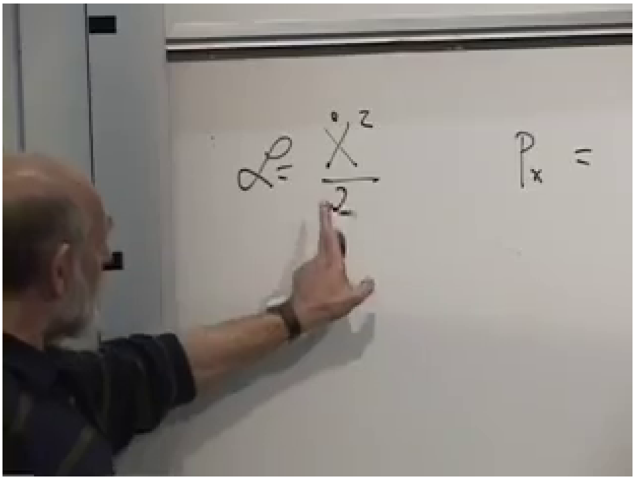

Figure 7.3: The mixed-derivative cancellation at 00:43:55. The q_i and p_i contributions enter with opposite signs.

Theorem 7.1 (Liouville's theorem)**.** *For a smooth Hamiltonian system, the time evolution preserves the canonical phase-space volume element* $d\Gamma = \prod_i dq_i\, dp_i$.

Proof. Equation (7.10) says that the Hamiltonian velocity field has zero divergence. The continuity equation then transports an initially uniform density without changing it. Equivalently, the Jacobian of the time-evolution map has unit determinant, so every transported region keeps its phase-space volume. □

This is the promised strengthening of reversibility. A one-to-one flow need not preserve volume in an arbitrary set of coordinates, but a Hamiltonian flow in canonical coordinates does so because of the crossed derivatives and the minus sign in Hamilton's equations.

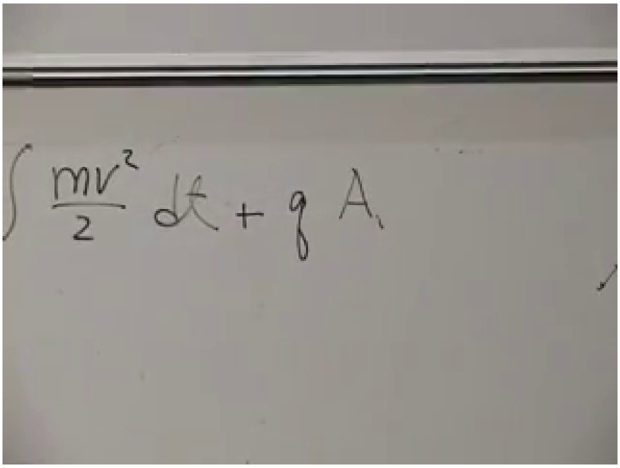

Figure 7.4: The reciprocal scaling of coordinate and canonical momentum at 00:49:45: $y = \alpha x$ while $p_y = p_x/\alpha$.

7.4 Canonical Area, Not Distance

For one degree of freedom, Liouville volume is area in the q-p plane. A simple change of coordinate shows why neither factor is separately invariant. Take a free particle of unit mass,

$$L = \frac{\dot{x}^2}{2}, \qquad p_x = \frac{\partial L}{\partial \dot{x}} = \dot{x}. \tag{7.11}$$

Introduce a rescaled coordinate

$$y = \alpha x, \qquad x = \frac{y}{\alpha}, \qquad \dot{x} = \frac{\dot{y}}{\alpha}. \tag{7.12}$$

Then

$$L = \frac{\dot{y}^2}{2\alpha^2}, \qquad p_y = \frac{\partial L}{\partial \dot{y}} = \frac{\dot{y}}{\alpha^2} = \frac{p_x}{\alpha}. \tag{7.13}$$

The coordinate interval stretches while the conjugate momentum interval shrinks:

$$\Delta y \, \Delta p_y = (\alpha \Delta x)\left(\frac{\Delta p_x}{\alpha}\right) = \Delta x \, \Delta p_x. \tag{7.14}$$

This transformation is canonical. It changes the appearance of a cell but not its symplectic area.

Question from the audience. Is α a time-evolution parameter, and does every coordinate transformation preserve area?[a]

Response. Here α is only a fixed change of scale, such as a change of units. The momentum must be transformed as the variable canonically conjugate to the new coordinate. An arbitrary independent reshuffling of q and p need not preserve area; a canonical transformation does.

[a]Lecture timestamp: 00:54:24–00:55:14.

Hamiltonian time evolution is itself a canonical transformation. It can move and deform a phase-space cell while preserving the canonical area element. That statement is entirely compatible with energy conversion. A falling ball trades potential energy for kinetic energy, so its position and momentum both change while its total energy and the area of an ensemble are preserved.

Question from the audience. Does a falling ball violate conservation because its momentum grows?[a]

Response. No. Potential energy decreases while kinetic energy increases. Liouville's theorem concerns the area occupied by a collection of nearby states, not the constancy of either coordinate or momentum for one state.

[a]Lecture timestamp: 00:53:27–00:54:23.

The same area already foreshadows quantum mechanics. Planck's constant has the dimensions of position times momentum, and the uncertainty principle sets a minimum scale for a quantum phase-space cell. The classical theorem does not impose that minimum; it supplies the canonical geometry in which the quantum statement will live.

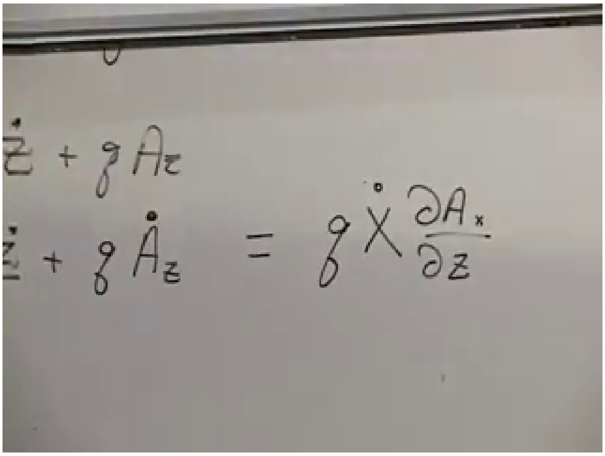

Figure 7.5: A compact patch and its filamented image at 01:03:40. The drawing illustrates deformation under chaotic evolution, not an increase of exact phase-space volume.

Question from the audience. Why is Planck's constant associated with phase-space area rather than with one axis?[a]

Response. Because its dimensions are action, qp, and the uncertainty relation constrains the product of conjugate uncertainties. A rescaling can enlarge q while shrinking p, but their canonical product is the invariant quantity.

[a] Lecture timestamp: 00:52:26–00:52:55.

7.5 Chaos, Probability, and Coarse-Graining

Liouville's theorem does not say that a phase-space patch remains simple. In a chaotic system the patch may stretch exponentially, fold, and return across the same macroscopic region in thinner and thinner filaments. Exact volume is preserved throughout this process.

Phase-space area also supplies a natural invariant measure. Suppose a chaotic trajectory explores an accessible component of an energy surface and define probability by the fraction of a long observation time spent in a region. When the motion is ergodic with respect to the Liouville measure, relative occupation probabilities are proportional to invariant phase-space volume:

$$\frac{P(\Omega_1)}{P(\Omega_2)} = \frac{\mu_L(\Omega_1)}{\mu_L(\Omega_2)}. \tag{7.15}$$

Chaos by itself does not prove ergodicity, but it explains why this statistical description can become useful even when the microscopic equations are fully deterministic.

Question from the audience. Before quantum mechanics, what physical meaning does the phase-space area have?[a]

Response. It supplies the invariant measure used in statistical mechanics. Under the appropriate ergodic assumption, equal Liouville volumes carry equal long-time statistical weight, so relative areas become relative occupation probabilities.

[a]Lecture timestamp: 00:57:50–00:59:43.

Question from the audience. Is this probability stochastic, or is chaotic mechanics still deterministic?[a]

Response. The microscopic equations remain deterministic. The statistical description enters because finite precision cannot track the ever-finer distinctions among nearby initial conditions. Probability records long-time occupation or incomplete knowledge, not an indeterministic force inserted into the equations.

[a]Lecture timestamp: 00:59:44–01:02:14.

To see the role of finite precision, partition phase space into cells of the smallest size an apparatus can distinguish. These cells are fixed by the measurement convention; they are not carried along and distorted with the Hamiltonian flow. An initially compact patch may fit into a small set of cells. After it has become finely filamented, it intersects many more fixed cells.

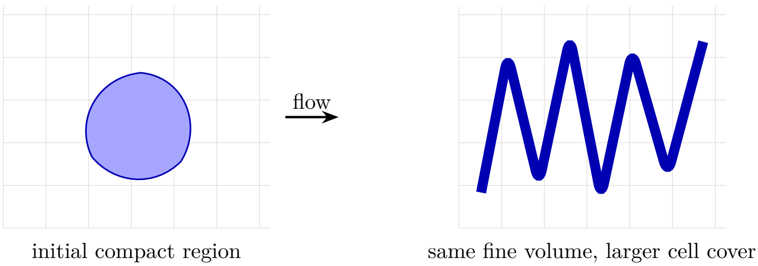

Figure 7.6: Fine-grained volume and coarse-grained coverage. Stretching and folding preserve exact volume, while finite-resolution cells can report a larger occupied region.

Question from the audience. Are the later coarse-graining cells the images of the initial cells under time evolution?[a]

Response. No. The coarse cells represent what the apparatus can distinguish at each time. If they were transported with the exact flow, they would stretch into the same fine filaments and no coarse growth would appear. The physically relevant comparison uses the same finite resolution before and after evolution.

[a] Lecture timestamp: 01:06:18–01:07:24.

The exact set still has its original volume. With infinitely precise calculation and measurement, no expansion would be found. With fixed finite resolution, however, the union of all cells intersecting the filaments can be much larger. Looking with "fuzzy eyes" fills in separations too fine to resolve.

Question from the audience. In what sense, then, can the phase-space area increase? [a]

Response. Only the coarse-grained area increases. The fine-grained set generated by Hamilton's equations retains exactly its original volume. The larger value counts every resolution cell touched by the thin filaments, including portions of those cells that the exact set does not occupy.

[a] Lecture timestamp: 01:07:39–01:11:26.

This distinction points toward the second law of thermodynamics. Entropy measures the logarithm of the detectable phase-space volume compatible with our macroscopic knowledge. With a reference volume Ω_0, one may write

$$S = k_B \log\left(\frac{\Omega_{\mathrm{cg}}}{\Omega_0}\right). \tag{7.16}$$

Microscopic information has not been destroyed. It has been hidden in correlations and filaments finer than the available resolution.

7.6 A Velocity-Dependent Fundamental Force

We now turn to the motion of a particle of mass m and charge q in a prescribed magnetic field. Its force is

$$\boxed{\boldsymbol{F} = q\,\boldsymbol{v} \times \boldsymbol{B}.} \tag{7.17}$$

Unlike the forces considered earlier, this one depends on velocity. Friction also depends on velocity, but friction is an effective dissipative force in a reduced description. The magnetic force is fundamental within the present approximation and follows from an action principle.

The cross product has components

$$\begin{aligned}
(\boldsymbol{v} \times \boldsymbol{B})_x &= v_y B_z - v_z B_y,\\
(\boldsymbol{v} \times \boldsymbol{B})_y &= v_z B_x - v_x B_z,\\
(\boldsymbol{v} \times \boldsymbol{B})_z &= v_x B_y - v_y B_x.
\end{aligned} \tag{7.18}$$

To construct a Lagrangian we introduce the vector potential $\boldsymbol{A}(\boldsymbol{x})$, defined by

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{7.19}$$

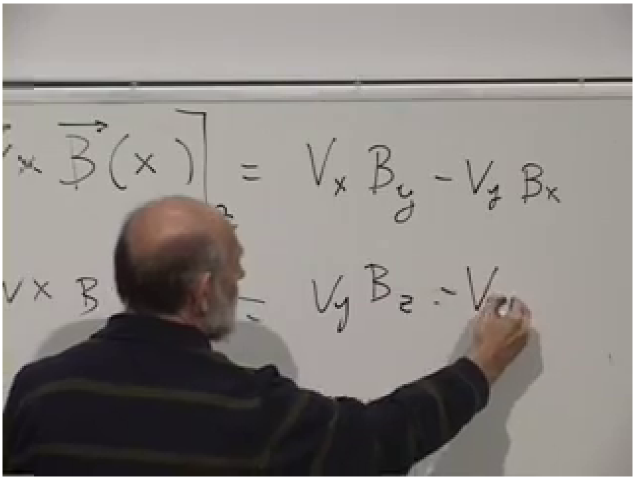

Figure 7.7: Components of $\boldsymbol{v} \times \boldsymbol{B}$ at 01:18:40. The upper line isolates the z-component used in the derivation below.

In components,

$$
\begin{aligned}
B_x &= \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}, \\
B_y &= \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}, \\
B_z &= \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}.
\end{aligned} \tag{7.20}
$$

Writing a field as a curl automatically gives $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$, the absence of magnetic monopole sources in the classical theory being used here.

The z-component of the magnetic force can now be expressed directly in terms of $\boldsymbol{A}$:

$$
\begin{aligned}
F_z &= q(v_x B_y - v_y B_x) \\
&= q v_x \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + q v_y \left(\frac{\partial A_y}{\partial z} - \frac{\partial A_z}{\partial y} \right).
\end{aligned} \tag{7.21}
$$

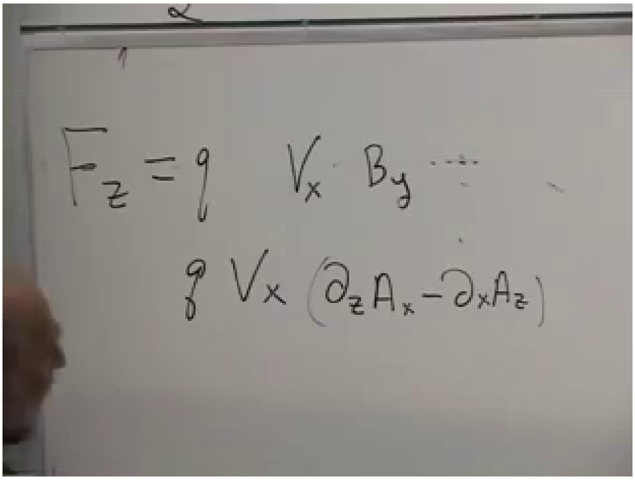

Figure 7.8: The z-component of the magnetic force rewritten with derivatives of the vector potential at 01:24:40.

Question from the audience. Why introduce $\boldsymbol{A}$ when Newton's equation can be written directly with $\boldsymbol{B}$?[a]

Response. The magnetic field is sufficient for the force law, but we want an action, canonical momenta, and a Hamiltonian. The vector potential is the quantity that couples locally to the velocity in the Lagrangian. Its curl then makes the equations of motion depend on $\boldsymbol{B}$.

[a] Lecture timestamp: 01:19:06–01:21:15.

7.7 The Magnetic Action and Canonical Momentum

The free-particle action is supplemented by a line integral along the particle's path:

$$S = \int \frac{1}{2} m v^2 \, dt + q \int_\gamma \boldsymbol{A} \cdot d\boldsymbol{x}. \tag{7.22}$$

Since $d\boldsymbol{x} = \dot{\boldsymbol{x}} \, dt$, the action takes the ordinary time-integral

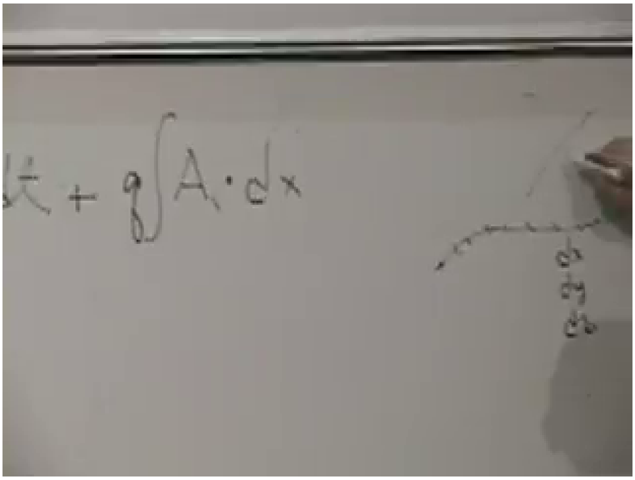

Figure 7.9: The action as a kinetic term plus a line integral of the vector potential at 01:28:50. The small path sketch emphasizes that the second term depends on the route through space.

form

$$S = \int L\,dt, \qquad L = \frac{1}{2}m\dot{\boldsymbol{x}}^2 + q\,\boldsymbol{A}(\boldsymbol{x})\cdot\dot{\boldsymbol{x}}. \tag{7.23}$$

We use units in which the coupling is written $q\boldsymbol{A}\cdot\boldsymbol{v}$. Other electromagnetic unit conventions may place a factor of $1/c$ in this term.

Question from the audience. How can the two terms in the action have the same units? [a]

Response. The action has units of energy times time. In the chosen convention $q\boldsymbol{A}$ has units of momentum, so $q\boldsymbol{A}\cdot d\boldsymbol{x}$ has units of action. A different convention may move a factor of c into the definition of the coupling, without changing the mechanics.

[a] Lecture timestamp: 01:29:51–01:31:17.

The canonical momentum is no longer simply the mechanical

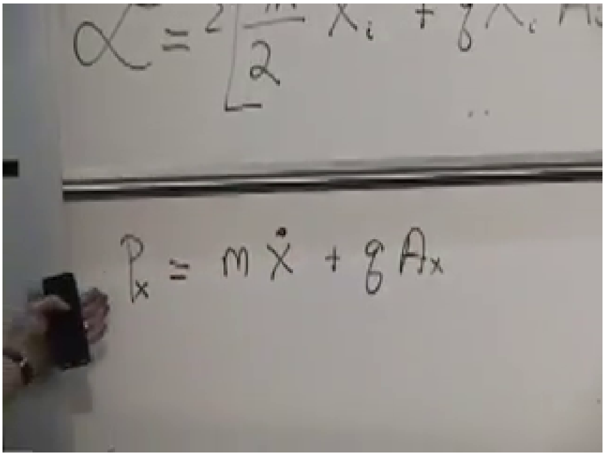

Figure 7.10: Canonical momentum in the x-direction at 01:35:20: $p_x = m\dot{x} + qA_x$.

momentum:

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m\dot{x}_i + qA_i. \tag{7.24}$$

Thus

$$\boldsymbol{p}_{\text{mech}} = m\boldsymbol{v} = \boldsymbol{p} - q\boldsymbol{A}. \tag{7.25}$$

This distinction is essential. Hamilton's equations use the canonical momentum $\boldsymbol{p}$; kinetic energy and the Lorentz force are most naturally expressed through the mechanical combination $\boldsymbol{p} - q\boldsymbol{A}$.

7.8 Recovering the Lorentz Force

We must verify that the proposed Lagrangian gives the correct equation of motion. Work out the z-component explicitly. The Euler–Lagrange equation is

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{z}} = \frac{\partial L}{\partial z}. \tag{7.26}$$

The left-hand side is

$$\frac{d}{dt}(m\dot{z} + qA_z) = m\ddot{z} + q\frac{dA_z}{dt}. \tag{7.27}$$

The right-hand side receives contributions from all three components of the vector potential:

$$\frac{\partial L}{\partial z} = q\dot{x}\frac{\partial A_x}{\partial z} + q\dot{y}\frac{\partial A_y}{\partial z} + q\dot{z}\frac{\partial A_z}{\partial z}. \tag{7.28}$$

The field is assumed static, meaning it has no explicit time dependence. It may still vary from point to point. Because the particle moves through that spatially varying field,

$$\frac{dA_z}{dt} = \dot{x}\frac{\partial A_z}{\partial x} + \dot{y}\frac{\partial A_z}{\partial y} + \dot{z}\frac{\partial A_z}{\partial z}. \tag{7.29}$$

Question from the audience. If the magnetic field is static, why is dA_z/dt not zero?[a]

Response. Static means no explicit dependence on time, not uniform in space. The particle samples different values of $\boldsymbol{A}(\boldsymbol{x})$ as it moves, so the value of A_z along the trajectory changes by the spatial chain rule.

[a]Lecture timestamp: 01:39:13–01:39:53.

Substitute Eq. (7.29) into the left-hand side of Eq. (7.26). The terms $q\dot{z}\,\partial A_z/\partial z$ cancel, leaving

$$\begin{aligned} m\ddot{z} &= q\dot{x}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + q\dot{y}\left(\frac{\partial A_y}{\partial z} - \frac{\partial A_z}{\partial y}\right) \\ &= q(v_x B_y - v_y B_x) = q(\boldsymbol{v}\times\boldsymbol{B})_z. \end{aligned} \tag{7.30}$$

The other components follow by cyclic permutation, and therefore

$$m\ddot{\boldsymbol{x}} = q\,\dot{\boldsymbol{x}}\times\boldsymbol{B}. \tag{7.31}$$

Question from the audience. Is the vector potential uniquely determined by the magnetic field?[a]

Response. No. For a static gauge transformation $\boldsymbol{A}\mapsto\boldsymbol{A}+\boldsymbol{\nabla}\chi$, the curl is unchanged. The Lagrangian changes by the total derivative $q\,d\chi/dt$, which changes the action only by endpoint terms and leaves the equations of motion unchanged.

[a]Lecture timestamp: 01:44:34–01:45:10.

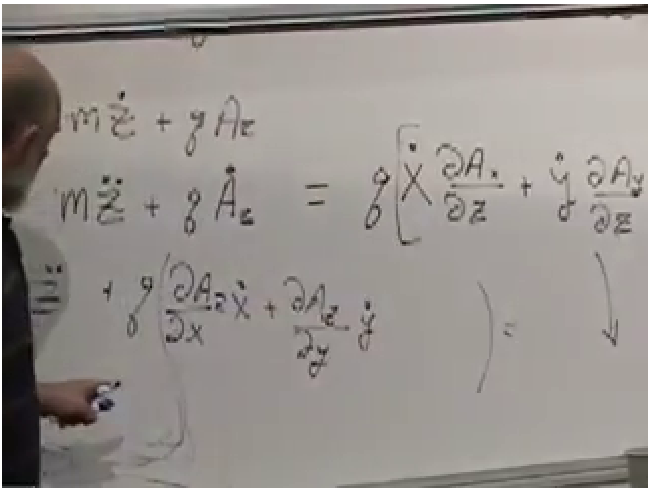

Figure 7.11: The two sides of the z-component Euler–Lagrange equation at 01:41:50. Terms proportional to $\dot{z}\,\partial A_z/\partial z$ occur on both sides and cancel.

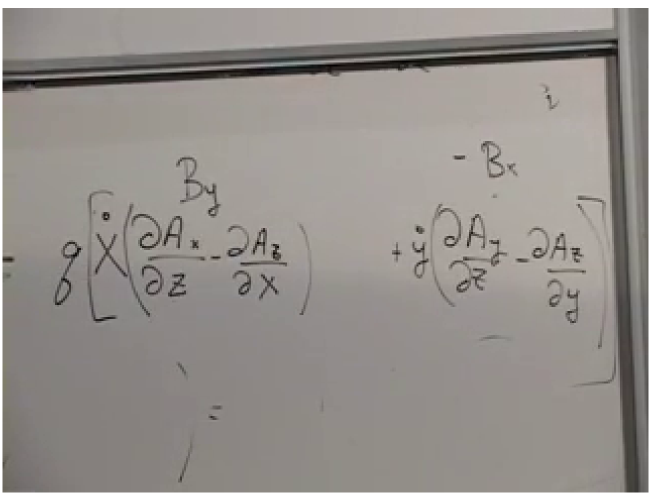

Figure 7.12: The surviving derivative combinations at 01:44:20, identified as B_y and $-B_x$, complete the z-component of the Lorentz force.

The contrast with friction is now sharp. A frictional force has a component opposite the velocity and changes the speed. A magnetic force is perpendicular to both $\boldsymbol{v}$ and $\boldsymbol{B}$:

$$\boldsymbol{F} \cdot \boldsymbol{v} = q(\boldsymbol{v} \times \boldsymbol{B}) \cdot \boldsymbol{v} = 0. \tag{7.32}$$

It changes the direction of motion without changing the speed.

Question from the audience. Why is a velocity-dependent magnetic force so different from friction?[a]

Response. The decisive issue is direction. Friction acts along the velocity axis and therefore changes kinetic energy. The magnetic force is perpendicular to the velocity, so its instantaneous power vanishes. It bends the trajectory but does not speed the particle up or slow it down.

[a] Lecture timestamp: 01:45:15–01:46:28.

7.9 Energy and the Hamiltonian

Assume that $\boldsymbol{A}$ has no explicit time dependence. The Hamiltonian is

$$H = \sum_i p_i \dot{x}_i - L. \tag{7.33}$$

First express it in terms of velocities:

$$\begin{aligned} H &= \sum_i (m\dot{x}_i + qA_i)\dot{x}_i - \left(\frac{1}{2} m \sum_i \dot{x}_i^2 + q \sum_i A_i \dot{x}_i \right) \\ &= \frac{1}{2} m \sum_i \dot{x}_i^2 = \frac{1}{2} m v^2. \end{aligned} \tag{7.34}$$

The two copies of the magnetic coupling cancel.

This agrees with the force argument: a static magnetic field does no work. In a uniform field perpendicular to the velocity, the particle moves in a circle at constant speed. Its direction continually changes, so it has an acceleration, but its kinetic energy remains constant.

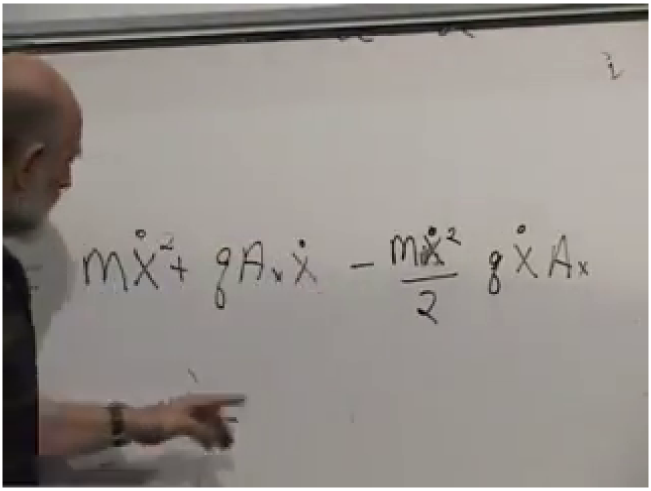

Figure 7.13: The x-component of the Legendre transform at 01:54:20. The $qA_x\dot{x}$ terms cancel, leaving $m\dot{x}^2/2$.

To use Hamilton's equations we must eliminate the velocities. From Eq. (7.24),

$$\dot{x}_i = \frac{p_i - qA_i}{m}. \tag{7.35}$$

The Hamiltonian in canonical variables is therefore

$$\boxed{H(\boldsymbol{x}, \boldsymbol{p}) = \frac{1}{2m}\big(\boldsymbol{p} - q\boldsymbol{A}(\boldsymbol{x})\big)^2.} \tag{7.36}$$

It looks more complicated than the kinetic energy because the canonical momentum contains the vector potential. The combination $\boldsymbol{p} - q\boldsymbol{A} = m\boldsymbol{v}$ is the mechanical momentum.

Question from the audience. An accelerated charge radiates. How can its kinetic energy remain constant?[a]

Response. Radiation is being neglected in this particle-in-an-external-field model. A complete treatment must include the electromagnetic field as a dynamical Hamiltonian system carrying its own energy. Radiation reaction is a higher-order correction to the approximation used here.

[a]Lecture timestamp: 01:57:29–01:58:29.

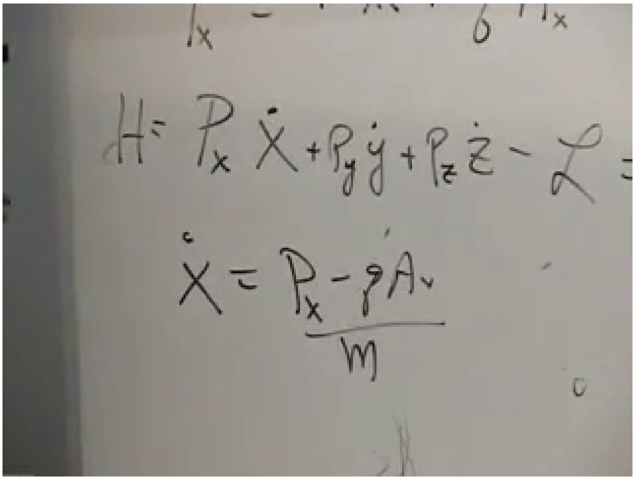

Figure 7.14: The Hamiltonian definition and the inversion $\dot{x} = (p_x - qA_x)/m$ at 01:57:00.

As a final check, Hamilton's equations give

$$\dot{x}_i = \frac{\partial H}{\partial p_i} = \frac{p_i - qA_i}{m} = v_i, \tag{7.37}$$

and

$$\dot{p}_i = -\frac{\partial H}{\partial x_i} = qv_j \frac{\partial A_j}{\partial x_i}. \tag{7.38}$$

Differentiating the mechanical momentum gives

$$\begin{aligned} m\dot{v}_i &= \dot{p}_i - q\frac{dA_i}{dt} \\ &= qv_j \left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \right) = q(\boldsymbol{v} \times \boldsymbol{B})_i. \end{aligned} \tag{7.39}$$

Thus Newtonian, Lagrangian, and Hamiltonian descriptions all produce the same magnetic motion. The vector potential changes the canonical bookkeeping, but the observable force depends only on its curl.

CHAPTER 8

GAUGE INVARIANCE, CYCLOTRON MOTION, AND POISSON BRACKETS

A charged particle in an electromagnetic field is a small system with an unusually rich canonical structure. Newton's equation is simple, but the momentum that enters Hamiltonian mechanics is not merely $m\boldsymbol{v}$. The action requires a vector potential that is not unique. A uniform magnetic field turns translation invariance into a statement about the center of a cyclotron orbit, and an added electric field produces a steady drift perpendicular to the applied force.

We shall develop those facts in the order in which they arise. The final part of the chapter changes language once more, from Hamilton's equations to Poisson brackets. That algebraic formulation is more abstract than Newton's law, but it is the form that most clearly anticipates quantum mechanics.

8.1 Four Forms of the Same Mechanics

For the present discussion the electromagnetic field is prescribed. It may vary from place to place, but it does not depend explicitly on time, and we do not yet solve equations for the field itself. The only dynamical object is a particle of mass m and charge q.

It is useful to place four familiar descriptions of mechanics side by side:

1. Newton's equations of motion;
2. the principle of stationary action;
3. the Euler–Lagrange equations;
4. Hamilton's equations in phase space.

They are equivalent when each is available, but they expose different features of a theory. An equation such as $F = ma$ tells us how a trajectory moves. It does not, by its form alone, tell us why energy or momentum is conserved. The action is deeper, but one normally uses it by taking a variation and obtaining differential equations. The Lagrangian is powerful and direct. The Hamiltonian is often the least intuitive, yet it displays the flow of whole ensembles through phase space, makes information conservation visible, and leads naturally toward quantum mechanics.

The charged particle will make both the power and the strangeness of these descriptions concrete. We restrict the motion to the x-y plane. The magnetic field is perpendicular to that plane, while an electric field may lie within it.

8.2 The Lorentz Force and the Potentials

The nonrelativistic equation of motion is

$$m\ddot{\boldsymbol{x}} = q\boldsymbol{E} + q\,\dot{\boldsymbol{x}} \times \boldsymbol{B}. \tag{8.1}$$

We use an electromagnetic convention in which no explicit factor of c appears. Other conventions insert $1/c$ in the magnetic term; that change does not alter the argument.

The magnetic force depends on velocity, but it is perpendicular to velocity:

$$\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B}) = 0. \tag{8.2}$$

Consequently

$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = \boldsymbol{v} \cdot q(\boldsymbol{v} \times \boldsymbol{B}) = 0. \tag{8.3}$$

A magnetic field bends a trajectory without changing its speed. This is the precise content of the statement that magnetic fields do no work. It also explains why planar motion remains planar: with $\boldsymbol{v}$ in the plane and $\boldsymbol{B}$ perpendicular to it, $\boldsymbol{v} \times \boldsymbol{B}$ lies in the plane.

To pass from the force law to an action, we need potentials. For a region in which the magnetic field has no monopole source,

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0, \tag{8.4}$$

we may write, at least locally,

$$\boxed{\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}.} \tag{8.5}$$

The local qualification matters in topologically nontrivial settings and in theories with magnetic monopoles. For the ordinary fields considered here, one vector potential is sufficient.

Question from the audience. Why does writing $\boldsymbol{B}$ as a curl imply that magnetic charge is absent?[a]

Response. The divergence of every curl vanishes: $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0$. Electric field lines may begin or end on electric charge because $\boldsymbol{\nabla} \cdot \boldsymbol{E}$ is proportional to charge density. In the classical electromagnetic theory being used here, there is no corresponding magnetic charge density. Magnetic monopoles are a possible extension of the theory, but none is part of this problem.

[a] Lecture timestamp: 00:11:34–00:13:18.

The electrostatic field is represented by a scalar potential,

$$\boldsymbol{E} = -\boldsymbol{\nabla} V, \qquad U = qV. \tag{8.6}$$

Thus V is energy per unit charge, while U is the potential energy of this particular particle. Doubling q doubles both the electric force and the potential energy in a fixed external field.

Question from the audience. What does the symbol V measure? Is it already the particle's potential energy?[a]

Response. V is the electric potential, measured in volts, or energy per unit charge. The particle's potential energy is $U = qV$. This distinction keeps the prescribed field separate from the charge used to probe it.

[a] Lecture timestamp: 00:15:43–00:16:31.

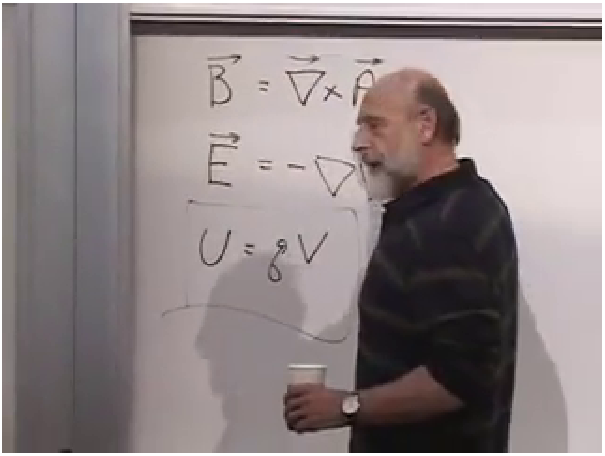

Figure 8.1: The vector potential, the electric potential, and the force law at 00:17:19. The action requires $\boldsymbol{A}$, although the Newtonian equation can be written entirely in terms of $\boldsymbol{B}$.

8.3 The Electromagnetic Action and Gauge Redundancy

For planar motion, the action is

$$S = \int \frac{1}{2} m(\dot{x}^2 + \dot{y}^2)\, dt + q \int_\gamma A_i\, dx_i - q \int V\, dt. \tag{8.7}$$

The middle term is a line integral along the path γ. Divide and multiply its infinitesimal displacement by dt:

$$A_i\, dx_i = A_i \dot{x}_i\, dt = \boldsymbol{A} \cdot \boldsymbol{v}\, dt. \tag{8.8}$$

The action therefore has the ordinary form $S = \int L\, dt$, with

$$\boxed{L = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2) + q(A_x \dot{x} + A_y \dot{y}) - qV(x, y).} \tag{8.9}$$

The combination $A_i\, dx_i - V\, dt$ hints at a four-dimensional electromagnetic potential. We do not need special relativity

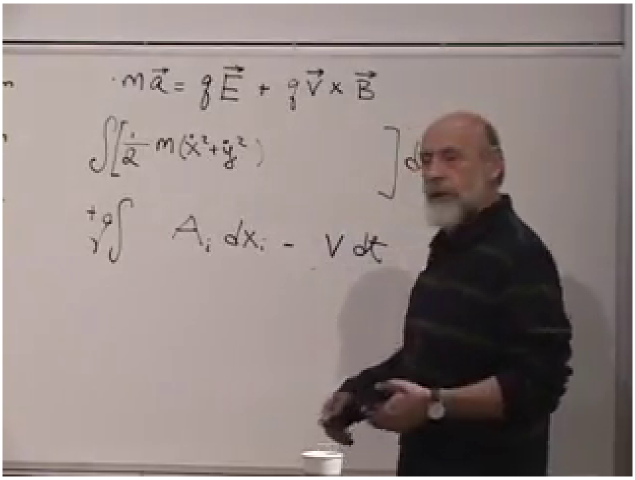

Figure 8.2: The electromagnetic action at 00:22:37. The magnetic coupling is a line integral $q\int \boldsymbol{A}\cdot d\boldsymbol{x}$, which becomes $q\boldsymbol{A}\cdot\boldsymbol{v}$ in the Lagrangian.

here, but the formula is already arranging the scalar and vector potentials into the form they will take in space-time.

Question from the audience. Is the repeated index in $A_i\,dx_i$ a sum, and is the same true of the scalar-potential term?[a]

Response. Yes for the vector term: $A_i\,dx_i = A_x\,dx + A_y\,dy$ in the plane. The quantity V is a scalar, so there is no component sum in $V\,dt$. To avoid confusing electric potential V with velocity, we shall write the latter as $\boldsymbol{v}$.

[a] Lecture timestamp: 00:23:36–00:24:49.

The action has introduced more description than the force law seemed to need. The reason is that $\boldsymbol{A}$ is not unique. For any smooth scalar function $\lambda(\boldsymbol{x})$,

$$\boldsymbol{A} \longrightarrow \boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\lambda \tag{8.10}$$

leaves the magnetic field unchanged, since mixed partial derivatives commute:

$$\nabla \times \boldsymbol{A}' = \nabla \times \boldsymbol{A} + \nabla \times \nabla\lambda = \boldsymbol{B}. \tag{8.11}$$

In the time-independent setting, the Lagrangian changes by a total derivative,

$$L' = L + q\nabla\lambda \cdot \dot{\boldsymbol{x}} = L + q\frac{d\lambda}{dt}. \tag{8.12}$$

Hence

$$S' = S + q\lambda(\boldsymbol{x}_f) - q\lambda(\boldsymbol{x}_i). \tag{8.13}$$

The endpoint contribution does not affect a variation with fixed endpoints. The Lagrangian changes, but the trajectory does not. This is gauge invariance: different mathematical descriptions represent the same physical field.

The redundancy is not a defect that can simply be discarded. We could stay with the Lorentz-force equation and never mention $\boldsymbol{A}$, but the local action, canonical momentum, and Hamiltonian all require it. Modern physics often reaches its most useful formulations by introducing such redundant variables and then identifying the transformations that leave observables unchanged.

8.4 Recovering the Lorentz Force

We should not accept the Lagrangian merely because it has the right-looking symbols. Let us derive the x-component of the equation of motion. First,

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + qA_x. \tag{8.14}$$

This is our first warning that canonical momentum and mechanical momentum are different:

$$\boldsymbol{p}_{\text{can}} = m\boldsymbol{v} + q\boldsymbol{A}, \qquad \boldsymbol{p}_{\text{mech}} = m\boldsymbol{v}. \tag{8.15}$$

The canonical momentum changes when $\boldsymbol{A}$ changes gauge. It is a phase space coordinate adapted to the chosen Lagrangian, not by itself a gauge-invariant observable.

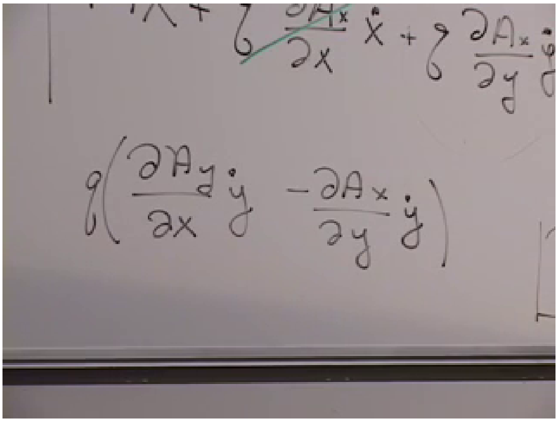

Figure 8.3: The surviving curl combination at 00:38:20. The separate derivatives of $\boldsymbol{A}$ combine into the gauge-invariant field B_z.

The Euler–Lagrange equation gives

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}. \tag{8.16}$$

Although A_x has no explicit time dependence, its value at the moving particle changes. The chain rule gives

$$\frac{dA_x}{dt} = \frac{\partial A_x}{\partial x}\dot{x} + \frac{\partial A_x}{\partial y}\dot{y}. \tag{8.17}$$

Therefore

$$m\ddot{x} + q\frac{\partial A_x}{\partial x}\dot{x} + q\frac{\partial A_x}{\partial y}\dot{y} = q\frac{\partial A_x}{\partial x}\dot{x} + q\frac{\partial A_y}{\partial x}\dot{y} - q\frac{\partial V}{\partial x}. \tag{8.18}$$

The common terms cancel, leaving

$$\begin{aligned} m\ddot{x} &= q\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\dot{y} - q\frac{\partial V}{\partial x} \\ &= qB_z\dot{y} + qE_x. \end{aligned} \tag{8.19}$$

The vector potential has disappeared from the final equation except through its curl. Thus the Lagrangian uses a redundant variable, while its Euler–Lagrange equation returns the gauge-invariant force law.

8.5 A Uniform Magnetic Field

We now use the canonical formalism rather than merely checking it. Fix ordinary right-handed axes with $+\hat{\boldsymbol{z}}$ pointing out of the board, and write the magnetic field into the board as

$$\boldsymbol{B} = -B\hat{\boldsymbol{z}}, \qquad B > 0. \tag{8.20}$$

Then $B_z = -B$. Two convenient vector potentials for this same field are

$$\text{gauge I:} \qquad A_x = By, \qquad\qquad A_y = 0, \tag{8.21}$$

$$\text{gauge II:} \qquad A_x = 0, \qquad\qquad A_y = -Bx. \tag{8.22}$$

Indeed, each gives $\partial_x A_y - \partial_y A_x = -B$. A symmetric choice,

$$A_x = \frac{B}{2}y, \qquad A_y = -\frac{B}{2}x, \tag{8.23}$$

does the same. No experiment can distinguish these gauges.

Set the electric field to zero and begin with gauge I. The Lagrangian is

$$L_1 = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + qBy\dot{x}, \tag{8.24}$$

and the canonical momenta are

$$p_x = m\dot{x} + qBy, \qquad p_y = m\dot{y}. \tag{8.25}$$

Question from the audience. Which of these two canonical momenta is conserved? [a]

Response. The Lagrangian depends on y but not on x. A translation $x \mapsto x + \epsilon$ changes no term appearing in L_1, so x is cyclic and p_x is conserved. A translation in y changes the term $qBy\dot{x}$, so $p_y = m\dot{y}$ is not the conserved canonical quantity in this gauge.

[a] Lecture timestamp: 00:46:18–00:48:38.

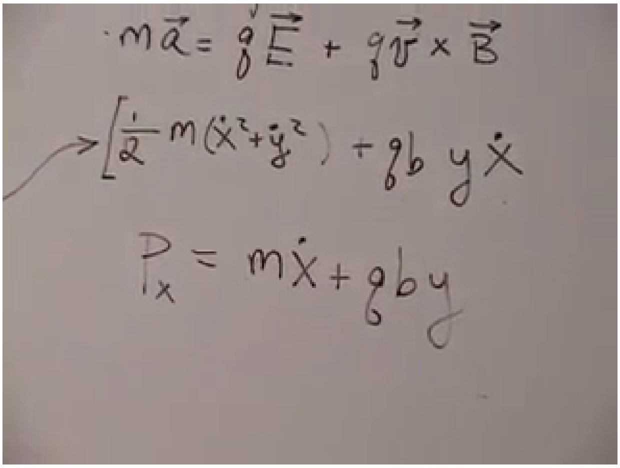

Figure 8.4: Gauge I at 00:46:20. The Lagrangian contains $qBy\dot{x}$, so the canonical momentum is $p_x = m\dot{x} + qBy$.

Choose the conserved value $p_x = 0$ for the moment. Then

$$\boxed{\dot{x} = -\frac{qB}{m}y.} \tag{8.26}$$

Now use gauge II. Its Lagrangian is

$$L_2 = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - qBx\dot{y}, \tag{8.27}$$

and y is cyclic. Thus

$$p_y = m\dot{y} - qBx \tag{8.28}$$

is conserved. Choosing $p_y = 0$ gives

$$\boxed{\dot{y} = \frac{qB}{m}x.} \tag{8.29}$$

It may seem peculiar to use one gauge to expose the x-relation and another to expose the y-relation. What is being combined is not two incompatible descriptions at one instant, but two gauge-independent statements about the same physical orbit. Each gauge makes a different translation symmetry manifest.

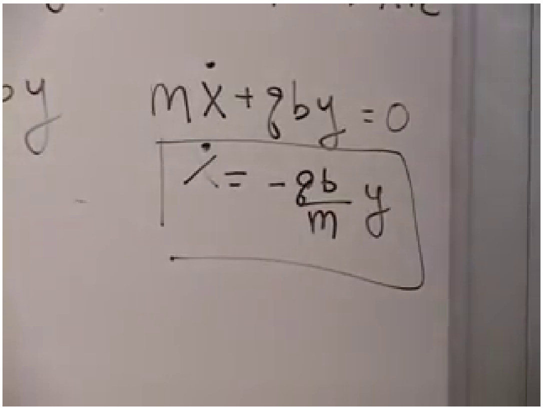

Figure 8.5: The first velocity relation at 00:50:10, obtained from the conserved canonical momentum in gauge I.

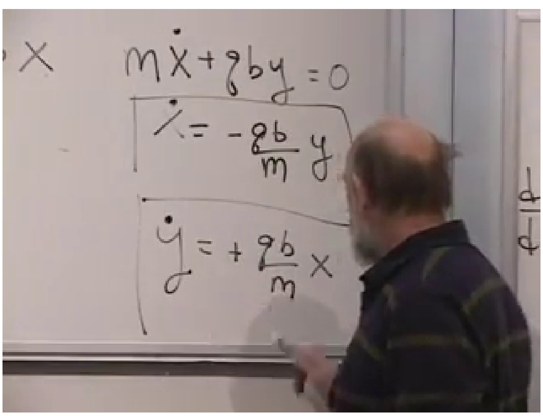

Figure 8.6: The two velocity relations together at 00:52:00. Their form already identifies uniform circular motion.

8.5.1 Cyclotron motion and the charge-to-mass ratio

The pair of first-order equations is solved by

$$x = r\cos(\omega_c t), \qquad y = r\sin(\omega_c t), \tag{8.30}$$

because

$$\dot{x} = -\omega_c y, \qquad \dot{y} = \omega_c x. \tag{8.31}$$

Comparison with Eqs. (8.26) and (8.29) gives the signed cyclotron frequency

$$\boxed{\omega_c = \frac{qB}{m}}, \qquad |\omega_c| = \frac{|q|B}{m}. \tag{8.32}$$

The sign fixes the direction of rotation; the magnitude fixes the number of radians swept out per unit time. A larger charge bends the trajectory more rapidly, while a larger mass bends it more slowly.

Historically, magnetic deflection experiments naturally measured q/m. The trajectory alone did not separately reveal the electron's charge and mass; another kind of measurement was needed to disentangle them.

Question from the audience. Does the sign of the cyclotron frequency matter? [a]

Response. Its magnitude is $|q|B/m$. The sign records orientation. Reversing the charge or the magnetic field reverses the direction in which the orbit is traversed, without changing its radius or speed.

[a] Lecture timestamp: 00:54:19–00:55:24.

8.5.2 What the conserved momenta measure

A uniform field does not privilege the origin, so the circle may be centered at any fixed point (x_0, y_0):

$$x = x_0 + r\cos(\omega_c t), \qquad y = y_0 + r\sin(\omega_c t). \tag{8.33}$$

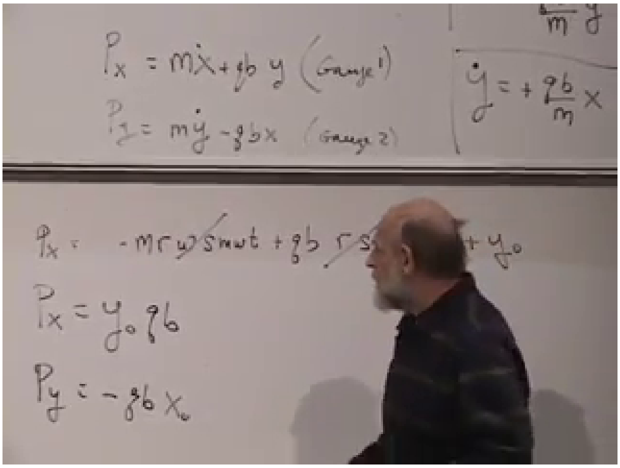

Figure 8.7: The shifted circular orbit and its conserved canonical momenta at 01:01:20. The board gives $p_x = qBy_0$ and $p_y = -qBx_0$.

Substitute this orbit into the two conserved canonical momenta. In gauge I,

$$\begin{aligned} p_x &= m\dot{x} + qBy \\ &= -m\omega_c r\sin(\omega_c t) + qB\big[y_0 + r\sin(\omega_c t)\big] \\ &= qBy_0. \end{aligned} \tag{8.34}$$

In gauge II,

$$\begin{aligned} p_y &= m\dot{y} - qBx \\ &= m\omega_c r\cos(\omega_c t) - qB\big[x_0 + r\cos(\omega_c t)\big] \\ &= -qBx_0. \end{aligned} \tag{8.35}$$

The oscillating terms cancel because $m\omega_c = qB$. The strange conserved momenta have a simple physical meaning: they encode the fixed guiding center of the orbit.

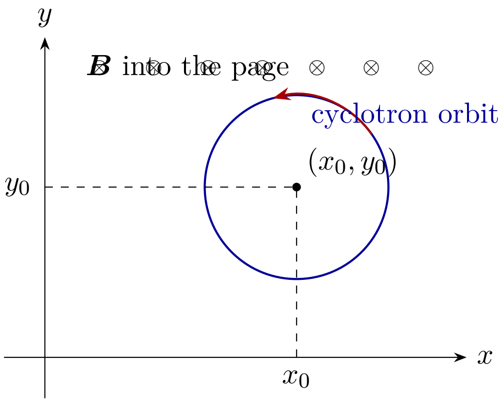

Figure 8.8: Reconstruction of the geometry used at 00:57:52–01:01:35. The canonical constants identify the orbit center rather than the instantaneous mechanical momentum.

Question from the audience. Are the dimensions of qBy_0 really those of momentum? [a]

Response. Yes, provided one uses one electromagnetic unit convention consistently. In the convention of the present Lagrangian, $q\boldsymbol{A}$ has the dimensions of momentum. Since a uniform-field potential scales as $A \sim B \times$ length, the product qBy_0 has the same dimensions as $m\dot{x}$.

[a] Lecture timestamp: 01:02:19–01:02:45.

8.6 Electric Drift and the Hall Response

Restore a constant electric field in the $+x$-direction. Choose

$$V(x) = -Ex, \qquad \boldsymbol{E} = E\hat{\boldsymbol{x}}. \tag{8.36}$$

With the field convention in Eq. (8.20), the component equations are

$$m\ddot{x} + qB\dot{y} = qE, \tag{8.37}$$

$$m\ddot{y} - qB\dot{x} = 0. \tag{8.38}$$

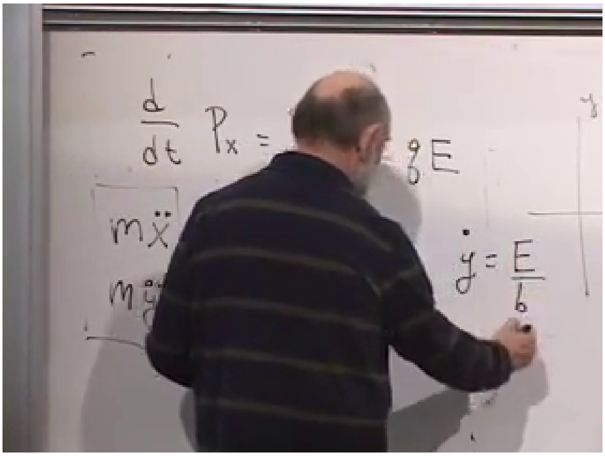

Figure 8.9: The no-acceleration solution at 01:08:20. The electric force points along x, while the steady drift has $\dot{y} = E/B$.

The electric potential depends on x, so x-translation invariance is lost and p_x is no longer conserved:

$$\dot{p}_x = \frac{\partial L}{\partial x} = qE. \tag{8.39}$$

Rather than solve the most general orbit, look first for a trajectory with no acceleration. Setting $\ddot{x} = \ddot{y} = 0$ in Eqs. (8.37)–(8.38) gives

$$\dot{x} = 0, \qquad \boxed{\dot{y} = \frac{E}{B}.} \tag{8.40}$$

The coordinate-free form is

$$\boxed{\boldsymbol{v}_D = \frac{\boldsymbol{E} \times \boldsymbol{B}}{B^2}.} \tag{8.41}$$

For $\boldsymbol{E} = E\hat{\boldsymbol{x}}$ and $\boldsymbol{B} = -B\hat{\boldsymbol{z}}$, this vector points along $+\hat{\boldsymbol{y}}$. The charge cancels: particles of different charge magnitude share the same $\boldsymbol{E} \times \boldsymbol{B}$ drift. The full crossed-field orbit is a cyclotron motion whose guiding center moves with $\boldsymbol{v}_D$.

With only a magnetic field, the no-acceleration solution is the zero-radius orbit: the particle simply sits still. Turning on $\boldsymbol{E}$ changes that solution into steady transverse motion. A distribution of charges therefore develops a sideways flow. This is the elementary mechanical structure behind the Hall response.

Question from the audience. Why does a force applied along x produce motion along y?[a]

Response. A trial velocity along y generates a magnetic force along x. At the special speed E/B, that magnetic force exactly cancels the electric force. The net force and acceleration vanish, so the transverse velocity persists.

[a]Lecture timestamp: 01:07:07–01:10:45.

Question from the audience. Is this related to Hall-effect devices used with motors or disk drives?[a]

Response. Hall sensors do exploit transverse electromagnetic response and are commonly used to sense magnetic fields, position, or rotor commutation. The detailed engineering of a motor is beyond this mechanics problem, so we should not infer a particular mechanism without examining the device.

[a]Lecture timestamp: 01:10:47–01:11:38.

The same kind of perpendicular response appears in gyroscopic precession. A torque that one might naively expect to tip a spinning body instead produces motion around a transverse direction. The physical systems differ, but their coupled first-order equations have a closely related structure.

8.6.1 Completing the Hamiltonian exercise

The charged-particle Hamiltonian follows directly from the exercise posed in the lecture. From

$$\boldsymbol{p} = m\boldsymbol{v} + q\boldsymbol{A} \tag{8.42}$$

we have

$$\boldsymbol{v} = \frac{\boldsymbol{p} - q\boldsymbol{A}}{m}. \tag{8.43}$$

The Legendre transform gives

$$\begin{aligned} H &= \boldsymbol{p} \cdot \boldsymbol{v} - L \\ &= \frac{(\boldsymbol{p} - q\boldsymbol{A})^2}{2m} + qV. \end{aligned} \tag{8.44}$$

Thus kinetic energy depends on the gauge-invariant mechanical combination $\boldsymbol{p} - q\boldsymbol{A} = m\boldsymbol{v}$, even though Hamilton's equations use the canonical coordinate $\boldsymbol{p}$. Applying those equations reproduces the Lorentz force, as it must.

8.7 Poisson Brackets

We now leave the charged-particle example and abstract the structure of Hamiltonian mechanics. For canonical coordinates (q_i, p_i), Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{8.45}$$

There are twice as many first-order equations as there are configuration coordinates, in place of the same number of second-order Lagrange equations.

Let $A(q_i, p_i)$ be any function on phase space with no explicit time dependence. Along a trajectory,

$$\begin{aligned} \dot{A} &= \sum_i \left(\frac{\partial A}{\partial q_i}\dot{q}_i + \frac{\partial A}{\partial p_i}\dot{p}_i \right) \\ &= \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial H}{\partial q_i} \right). \end{aligned} \tag{8.46}$$

This combination motivates a definition. For any two phase-space functions A and B, their Poisson bracket is

$$\boxed{\{A, B\} = \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i} \right).} \tag{8.47}$$

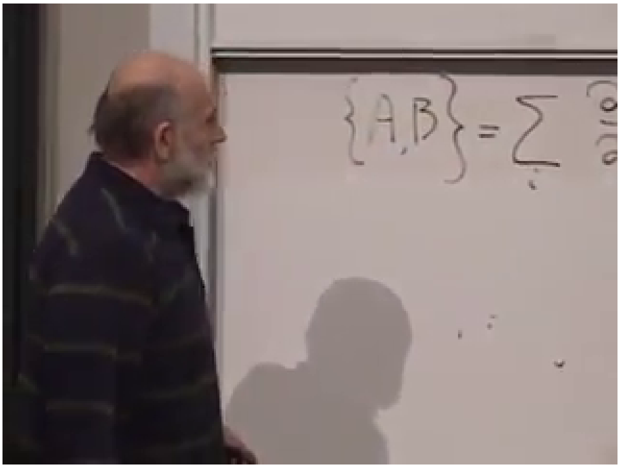

Figure 8.10: The Poisson-bracket definition at 01:20:40. The relative minus sign is the source of its antisymmetry.

Equation (8.46) is now simply

$$\boxed{\dot{A} = \{A, H\}.} \tag{8.48}$$

If A has explicit time dependence, the complete statement is

$$\frac{dA}{dt} = \frac{\partial A}{\partial t} + \{A, H\}. \tag{8.49}$$

The bracket formula contains both Hamilton equations. With one canonical pair (Q, P),

$$\dot{Q} = \{Q, H\} = \frac{\partial H}{\partial P}, \tag{8.50}$$

$$\dot{P} = \{P, H\} = -\frac{\partial H}{\partial Q}. \tag{8.51}$$

So Poisson brackets do not introduce new dynamics. They compress the old dynamics into an algebraic operation on functions.

Question from the audience. What happens if A is the constant function 1, or if it is simply P or Q?[a]

Response. A constant has zero derivatives, so $\{1, H\} = 0$. Choosing $A = P$ gives $\dot{P} = -\partial H/\partial Q$; choosing $A = Q$ gives $\dot{Q} = \partial H/\partial P$. The simplest choices recover the two Hamilton equations directly.

[a]Lecture timestamp: 01:22:28–01:25:34.

8.7.1 The fundamental algebra

Interchanging the two entries changes the sign:

$$\{A, B\} = -\{B, A\}. \tag{8.52}$$

In particular, any function has zero bracket with itself. For the canonical variables,

$$\boxed{\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij}.} \tag{8.53}$$

Consequently $\{p_i, q_j\} = -\delta_{ij}$: reversing the entries reverses the sign.

Question from the audience. Can one see immediately that $\{P, P\} = 0$, without expanding the definition?[a]

Response. Yes. Antisymmetry says $\{P, P\} = -\{P, P\}$. The only quantity equal to its own negative is zero. The same argument gives $\{Q, Q\} = 0$.

[a]Lecture timestamp: 01:27:06–01:28:05.

For one canonical pair, bracketing with P or Q acts as differentiation:

$$\{P, F\} = -\frac{\partial F}{\partial Q}, \qquad \{Q, F\} = \frac{\partial F}{\partial P}. \tag{8.54}$$

More generally,

$$\{p_i, F\} = -\frac{\partial F}{\partial q_i}, \qquad \{q_i, F\} = \frac{\partial F}{\partial p_i}. \tag{8.55}$$

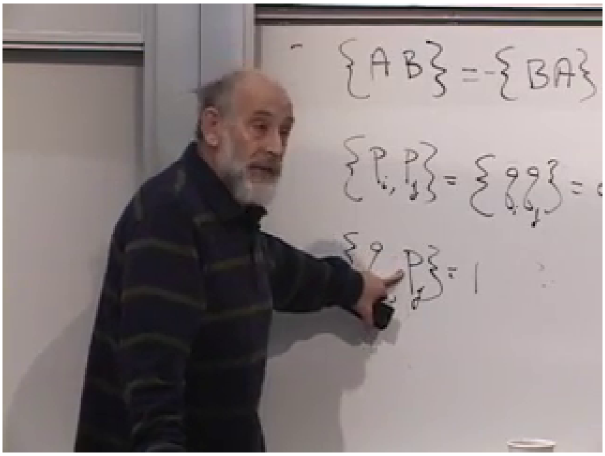

Figure 8.11: Antisymmetry and the fundamental canonical brackets at 01:29:40. The Kronecker delta is one for matching indices and zero otherwise.

This is the symplectic asymmetry of classical mechanics: exchanging a coordinate with its momentum brings in a minus sign.

The bracket is linear in each entry. For constants α, β,

$$\{\alpha A + \beta C, B\} = \alpha\{A, B\} + \beta\{C, B\}. \tag{8.56}$$

It also obeys the ordinary product rule in either entry. Expanding the derivatives of AB gives

$$\begin{aligned}\{AB, C\} &= \frac{\partial(AB)}{\partial q_i}\frac{\partial C}{\partial p_i} \\ &\quad - \frac{\partial(AB)}{\partial p_i}\frac{\partial C}{\partial q_i} \\ &= A\left(\frac{\partial B}{\partial q_i}\frac{\partial C}{\partial p_i} - \frac{\partial B}{\partial p_i}\frac{\partial C}{\partial q_i}\right) \\ &\quad + B\left(\frac{\partial A}{\partial q_i}\frac{\partial C}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial C}{\partial q_i}\right) \\ &= A\{B, C\} + B\{A, C\}. \end{aligned} \tag{8.57}$$

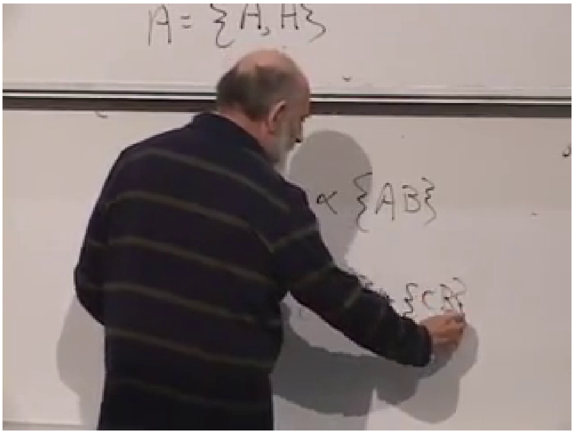

Figure 8.12: Linearity of the bracket at 01:36:20. Multiplication by a constant and addition pass through the bracket.

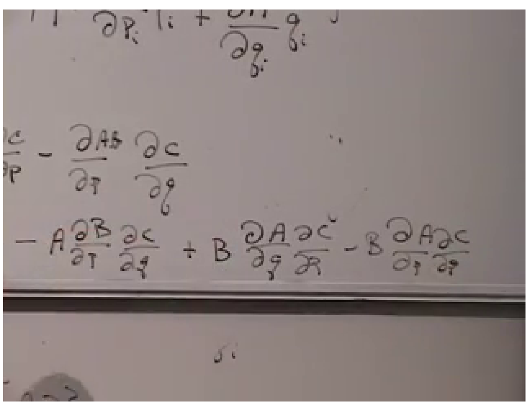

Figure 8.13: The product-rule expansion at 01:39:00. Grouping the terms yields $\{AB, C\} = A\{B, C\} + B\{A, C\}$.

Repeated indices are summed in this calculation.

Question from the audience. Besides the fundamental brackets, what rules are needed to calculate brackets of general functions?[a]

Response. Antisymmetry fixes the order, linearity handles sums and constant multiples, and the product rule handles products. Together with $\{q_i, p_j\} = \delta_{ij}$, these rules determine brackets of polynomials and, by suitable limits, the ordinary smooth functions used in mechanics.

[a]Lecture timestamp: 01:33:48–01:40:48.

We can therefore state classical mechanics in a compact way. Begin with canonical variables obeying the fundamental Poisson brackets, specify a Hamiltonian, and evolve every observable by $\dot{A} = \{A, H\}$. This formulation follows from Hamilton's equations, but it makes the algebraic structure stand on its own.

The reason for carrying the abstraction this far is not merely elegance. In quantum mechanics, Poisson brackets are replaced in a precise way by commutators, and the same algebraic pattern becomes the law of motion for operators. Heisenberg and Dirac made that connection central. What looks like an ornate reformulation of classical mechanics here will become one of the shortest routes into quantum dynamics.

CHAPTER 9

CANONICAL TRANSFORMATIONS, GENERATORS, AND SYMMETRY

Phase space is not merely a list of positions and momenta. It has a structure, and the structure is what remains meaningful when the coordinates used to describe it are changed. The Poisson bracket is the local expression of that structure. It tells us which changes of phase-space coordinates are allowed, how functions flow under those changes, and why a symmetry carries a conserved quantity with it.

9.1 Structure and Flow on Phase Space

There are many kinds of structure one can place on a space. A Riemannian metric, for example, supplies a distance between neighboring points. A Poisson structure is different: it supplies an antisymmetric bracket between functions. For mechanics the coordinates of the space are not only $q_1, \ldots, q_n$, but the full set

$$(q_1, \ldots, q_n, p_1, \ldots, p_n).$$

The question is therefore richer than a change of coordinates in ordinary space. We want to know which transformations may mix the q's and p's without destroying the form of mechanics.

It is useful to think first about familiar transformations. A finite rotation is built from many small rotations, and a finite translation from many small translations. The intermediate configurations trace a flow. The flow need not be the actual motion of a physical system; it can simply carry one description into another.

The same idea applies to phase space. Hamiltonian evolution is one flow: a phase point moves with time under the influence of H. Rotations, translations, and less familiar transformations that mix positions and momenta define other flows. Poisson brackets give one language for all of them. Newton's equations are the most concrete end of this story; the Poisson formulation is the abstract end needed here.

Ordinary mechanical systems most visibly exhibit translations and rotations, although special systems can hide additional symmetries. More advanced physics adds internal symmetries that are not motions of ordinary space at all. The common question is structural: what can be transformed while the laws retain their form? Hamilton and Poisson supplied the classical language in which that question can be asked directly on phase space.

9.2 The Poisson Algebra

Let A, B, and C be functions on phase space. The first defining property is antisymmetry,

$$\{A, B\} = -\{B, A\}. \tag{9.1}$$

The second is linearity in each entry. For ordinary numbers α and β,

$$\{\alpha A + \beta B, C\} = \alpha\{A, C\} + \beta\{B, C\}. \tag{9.2}$$

Question from the audience. Should the first relation have a minus sign when A and B are interchanged?[a]

Response. Yes. That minus sign is the essential point: $\{A, B\} = -\{B, A\}$. Without it there would be no distinction between the two orders.

[a]Lecture timestamp: 00:08:09–00:08:18.

The bracket also obeys the product rule,

$$\{AB, C\} = A\{B, C\} + B\{A, C\}. \tag{9.3}$$

By antisymmetry the corresponding rule in the second entry follows as well.

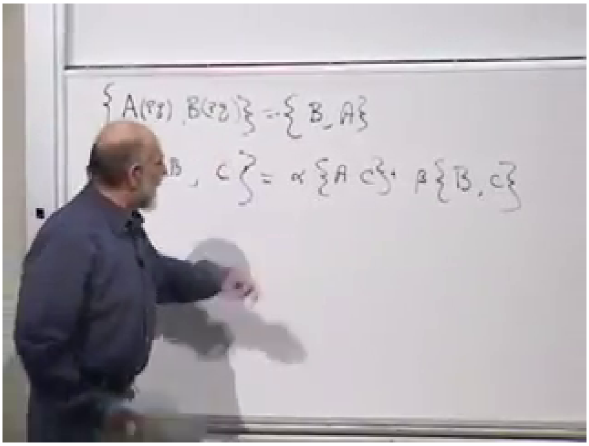

Figure 9.1: Antisymmetry and linearity on the board at 00:09:12. The minus sign in the first rule is the feature that fixes the order of every later bracket.

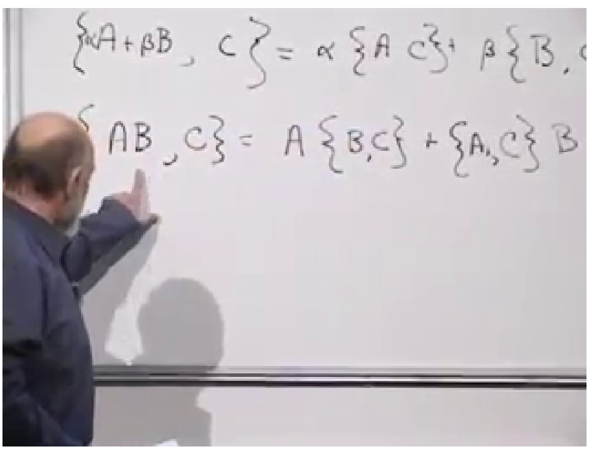

Figure 9.2: The product rule at 00:11:18. In classical mechanics the functions outside the bracket multiply as ordinary commuting numbers.

The order of multiplication outside a bracket is immaterial: $A\{B,C\} = \{B,C\}A$. The order *inside* the bracket is not immaterial, because reversing it changes the sign. This distinction is easy to overlook in classical mechanics, where functions commute, but it anticipates the ordering questions that become unavoidable in quantum mechanics.

Question from the audience. What sort of objects are A and B? Must they be real-valued scalars?[a]

Response. They are numerical-valued functions of all the phase-space coordinates. They may be added and multiplied pointwise, and complex-valued functions are allowed when useful. Here *numerical* only means that their ordinary multiplication is commutative; it is separate from the physicist's use of *scalar* to mean invariant under spatial rotations.

[a] Lecture timestamp: 00:09:26–00:14:13.

For canonical coordinates the concrete derivative formula is

$$\{A,B\} = \sum_{i=1}^{n} \left(\frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i} \right). \tag{9.4}$$

We assume throughout that the functions are smooth enough for the indicated derivatives and limiting arguments.

9.3 Canonical Brackets as Derivatives

Antisymmetry immediately gives every self-bracket:

$$\{p_i,p_i\} = 0, \qquad \{q_i,q_i\} = 0.$$

The derivative definition gives the complete canonical relations,

$$\{q_i,q_j\} = 0, \qquad \{p_i,p_j\} = 0, \qquad \{q_i,p_j\} = \delta_{ij}. \tag{9.5}$$

Consequently $\{p_i,q_j\} = -\delta_{ij}$. The Kronecker delta is one when $i = j$ and zero otherwise.

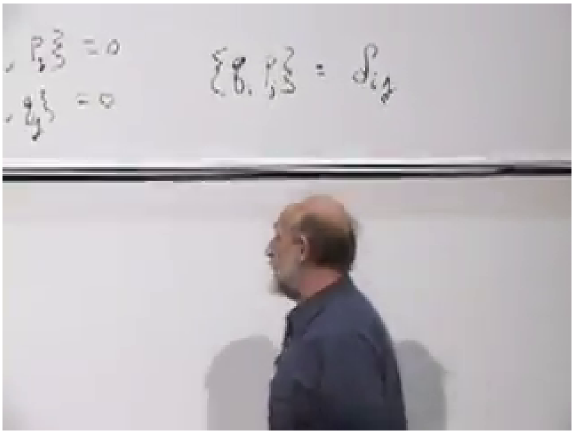

Figure 9.3: The canonical brackets at 00:17:25. The mixed bracket is the Kronecker delta; brackets among two coordinates or two momenta vanish.

Antisymmetry determines a self-bracket because it equals its own negative. For two different variables it says only $\{q, p\} = -\{p, q\}$; the normalization $\{q, p\} = 1$ comes from the canonical derivative definition.

Two useful zeroes follow at once:

$$\{F(q), K(q)\} = 0, \qquad \{G(p), L(p)\} = 0. \tag{9.6}$$

Each term in a Poisson bracket contains one derivative with respect to a coordinate and one with respect to a momentum. A nonzero contribution therefore needs a coordinate dependence on one side to meet the conjugate momentum dependence on the other.

The fundamental brackets and the product rule turn bracketing into differentiation. For one canonical pair,

$$\{F(q), p\} = \frac{dF}{dq}. \tag{9.7}$$

Rather than simply reading this from Eq. (9.4), let us recover it from the algebra.

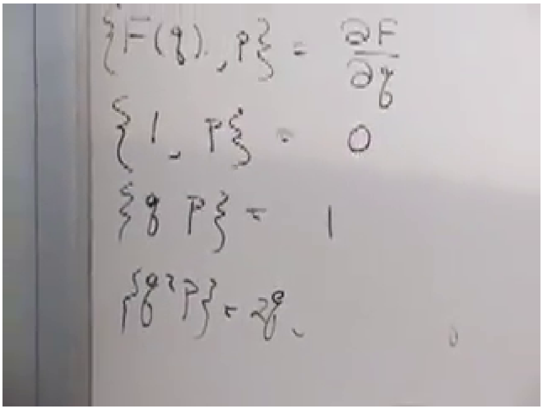

Figure 9.4: The monomial proof at 00:22:43: the cases 1, q, and q^2 reproduce their ordinary derivatives.

For the constant function,

$$\{1^2, p\} = 1\{1, p\} + \{1, p\}1 = 2\{1, p\}. \tag{9.8}$$

Since $1^2 = 1$, this says $\{1, p\} = 2\{1, p\}$, hence $\{1, p\} = 0$. Next,

$$\{q, p\} = 1.$$

For the next monomial,

$$\{q^2, p\} = q\{q, p\} + \{q, p\}q = 2q. \tag{9.9}$$

Induction gives $\{q^r, p\} = rq^{r-1}$. Linearity extends the result to polynomials and, under the smoothness assumption, to the functions used in mechanics. In many variables the paired identities hold for a general phase-space function:

$$\{F(\boldsymbol{q}, \boldsymbol{p}), p_i\} = \frac{\partial F}{\partial q_i}, \qquad \{q_i, F(\boldsymbol{q}, \boldsymbol{p})\} = \frac{\partial F}{\partial p_i}. \tag{9.10}$$

The order is deliberate: p_i goes on the right to differentiate with respect to q_i, whereas q_i goes on the left to differentiate with respect to p_i.

Question from the audience. Does the extension from polynomials require continuity of the bracket and of the functions?[a]

Response. Yes. We are assuming smooth, multiply differentiable functions and the usual continuity needed to pass from polynomial approximations to their limits.

[a]Lecture timestamp: 00:24:11–00:24:38.

9.4 Hamiltonian Evolution Is One Flow

Dynamics enters through one further relation:

$$\dot{A} = \{A, H\}. \tag{9.11}$$

A phase point follows an orbit determined by H. As the point moves, a function $A(q, p)$ changes along that orbit, and Eq. (9.11) is its directional derivative along the Hamiltonian flow.

Question from the audience. From the abstract viewpoint, are the p's and q's being assumed to depend on time?[a]

Response. Apply the same rule to the coordinates themselves: $\dot{q}_i = \{q_i, H\}$ and $\dot{p}_i = \{p_i, H\}$. Their time dependence is then defined by the Hamiltonian flow.

[a]Lecture timestamp: 00:27:20–00:27:52.

For a free particle in one dimension,

$$H = \frac{p^2}{2m}. \tag{9.12}$$

The momentum equation is

$$\dot{p} = \left\{p, \frac{p^2}{2m}\right\} = 0, \tag{9.13}$$

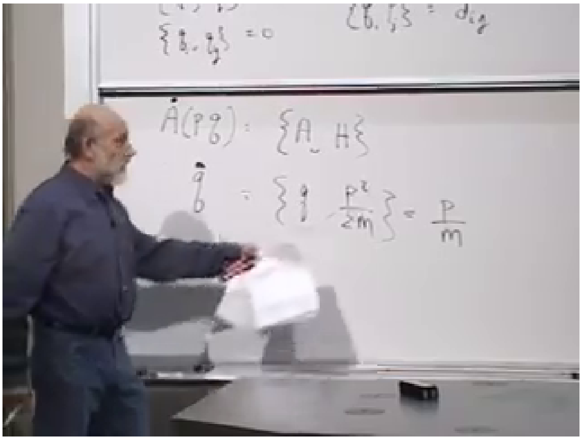

Figure 9.5: The free-particle calculation at 00:29:22. The Poisson algebra gives $\dot{p} = 0$ and $\dot{q} = p/m$ directly.

because both entries depend only on p. The coordinate equation is

$$\begin{aligned} \dot{q} &= \left\{ q, \frac{p^2}{2m} \right\} \\ &= \frac{1}{2m} \left(p\{q,p\} + \{q,p\}p \right) = \frac{p}{m}. \end{aligned} \tag{9.14}$$

Thus the abstract bracket algebra reproduces Hamilton's equations and the ordinary velocity–momentum relation.

Question from the audience. Is this Poisson formulation restricted to nonrelativistic mechanics?[a]

Response. No. The example $H = p^2/(2m)$ is nonrelativistic, but the canonical structure applies much more broadly: relativistic particles, classical fields, electromagnetism, and general relativity all admit the appropriate Hamiltonian form. Quantum mechanics replaces the classical bracket by its operator counterpart.

[a]Lecture timestamp: 00:29:57–00:30:45.

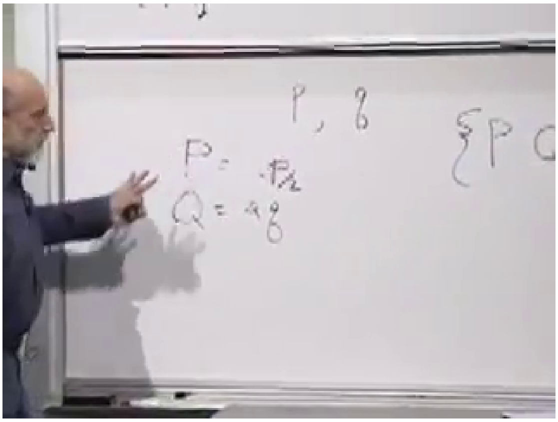

Figure 9.6: The compensated rescaling at 00:33:55. Stretching q while squeezing p preserves the canonical bracket.

9.5 Which Coordinate Changes Preserve Mechanics?

We now separate two questions. First, which changes of phase-space coordinates preserve the Poisson structure? Second, among those changes, which leave a particular Hamiltonian invariant? The first question defines a canonical transformation; the second will define a symmetry.

Take one canonical pair (q, p). A simultaneous stretch,

$$Q = 2q, \qquad P = 2p, \tag{9.15}$$

fails because

$$\{Q, P\} = 4\{q, p\} = 4.$$

A compensated stretch and squeeze,

$$Q = 2q, \qquad P = \frac{1}{2}p, \tag{9.16}$$

succeeds:

$$\{Q, P\} = \left\{2q, \frac{1}{2}p\right\} = 1.$$

For one pair, the successful transformation preserves area in the (q, p)-plane. The double stretch does not. In many dimensions, preserving total volume is not enough: the transformation must preserve the full symplectic form

$$\omega = \sum_i dq_i \wedge dp_i,$$

or equivalently the complete canonical bracket relations.

Question from the audience. Is the squeeze-and-stretch condition basically conservation of area?[a]

Response. For one q, p pair, yes: canonical transformations are area-preserving maps of that plane. With several pairs, ordinary volume preservation is only a consequence; preservation of the full symplectic structure is the stronger condition.

[a] Lecture timestamp: 00:34:51–00:35:20.

A rotation of the phase-space plane also works:

$$P = \cos\theta\, p + \sin\theta\, q, \qquad Q = -\sin\theta\, p + \cos\theta\, q. \tag{9.17}$$

Self-brackets vanish by antisymmetry, so only the mixed bracket needs work:

$$\begin{aligned}\{Q, P\} &= \{-\sin\theta\, p + \cos\theta\, q,\ \cos\theta\, p + \sin\theta\, q\}\\ &= \cos^2\theta\,\{q, p\} - \sin^2\theta\,\{p, q\}\\ &= \cos^2\theta + \sin^2\theta = 1.\end{aligned} \tag{9.18}$$

Definition 9.1. A transformation

$$(q_i, p_i) \longmapsto (Q_i, P_i)$$

is canonical when

$$\{Q_i, Q_j\} = 0, \qquad \{P_i, P_j\} = 0, \qquad \{Q_i, P_j\} = \delta_{ij}.$$

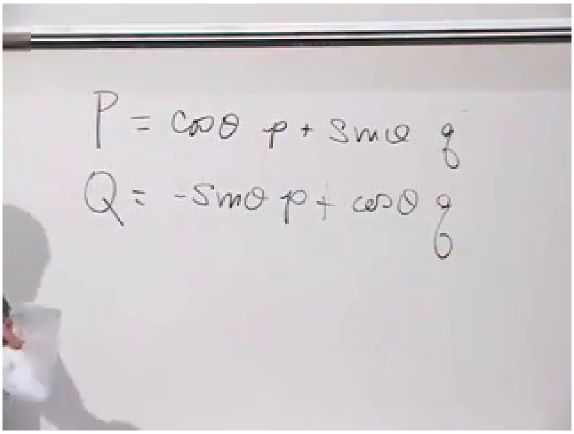

Figure 9.7: A phase-space rotation at 00:36:10. Positions and momenta are mixed, but the canonical structure survives.

Question from the audience. Are the equations of motion unchanged after a canonical transformation?[a]

Response. Their canonical form is unchanged, but the Hamiltonian must be re-expressed as a function of the new variables Q and P. The numerical formula for H need not look the same.

[a]Lecture timestamp: 00:40:04–00:40:20.

Question from the audience. Are canonical transformations examples of gauge transformations?[a]

Response. No. A gauge transformation relates redundant descriptions of the same physical state. A canonical transformation is an ordinary change of coordinates on phase space that preserves its Poisson structure.

[a]Lecture timestamp: 00:40:21–00:41:05.

Question from the audience. Can a canonical transformation carry one arbitrary physical system into another?[a]

Response. Here it means a change of phase-space coordinates for the same system. Phase spaces with the same number of degrees of freedom may be locally isomorphic, but a canonical re-description does not by itself turn a harmonic oscillator into an unrelated many-body system.

[a]Lecture timestamp: 00:41:42–00:44:52.

9.6 Infinitesimal Transformations and Their Generators

Question from the audience. Is there a more constructive test than checking all the new Poisson brackets one by one?[a]

Response. For an infinitesimal transformation there is. Build the change from a generator $G(q,p)$. Finite transformations connected to the identity can then be assembled from these small steps.

[a]Lecture timestamp: 00:45:07–00:46:31.

Write one small step as

$$Q = q + \delta q, \qquad P = p + \delta p, \tag{9.19}$$

where δq and δp are themselves functions of phase space. Infinitesimal means that terms quadratic and higher in these changes may be discarded. Expanding the transformed bracket gives

$$\begin{aligned}\{Q,P\} &= \{q+\delta q, p+\delta p\} \\ &= \{q,p\} + \{\delta q, p\} + \{q, \delta p\} + \{\delta q, \delta p\}.\end{aligned} \tag{9.20}$$

The last term is second order. Therefore the first-order canonical condition is

$$\{\delta q, p\} = -\{q, \delta p\}. \tag{9.21}$$

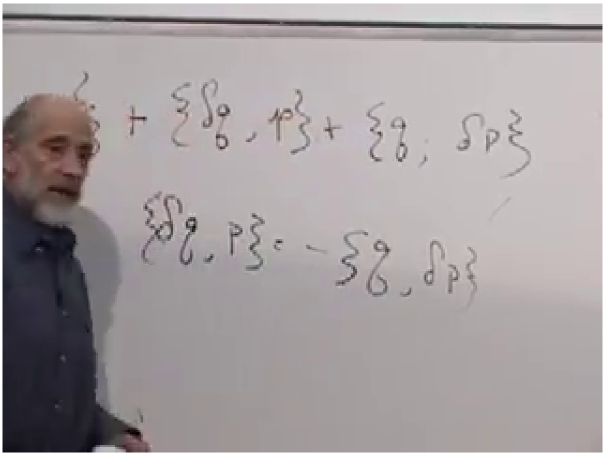

Figure 9.8: The first-order canonical condition at 00:50:44: $\{\delta q, p\} = -\{q, \delta p\}$.

Introduce a small parameter ϵ and a function $G(q, p)$. Define

$$\delta q = \epsilon\{q, G\}, \qquad \delta p = \epsilon\{p, G\}. \tag{9.22}$$

Using Eq. (9.10),

$$\delta q = \epsilon\frac{\partial G}{\partial p}, \qquad \delta p = -\epsilon\frac{\partial G}{\partial q}. \tag{9.23}$$

These formulas solve the canonical condition automatically:

$$\begin{aligned}
\{\delta q, p\} &= \epsilon\left\{\frac{\partial G}{\partial p}, p\right\} = \epsilon\frac{\partial^2 G}{\partial q\,\partial p},\\
-\{q, \delta p\} &= \epsilon\left\{q, \frac{\partial G}{\partial q}\right\} = \epsilon\frac{\partial^2 G}{\partial p\,\partial q}.
\end{aligned} \tag{9.24}$$

The equality is simply equality of mixed partial derivatives.

Question from the audience. Must the two variations use the bracket with G in the same order?[a]

Response. Yes: $\delta q = \epsilon\{q, G\}$ and $\delta p = \epsilon\{p, G\}$. The opposite signs in their derivative forms come from the canonical bracket itself, not from reversing the order.

[a]Lecture timestamp: 00:52:08–00:52:15.

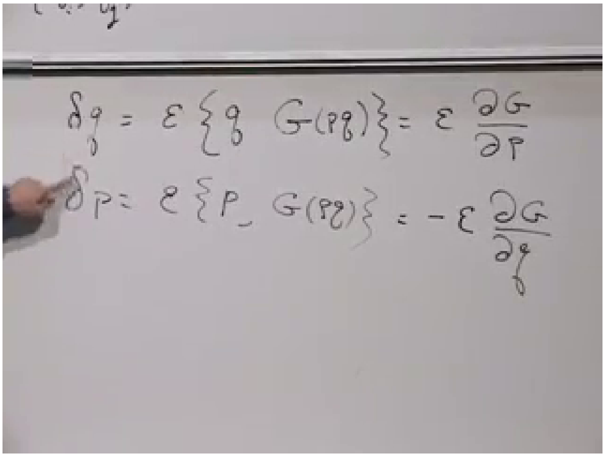

Figure 9.9: The generator equations at 00:53:55. A single function G determines both components of the infinitesimal phase-space flow.

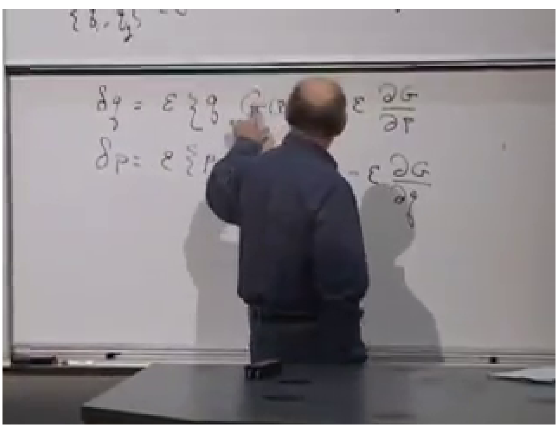

Figure 9.10: The mixed-derivative check at 00:55:50. It proves that a generator-defined infinitesimal flow is canonical.

The same generator controls every function. By the chain rule,

$$\begin{aligned}\delta A &= \frac{\partial A}{\partial q}\delta q + \frac{\partial A}{\partial p}\delta p \\ &= \epsilon\left(\frac{\partial A}{\partial q}\frac{\partial G}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial G}{\partial q}\right) \\ &= \epsilon\{A, G\}. \end{aligned} \tag{9.25}$$

For many degrees of freedom the sum over i is understood, and each δq_i may depend on every q_j and p_j, not only on the pair with the same index.

Hamiltonian evolution is the special choice

$$G = H, \qquad \epsilon = \delta t.$$

Thus time evolution is itself a canonical transformation. More generally, finite canonical transformations in the component connected to the identity are built by composing infinitesimal generator flows. This is the precise local statement needed here; global topology can introduce qualifications.

Question from the audience. If G generates a flow in the same way as H, is it just the Hamiltonian of another system?[a]

Response. Mathematically it may be treated that way. Any suitable G defines a Hamiltonian-like flow. Which function is the actual energy of the physical system is additional physical information, fixed by the system and ultimately by experiment.

[a]Lecture timestamp: 00:58:35–00:59:14.

9.7 Symmetry and Conservation

A canonical transformation preserves the form of mechanics. A symmetry of a particular system must do more: it must preserve that system's Hamiltonian. Suppose H generates the

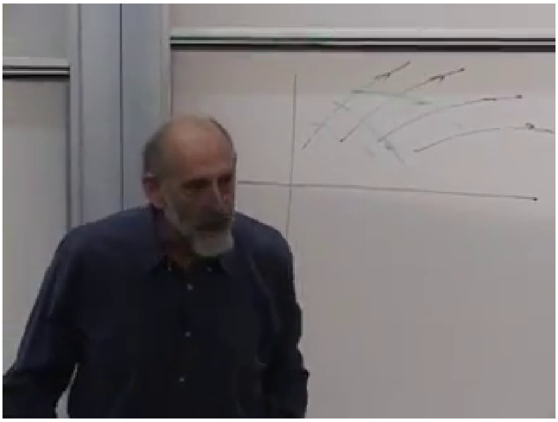

Figure 9.11: The phase-space flow picture at 01:02:10. The curved trajectories represent Hamiltonian motion; a second flow is a symmetry when it remains on the same energy surface.

physical time flow and G generates another canonical flow. In general the G-flow crosses surfaces of constant energy. It is a symmetry precisely when it runs along them.

Equation (9.25) applied to $A = H$ gives

$$\delta H = \epsilon\{H, G\}. \tag{9.26}$$

The symmetry condition is therefore

$$\{H, G\} = 0. \tag{9.27}$$

By antisymmetry, $\{G, H\} = 0$. If G has no explicit time dependence, Hamiltonian evolution gives

$$\dot{G} = \{G, H\} = 0. \tag{9.28}$$

This is the reciprocal core of Noether's theorem in Hamiltonian language. The G-flow does not change H, and the H-flow does not change G. A generator of a continuous symmetry is a conserved quantity.

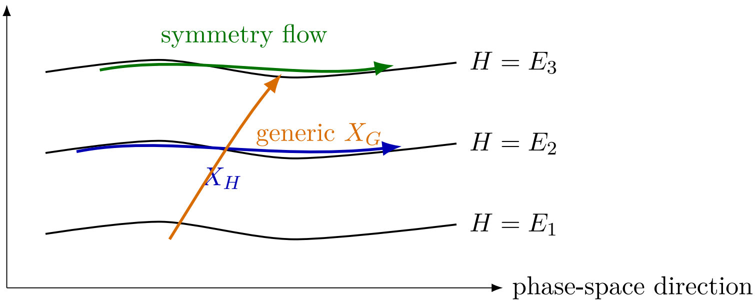

Figure 9.12: Reconstruction of the board geometry: a generic generator crosses energy surfaces, whereas a symmetry generator is tangent to them.

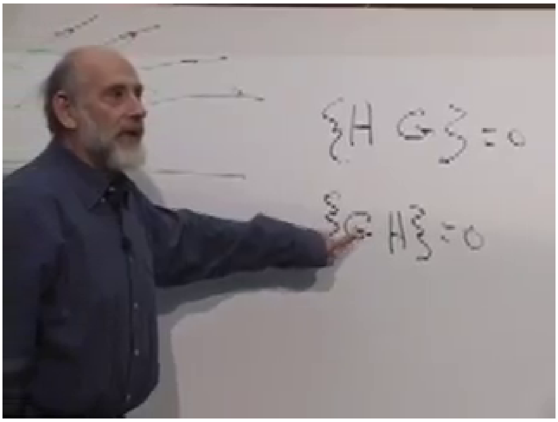

Figure 9.13: The symmetry condition at 01:05:44: $\{H, G\} = 0$ is equivalently $\{G, H\} = 0$.

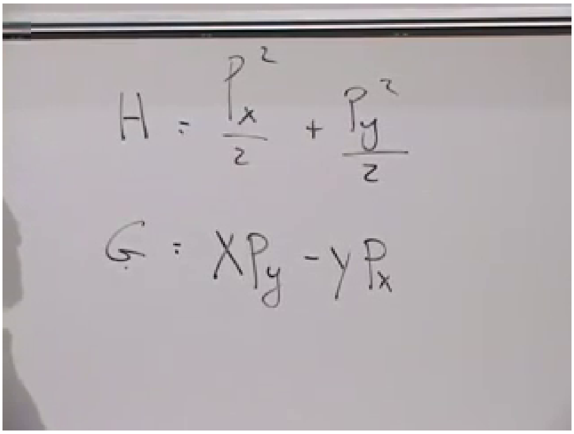

Figure 9.14: The free-particle Hamiltonian and angular-momentum generator at 01:09:35. The board sets $m = 1$; the derivation here retains m.

The first statement, that G leaves the energy unchanged, is $\{H, G\} = 0$. Antisymmetry gives $\{G, H\} = 0$, and the latter bracket is precisely $\dot{G}$. They are the same vanishing bracket read along the two flows.

Question from the audience. What is a concrete example of this abstract relation? [a]

Response. Rotational symmetry and angular momentum. For a rotationally invariant Hamiltonian, the angular-momentum generator has zero Poisson bracket with H, so angular momentum is conserved.

[a] Lecture timestamp: 01:07:56–01:08:40.

9.8 Angular Momentum and the Free Particle

Take a free particle in two dimensions:

$$H = \frac{p_x^2 + p_y^2}{2m}, \qquad G = L_z = xp_y - yp_x. \tag{9.29}$$

Only canonical pairs contribute:

$$\{x, p_x\} = 1, \qquad \{y, p_y\} = 1,$$

while $\{x, p_y\} = \{y, p_x\} = 0$. Expanding,

$$\begin{aligned} \{G, H\} &= \frac{1}{2m}\{xp_y - yp_x, p_x^2 + p_y^2\} \\ &= \frac{1}{2m}\{xp_y, p_x^2\} + \frac{1}{2m}\{xp_y, p_y^2\} \\ &\quad - \frac{1}{2m}\{yp_x, p_x^2\} - \frac{1}{2m}\{yp_x, p_y^2\}. \end{aligned} \tag{9.30}$$

The two cross-mismatched terms vanish. The remaining terms are

$$\begin{aligned} \{xp_y, p_x^2\} &= p_y\{x, p_x^2\} = 2p_xp_y, \\ \{yp_x, p_y^2\} &= p_x\{y, p_y^2\} = 2p_xp_y. \end{aligned} \tag{9.31}$$

They cancel:

$$\{G, H\} = \frac{1}{2m}\left(2p_xp_y - 2p_xp_y\right) = 0. \tag{9.32}$$

The same pair can be read in reverse. Invent a system whose Hamiltonian is the rotation generator,

$$H_{\text{rot}} = xp_y - yp_x. \tag{9.33}$$

Question from the audience. What would it physically mean to use the generator $xp_y - yp_x$ itself as a Hamiltonian? [a]

Response. Work out Hamilton's equations. The generator then produces uniform rotation: the position and momentum vectors each move around a circle.

[a] Lecture timestamp: 01:14:30–01:16:35.

Hamilton's equations give

$$\dot{x} = -y, \qquad \dot{y} = x, \qquad \dot{p}_x = -p_y, \qquad \dot{p}_y = p_x. \tag{9.34}$$

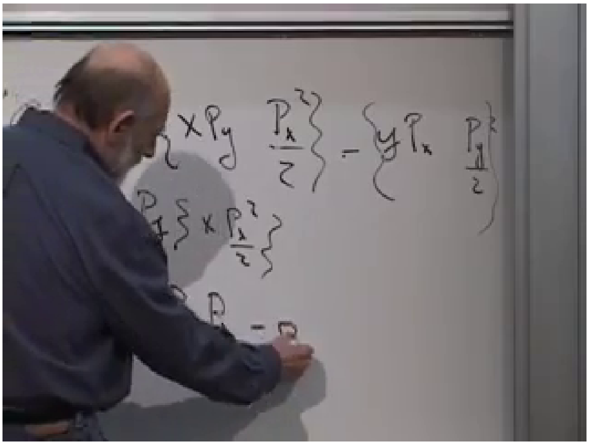

Figure 9.15: The cancellation at 01:12:50. The two surviving product-rule terms are equal and enter with opposite signs.

Both the position vector and the momentum vector rotate uniformly. In particular,

$$\ddot{x} = -x, \qquad \ddot{y} = -y,$$

so the configuration-space orbit is a circle.

Because $\{p_x^2 + p_y^2, H_{\mathrm{rot}}\} = 0$, the squared momentum is conserved in this invented system. The algebra does not privilege either member of a vanishing-bracket pair; physics tells us which one is the actual Hamiltonian.

Question from the audience. With many degrees of freedom, does each transformed coordinate depend only on the old pair carrying the same index? [a]

Response. No. There is one transformed Q_i and P_i for each index, but each may be a function of all the old q_j's and p_j's. The full set of canonical bracket relations constrains those coupled functions.

[a] Lecture timestamp: 01:17:01–01:18:09.

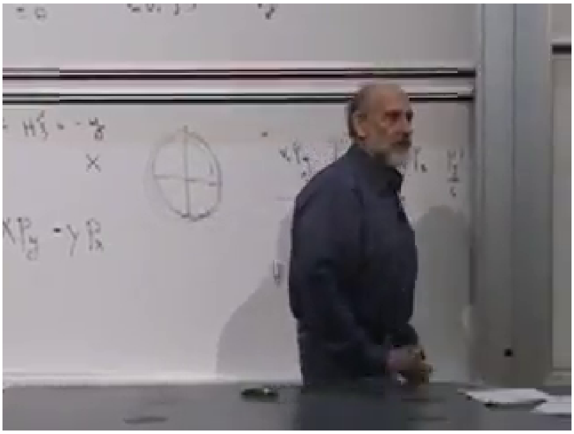

Figure 9.16: The circular orbit drawn at 01:16:35 after choosing angular momentum itself as the Hamiltonian.

Question from the audience. How is the momentum related to the velocity in Hamiltonian language? It is not always $m\dot{q}$. [a]

Response. Use the first Hamilton equation, $\dot{q}_i = \partial H/\partial p_i$, and solve it for the momenta when that relation is invertible. For $H = p^2/(2m) + U(q)$, it gives $p = m\dot{q}$. The second equation, $\dot{p} = -\partial U/\partial q$, then gives Newton's force law. More general Hamiltonians produce more general momentum–velocity relations.

[a] Lecture timestamp: 01:18:12–01:21:08.

The bracket that was introduced as an abstract algebra has now done three jobs. It generated time evolution, characterized the coordinate changes that preserve mechanics, and converted symmetry into conservation. In quantum mechanics the corresponding algebra becomes even more concrete: vanishing Poisson brackets give way to commuting operators, and generators act directly on states and observables. The classical structure already contains the map.